Promoter-Targeted activating RNAs for haploinsufficient genetic epilepsies: Design, mechanisms, and strategies for neuronal delivery

Joep van de Vondervoort

Promoter-Targeted activating RNAs for haploinsufficient genetic epilepsies: Design, mechanisms, and strategies for neuronal delivery

Boosting gene expression to solve the deficiency

Promotergerichte activerende RNA's voor genetische epilepsieën veroorzaakt door haploinsufficiëntie:
Ontwerp, mechanismen en strategieën voor neuronale afgifte

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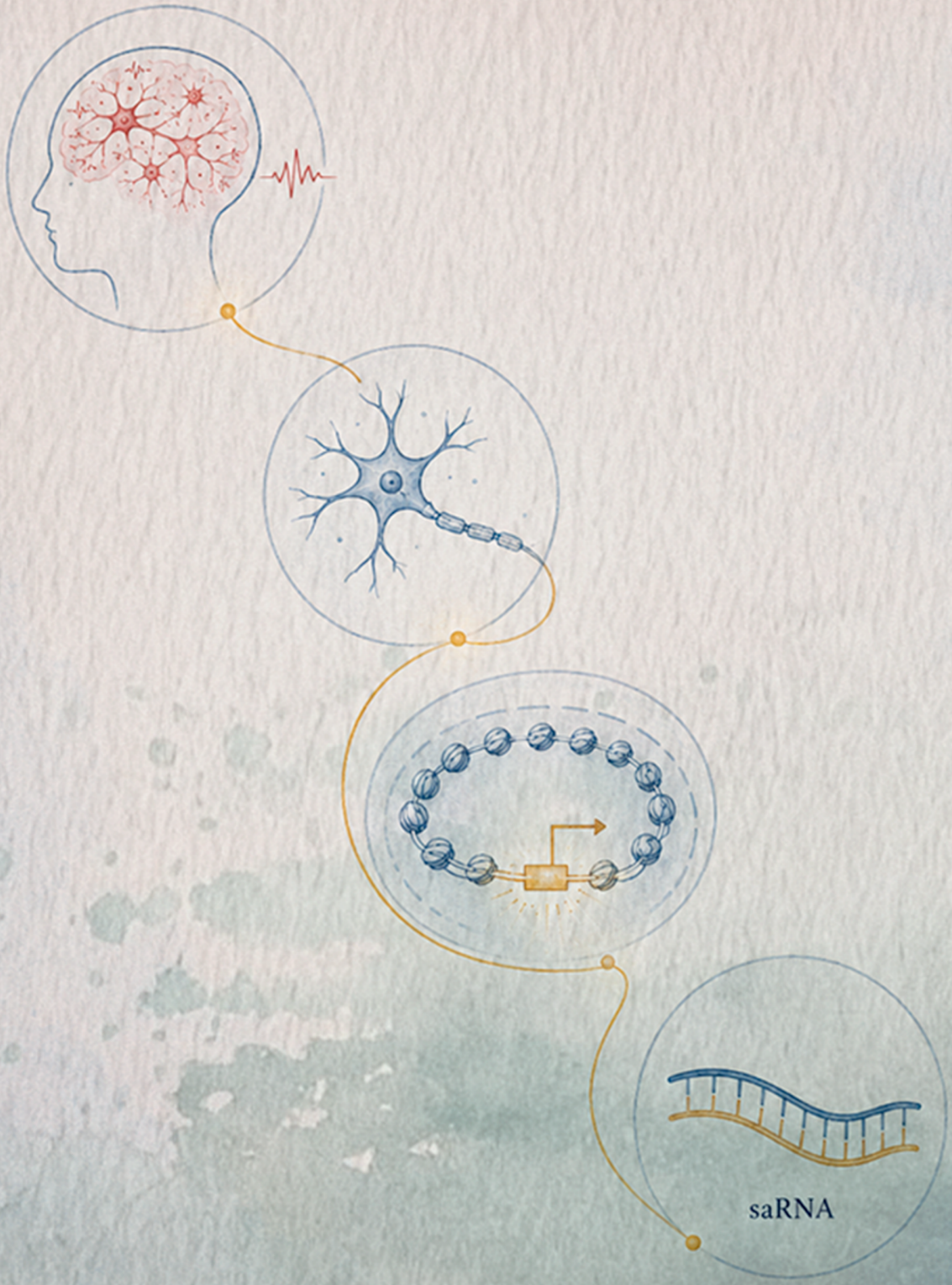
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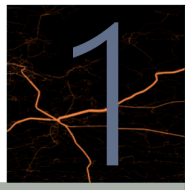

Chapter 1

General Introduction and outline of the thesis

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The clinical problem

Epilepsy is a heterogeneous neurological disorder characterized by recurrent seizures, that are caused by involuntary electrical discharges affecting part of or the entire brain. Clinical symptoms of seizures, including decreased consciousness, motor signs, behavioral changes, and sensory perceptions, are diverse as are the causes of epilepsy; ranging from structural brain lesions and inflammatory disorders, to established or presumed genetic etiologies. It has been estimated that around 50 million people worldwide are suffering from epilepsy [1]. Improper or delayed diagnosis and treatment, frequent pharmacoresistance, as well as an immense treatment gap between people with epilepsy in low- and high-income countries, contribute to epilepsy being a global health problem that needs novel ways for fast and reliable diagnosis and improved personalized treatment.

Among the spectrum of epilepsy syndromes, developmental and epileptic encephalopathies (DEEs) are a severe, early-onset subgroup in which both etiology and ongoing epileptic activity drive developmental slowing or regression and pharmacoresistant seizures. DEE is one of the most challenging epilepsies to treat, in part due to its variable expression and clinical heterogeneity that underlines the need for personalized treatment. Nowadays, genetics are instrumental to diagnose many of the DEE subtypes, each of which being defined by syndrome phenotype and etiology. For around 60% of DEE, a monogenic mutation in one of more than one hundred genes can be found in patients referred for Next Generation Sequencing DNA diagnostics. One of the most frequent genetic causes of monogenic DEE is Dravet syndrome (DS).

DS is a prototypical DEE with onset in the first year of life, multiple seizure types, cognitive and behavioral comorbidities, and increased SUDEP risk [2]. The majority of DS cases arise from de novo loss-of-function (LoF) variants in the *SCN1A gene*, leading to haploinsufficiency of the $Na_v1.1$ sodium channel in inhibitory interneurons and network disinhibition [3]. The identification of a clear disease causing LoF variant in *SCN1A* has been the clearest example of how progress in our genetic understanding of epilepsy etiology can improve the lives of patients. It has shown that early genetic diagnosis shortens the diagnostic odyssey, preventing unnecessary clinical follow-up and burdening diagnostics, prevents trial and error pharmacological treatment, and guides early optimal treatment, which in the case of DS includes the avoidance of sodium channel blockers, which are known to aggravate symptoms. Nevertheless, DS remains difficult to treat despite newer available antiseizure medications (ASM), such as stiripentol, cannabidiol, and fenfluramine. Seizures in many patients remain incompletely controlled and intellectual and behavioral comorbidities remain untreated, underscoring the need for disease-modifying strategies that address the gene-level cause rather than symptoms.

Gene therapies are among the most promising new treatment strategies. At the time of writing, only one gene therapy for DS is being tested in clinical trials, pioneered by STOKE therapeutics. Their solution involves a small RNA that can increase the correct use of already available pre-mRNA. This dependance on available pre-mRNA also limits its approach.

Therefore, other strategies that aim to boost expression from the intact allele (*i.e.*, increase *SCN1A* output in interneurons) need to be explored [4].

The studies described in this thesis therefore aimed to explore novel approaches to treat DS. We focused on modalities that can upregulate transcription, including CRISPR activation (CRISPRa), antisense approaches that relieve negative regulation (*e.g.*, NAT blockade), and small activating RNAs (saRNAs), which harness Argonaute-dependent RNA activation (RNAa) at promoters. Proof-of-concept that raising *Scn1a* can modify disease comes from antisense oligonucleotides that increase productive *Scn1a* transcript and reduce seizures and SUDEP in a DS mouse model, supporting the broader feasibility of gene-activation therapies [5]. saRNAs offer a compact, drug-like alternative that does not require large protein cargos. In particular, our work aimed to evaluate whether saRNA-mediated promoter activation can be deployed for neurogenetic epilepsies, using *SCN1A*/Dravet as the primary model, and to define (i) efficacy in human neural cells, (ii) mechanistic determinants of RNAa in this context, and (iii) delivery steps needed to move toward translation.

Background and rationale

From mechanism to target

Many studies, both *in vitro* and *in vivo*, have shown that loss of $Na_v1.1$ function in fast-spiking interneurons is sufficient to produce DS-like hyperexcitability, rooting the disorder in impaired inhibition [6]. Because excitatory neurons rely less on $Na_v1.1$, selectively restoring *SCN1A* in GABAergic interneurons should, in principle, rebalance networks; whereas further sodium-channel blockade can worsen DS. However, DS symptoms include more than just seizures and low expression of *SCN1A* in other cell types, including astrocytes, could have an influence on the full spectrum of disease. In this thesis we focused nevertheless on cell cultures that were either convenient for screening, or the AF22 cell line, which has been reported to represent mostly GABA-ergic neurons (see chapter 2).

RNA activation (RNAa) and saRNAs

RNA activation (RNAa) is a relatively novel technique that uses short double-stranded RNAs. Small activating RNAs (saRNAs) are approximately 21 nucleotides long and designed as duplex RNAs to target promoter-associated sequences, leading to increased gene transcription [7]. Since RNAa was first described in 2006, saRNAs have been reported for more than two dozen genes, but only a limited number have undergone rigorous mechanistic validation, and only one (MTL-CEBPA) has progressed to clinical testing [8], [9]. Its mechanism remains to be confirmed, but first experiments suggest that the saRNA guide-strand loads into AGO2 that enables recruitment of an activation hub, the RNA-induced transcriptional activation (RITA) complex (AGO2, RNA helicase A, CTR9), with increased

RNAPII engagement and promoter activity (Figure 1) [10]. RNAa typically shows delayed onset (24–48 h) and durability over days, consistent with chromatin engagement. While extensively characterized at *CDKN1A* (P21) and other loci in dividing cells, neuronal contexts remain underexplored, motivating the present work.

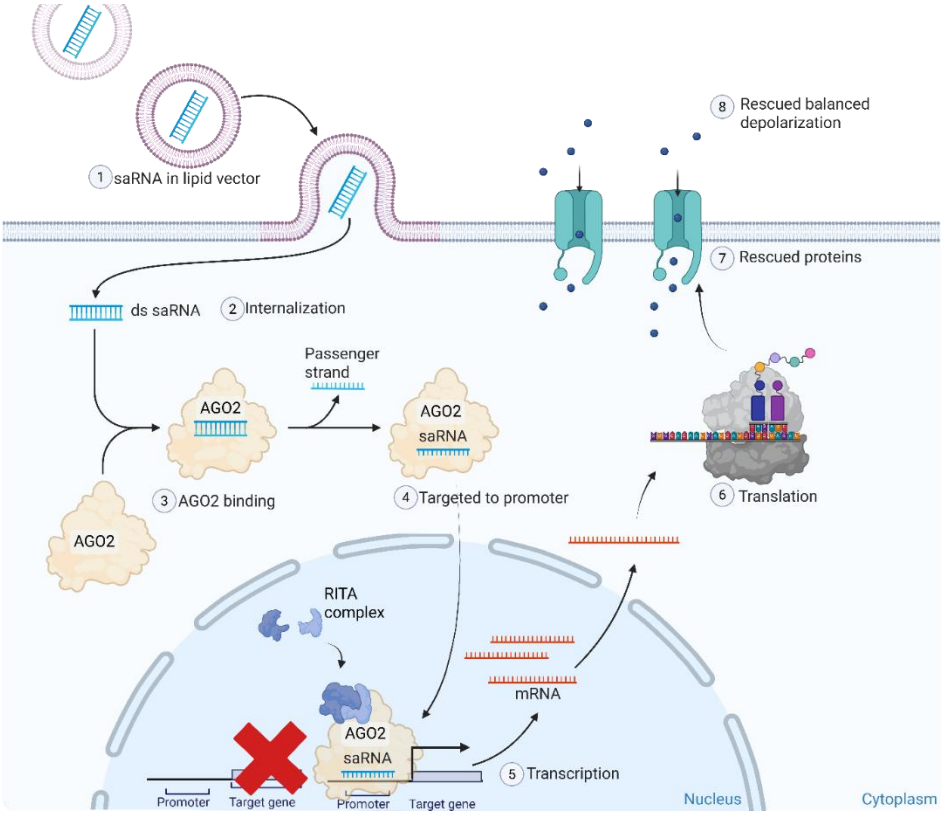

Figure 1. Hypothetical sequence of events underlying saRNA-mediated activation of SCN1A in human neurons. *(1) A double-stranded saRNA is formulated in a lipid-based delivery vehicle to facilitate cellular uptake. (2) The saRNA–lipid complex is internalized by the target cell. (3) In the cytoplasm, the duplex is recognized by AGO2; the passenger strand is discarded, yielding an AGO2–guide saRNA complex. (4) The AGO2–saRNA complex translocates to the nucleus and is directed to promoter-associated target sequences at the SCN1A locus, where it can nucleate a RITA-like activation complex (e.g. AGO2, RHA, CTR9). (5) Recruitment of this activating machinery increases RNA polymerase II–driven transcription of SCN1A, resulting in elevated mRNA. (6) The increased SCN1A* mRNA is exported and translated into $Na_V1.1$ protein. (7) Higher $Na_V1.1$ levels restore inhibitory interneuron channel density. (8) At the circuit level, this is expected to improve inhibitory control and rebalance depolarization/excitability. This figure represents a working model extrapolated from promoter-targeted *RNAa studies in human cells; several steps, particularly nuclear targeting and promoter engagement in neurons are experimentally tested in subsequent chapters rather than assumed. Created with Biorender.*

Cellular models to bridge feasibility and relevance

The major challenge for any RNA- or DNA therapeutic development for neuronal diseases is the lack of robust and easy to generate and use cellular models. Primary human neurons are notoriously difficult to culture and transfection is even more challenging at scale. Long-term neuroepithelial-like stem (lt-NES) cells such as AF22, derived from human pluripotent stem cells, offer proliferative neural progenitors that can be differentiated into post-mitotic neurons, enabling a two-stage evaluation (screening in progenitors, functional testing in neurons) in a human context [11]. In this thesis we also used a standard HeLa, a cervical cancer cell line, as a transfection-friendly comparator for initial screening and some mechanistic assays.

The delivery bottleneck

Any RNA therapeutic for the CNS must overcome uptake, endosomal escape, and nuclear access in post-mitotic neurons. Current transfection strategies that have been used for decades are optimal for dividing cells only. Fully differentiated neurons do not divide and therefore, transfection efficiencies are generally low. In vitro transfection techniques such as the use of lipofectamine, are too aggressive for clinical translation. Therefore, less toxic transfection methods that mimic natural transfer of macromolecules between cells should be explored. A promising delivery method uses lipid nanoparticles (LNPs) that are rapidly evolving with strategies to cross or engage the blood–brain barrier and transfect brain-resident cells, including ligand- or peptide-functionalized designs. These developments frame the delivery requirements for saRNAs and have been explored in this thesis in chapter 5.

Aims and outline of this thesis

Chapter 2: Designing and screening saRNAs to upregulate *SCN1A*.
We designed promoter-targeted saRNAs based on established RNAa rules and screened them in HeLa, AF22 progenitors and differentiated AF22 neurons. Readouts included qPCR for *SCN1A* mRNA and calcium imaging for neuronal activity assays to link transcriptional gains to cellular function.

Chapter 3: Mechanistic link to AGO2 in neural cells.
We tested the requirement for AGO2 using loss-of-function approaches and evaluated locus engagement specifically by ChIP-qPCR for AGO2 at the *SCN1A* promoter during saRNA treatment. These experiments asked whether the AGO2-anchored component of the RITA paradigm applies in human neural cells.

Chapter 4: Extending saRNA activation to additional epilepsy genes
We broadened saRNA design and testing to other haploinsufficient genes implicated in focal epilepsies and mTORopathies: *TSC1*, *TSC2*, *DEPDC5*, *NPRL2*, *NPRL3*: and the chromatin remodeler *CHD2*. Screens in HeLa were followed by validation in AF22 derivatives, asking

whether shared sequence/structural features predict responsiveness and whether promoter context constrains RNAa across loci.

Chapter 5: Delivery optimization in neurons
We compared optimized lipofection to LNP formulations tailored for neuronal uptake, quantifying uptake, viability, and on-target activation (qPCR) in differentiated AF22 neurons. Results are discussed in the context of current CNS-directed LNP advances and the need for eventual *in vivo* delivery.

Chapter 6: General discussion and translational outlook
We integrated efficacy, AGO2-mechanistic data, locus-to-locus variability, and delivery performance to outline a translational path for saRNA activation in DEEs and related haploinsufficient brain disorders and positioned saRNAs alongside antisense and AAV-based strategies.

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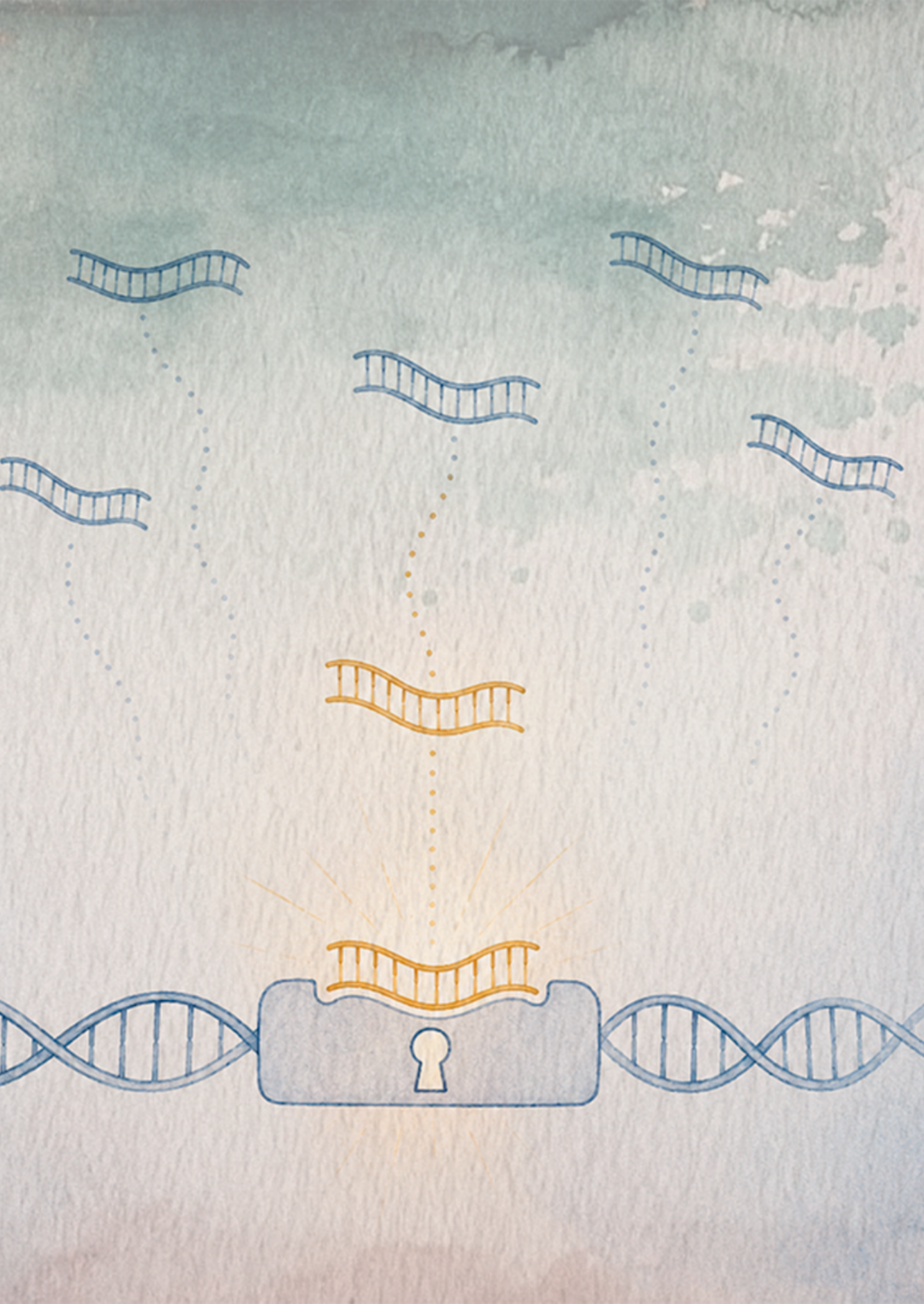

Chapter 2

Designing and screening saRNAs to upregulate SCN1A

Chapter 2

First proof of principle for small activating RNAs treating haploinsufficiency in severe monogenic brain disorders:

the case for Dravet syndrome

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Abstract

This study explores the potential of small activating RNAs (saRNAs) to upregulate *SCN1A* as a therapeutic strategy for Dravet syndrome, a severe monogenic epileptic encephalopathy caused by haploinsufficiency of the *SCN1A* gene. We evaluated a panel of ten *SCN1A*-targeting saRNA candidates (SCN1A-6 to SCN1A-15) in HeLa cells, with selected candidates subsequently tested in undifferentiated AF22 neural stem cells (NES) and differentiated AF22 neurons. Among them, saRNA-10 consistently induced strong *SCN1A* mRNA upregulation in HeLa and AF22 NES cells, while saRNA-12 and saRNA-14 also demonstrated significant upregulation, as confirmed by quantitative PCR and RNA sequencing. A more modest but reproducible increase was observed in differentiated AF22 neurons. Western blot analyses confirmed corresponding $Na_v1.1$ protein upregulation, validating successful translation of transcript changes.

Functional calcium imaging in differentiated AF22 neurons revealed subtle changes following saRNA treatment, with a slight reduction in calcium peak frequency. In contrast, siRNA-mediated *SCN1A* knockdown markedly reduced calcium transient frequency, slope, and width, confirming assay sensitivity to *SCN1A* levels. Pharmacological testing with veratridine (pan-sodium channel activator), PTZ (GABA_A antagonist), and a $Na_v1.1$-specific activator revealed intact sodium channel function but limited response to enhanced $Na_v1.1$ activity alone. This likely reflects the low inhibitory tone and compensatory homeostatic mechanisms within these mixed neuronal cultures, where excitatory neurons predominantly drive network activity.

Our findings demonstrate that saRNAs can robustly and specifically enhance *SCN1A* expression, particularly in proliferating cells, supporting their potential as a gene-upregulating therapeutic platform. While functional outcomes in wild-type neurons were modest, this suggests minimal risk of hyperexcitability from overexpression, enhancing the safety profile for clinical translation. These results provide proof-of-concept for saRNA-mediated *SCN1A* upregulation and lay the groundwork for evaluation in disease-relevant models where correcting *SCN1A* haploinsufficiency could demonstrate clear therapeutic benefit.

Keywords

'Small-activating RNAs', 'Dravet syndrome', '*SCN1A*', 'RNA-therapy', 'Epilepsy'

Introduction

Dravet syndrome (DS) is a severe neurodevelopmental and epileptic disorder that is characterized by recurrent seizures, cognitive and motor impairments, and a high risk of sudden unexpected death in epilepsy (SUDEP) [1]. The estimated incidence of DS ranges from 1:20,000 to 1:40,000, and the mortality rate associated with this syndrome is alarmingly high [2], [3]. Treating DS presents significant challenges due to the highly drug-resistant nature of the seizures, often requiring multiple anti-seizure medications (ASMs), while some ASMs may even exacerbate seizures.

Initial treatment approaches for DS involve broad-spectrum ASMs such as valproate (VPA) and clobazam (CLB). However, these are generally insufficient to control seizures effectively. Over time, three ASMs have been specifically approved for the treatment of seizures in DS patients: stiripentol (STP), cannabidiol (CBD), and fenfluramine, a serotonin-releasing drug (FFA). These "add-on" therapies, primarily used in conjunction with VPA/CLB, serve as escalation treatments, with the choice depending on factors such as availability in different countries, patient characteristics, and caregiver preferences. Although these compounds provide additional seizure control, they rarely lead to seizure freedom, and their long-term effectiveness is still under investigation [4]. Importantly, treatment is primarily focused on seizure reduction, and it remains unclear whether early treatment will improve neurodevelopment and other comorbidities. The absence of an effective treatment places a significant burden on society and the healthcare system, including the low quality of life of individual patient and their family. Apart from the high economic cost, the life-long care needed for Dravet syndrome patients has been reported to significantly affect physical and emotional health of parents and other day-to-day caregivers.

Therefore, novel treatments that directly address the underlying pathophysiological mechanism of the disease need to be explored. Targeted therapies may not only better suppress seizures but also modify the disease course of DS and its comorbidities. DS arises from loss-of-function (LoF) mutations in the *SCN1A* gene that encodes the $Na_v1.1$ voltage-gated sodium channel, resulting in abnormal neuronal excitability and seizures [4]. Experimental models using *Scn1a* knockout mice have demonstrated that a reduction in Na^+ ion currents leads to loss of function of inhibitory neurons and seizures [5].

Genetic therapies targeting the *SCN1A* gene have emerged as promising approaches for treating DS, showing encouraging results in various stages of clinical development. For instance, an antisense oligonucleotide (ASO) called STK-001 has been demonstrated to increase *Scn1a* mRNA levels, elevate $Na_v1.1$ protein expression, reduce seizures, and improve survival in the *Scn1a*+/- mouse model of DS using Targeted Augmentation of Nuclear Gene Output (TANGO) [6]. Interim results from the ongoing Phase 1/2a MONARCH (NCT04442295) and ADMIRAL (ISRCTN99651026) trials also show promising results, suggesting an effective and safe reduction of seizures. Another successful approach involves upregulating *Scn1a* levels in mouse models using a dCas9-VP160 complex. However, the

delivery of these constructs into the brain is often challenging, and their administration can be immunogenic [7], [8].

In the realm of innovative therapeutic strategies, small activating RNAs (saRNAs) emerge as a promising approach for treating Dravet Syndrome (DS). SaRNAs represent a novel class of small RNA molecules that are uniquely capable of targeting and inducing the expression of specific genes. The underlying mechanism of saRNAs involves their interaction with the Argonaute protein 2 (AGO2). This interaction guides the saRNA-AGO2 complex to the gene's promoter region, where binding occurs. Subsequently, this binding event facilitates the recruitment of the RNA-induced transcription activation (RITA) complex, which plays a pivotal role in stimulating the transcription of the targeted gene [9].

Since the discovery of saRNAs, numerous saRNAs have been investigated [10], [11]. In 2006, Li et al. discovered that short double-stranded RNAs targeting the promoter of P21 could upregulate both P21 mRNA and protein expression [12]. A saRNA-increasing expression of MTL-CEBPA, developed by MiNA Therapeutics, has shown significant success. In combination with the protein kinase inhibitor sorafenib, MTL-CEBPA has entered phase 2 trials for treating patients with advanced hepatocellular carcinoma (HCC) resulting from hepatitis B or C infection (NCT identifier: NCT04710641) [13], [14].

In this study, our objective was to develop a saRNA capable of upregulating *SCN1A* expression and potentially serving as a novel therapy for DS. We evaluated ten saRNA candidates (SCN1A-6 to SCN1A-15) targeting two different promoters upstream of the *SCN1A* gene and assessed their efficacy in vitro. The aim was to identify saRNA(s) capable of effectively increasing *SCN1A* mRNA and protein expression levels, with the ultimate goal of identifying a saRNA candidate that could be further developed as a therapy for DS.

Material and methods

Cell culture

HeLa cells were cultured in Dulbecco's modified Eagle's medium (DMEM) with high glucose supplemented with 10% of fetal bovine serum and 1% penicillin/streptomycin. The induced pluripotent stem cell (iPSC)-derived, long-term self-renewing neuro-epithelial stem cell line (AF22) was obtained from the human iPS core facility at Karolinska Institutet [15]. We used this cell line as a more representative model which has a *SCN1A* expression and spontaneous calcium activity upon differentiation. For AF22 NES cells, we used NES medium composed out of DMEM-F/12 medium + glutamax (Gibco) and supplemented this medium with 1% N2 supplement (Gibco) and 0.1% B27 (Gibco), 10 ng/mL Epidermal Growth Factor (EGF) (Thermo Fisher), 10 ng/mL basic Fibroblast Growth Factor (bFGF2) (Thermo Fisher), and 1% Penicillin/Streptomycin (Thermo Fisher). Undifferentiated AF22 cells were cultured on plates coated with 0.1% gelatin solution. AF22 cells were cultured in a differentiation medium containing DMEM/F12 + glutamax, 1% N2, 1% B27, and 1% Penicillin/Streptomycin to

encourage differentiation. The AF22 cells were differentiated after four weeks of being cultured in the differentiation medium on poly-ornithine and laminin (both 1:500) coated plates. All cells used in this study were grown at 37°C in a humidified 5% CO_2 incubator.

Transfection

One day before transfection, cells were seeded in a 12-wells plate to be at a ~70% confluency (~200.000- 250.000 cells) on the day of transfection. For transfections of HeLa cells, the medium was changed to Opti-MEM™ (Thermo Fisher) ~4 hours prior to transfection. Lipofectamine RNAiMAX (Thermo Fisher) was used to transfect the saRNAs and siRNA into the cells at a final concentration of 50 nM. Cells were harvested 72 hours post-transfection. The scrambled and P21 saRNAs were used as negative and positive controls, respectively [9,14,22]. We always confirmed if the transfection was successful by confirming the efficacy of the saRNA targeting P21 (supplementary 7).

Microscopy

Microscopy was performed using the Zeiss Cell Observer microscope (Zeiss) and the EVOS M5000 microscope (Thermo-Fisher) for the differentiated AF22 brightfield imaging. The Zeiss Cell observer was used for the localization of fluorescently labeled saRNAs 72 hours after transfection in HeLa and AF22 cells.

The Plan-APOCHROMAT 40x/1.3 Oil objective was used. Cells were seeded onto 4 Well Ibidi μ-Slides for microscopy. Before imaging, staining of the nuclei was performed with Hoechst 34580 (Sigma Aldrich) by adding 1 μg/ml per well.

For immunofluorescence microscopy, differentiated AF22 cells were stained for neuronal and astrocytic markers MAP2, NeuN, and GFAP. Initially, cells were permeabilized with PBS containing 0.2% Triton X-100 for at least 5 minutes. Subsequently, cells were blocked with 3% BSA in PBS containing 0.2% Triton X-100 for 30 minutes. Primary antibodies used were MAP2 (Abcam, polyclonal chicken IgY, ab92434; 1:1000), NeuN (Abcam, monoclonal mouse IgG2b, ab104224; 1:500), and GFAP (Abcam, polyclonal rabbit IgG, ab7260; 1:500), incubated for 1 hour at room temperature.

Following three 5-minute washes in PBS containing 0.2% Triton X-100, the cells were incubated with secondary antibodies Alexa 568 goat anti-chicken IgG (Fisher Scientific, A11041), Alexa 488 goat anti-mouse IgG (Fisher Scientific, A11029), and Alexa 647 goat anti-rabbit IgG (Fisher Scientific, A21245), each diluted 1:500 in PBS-Tx100-BSA, for 30 minutes at room temperature. Cells were then washed once with PBS containing 0.2% Triton X-100, stained with DAPI (1:2500 in PBS-Tx100-BSA) for several minutes, and finally washed twice with PBS.

Slides were air-dried and mounted using approximately 5 μL of Prolong Diamond Antifade Mountant (ThermoFisher Scientific, P36970) onto cleaned glass slides, prepared by rinsing

with 96% ethanol. Slides were covered with the stained coverslips placed upside down and allowed to cure at room temperature before imaging.

Images were captured using the ZEISS Axio Observer live cell microscope equipped with a Hamamatsu camera. Imaging was conducted using a 10x EC Plan-Neofluar objective (NA 0.30). Fluorescence was acquired through specific channels: DAPI (emission wavelength 414-450 nm), Alexa 568 (MAP2, emission wavelength 573-613 nm), and Alexa 647 (GFAP, emission wavelength 663-733 nm).

RT-PCR

One-step qPCR was performed using ViiA 7 Real-Time PCR System (Thermo Scientific) with the following run method: first 48°C for 15 minutes and 95°C for 10 minutes, following 40 cycles of 95°C for 15 seconds and 60°C for 1 second. Each reaction contained 3,75 µl diluted RNA, 5 µl Taqman RT-PCR mix (Applied Biosystems), 0,5 µlassay *SCN1A*-FAM (Hs00897353_m1, Thermo Scientific), 0,5 µl assay reference gene *GUSB*-VIC (Hs00939627_m1), Thermo Scientific), and 0,25 µl of Taqman Enzyme mix (Applied Biosystems) in a total volume of 10 µl. Ct values of genes of interest were normalized to Ct values of the reference gene: *GUSB*. ΔCt values were normalized against averages of ΔCt of the non-targeting control samples to calculate the ΔΔCt. Finally, the 2^-(ΔΔCt) formula was used to calculate the fold change. A one-way ANOVA test was performed on the ΔΔCt values to determine if the differences were significant.

Western blot

For protein extraction, cells were harvested by scraping them from the plate in 50 µL RIPA Lysis and Extraction buffer (Thermo Scientific) with cOmplete EDTA-free protease inhibitor cocktail (Roche). Lysed cells were centrifuged at 16000 rpm for 3 minutes at 4°C, after which the supernatant was transferred to a new Eppendorf Tube.

Western blot experiments were performed to measure relative protein expression after different treatments. For Western Blotting, approximately 30-50 µg of protein samples were diluted in Bolt LDS sample buffer (4X) (Invitrogen) and NuPage Sample Reducing agent (10X) (Invitrogen). Samples were boiled at 70°C for 10 minutes. Next, samples were loaded on a NuPage 3-8% Tris-Acetate gel (Invitrogen), and the gel was run at 150 V for 50 minutes. Afterward, according to the manufacturer's instructions, the gel was transferred onto a nitrocellulose membrane using the Trans-Blot Turbo Mini Transfer Pack (Bio-Rad Laboratories). Subsequently, the membrane was blocked for 1 hour at room temperature with blocking buffer (5% milk in PBST) and afterward incubated overnight at 4°C with primary antibody against $Na_v1.1$, Rabbit Anti- $Na_v1.1$ (Alomone labs) diluted in blocking buffer (1:200). Next, the membrane was incubated with primary antibody against Tubulin, Mouse anti-alpha-Tubulin (Invitrogen) diluted in blocking buffer (1:4000), for 1 hour at room temperature. Afterward, the membrane was incubated for 1.5 hours at room temperature with the appropriate secondary antibodies (Goat anti-Rabbit and anti-mouse IgG, Life

Technologies) diluted in blocking buffer (1:3000). Protein bands were visualized using the Typhoon NIR Biomolecular imager (GE Amersham) and the molecular weight of proteins was estimated by comparison with the Spectra multicolor high-range protein ladder (Thermo Scientific).

Calcium imaging

For calcium imaging, differentiated AF22 cells were loaded with 5,21 μM Fluoro-4 AM (ThermoFisher Scientific), diluted in Hank's Balanced Salt Solution (HBSS), and incubated for 30 minutes at 37°C. Subsequently, the loading medium was replaced either with fresh HBSS or HBSS supplemented with chemical treatments: $Na_V1.1$ activator BI-7150 (Boehringer Ingelheim), Pentylenetetrazole (PTZ, Sigma-Aldrich), or Veratridine (Tocris).

Imaging was conducted using the Leica THUNDER Imager Live Cell microscope, maintaining cells at optimal physiological conditions (37°C, 5% CO2). Recordings lasted 10 minutes, capturing images every 2.5 seconds, ensuring high temporal resolution while mitigating phototoxic effects and dye bleaching. Fluorescence excitation was performed using a 506 nm LED, with emission signals collected at 10x magnification.

Cell segmentation and identification were performed using Cellpose [16], a deep learning-based algorithm optimized for accurate detection of cell boundaries in fluorescence microscopy videos. Calcium dynamics were quantified using FIJI (ImageJ). Subsequent analysis of calcium transients employed a custom Python script (available on GitHub: JoepVondervoort/Calcium-imaging-analysis). Peak identification utilized the SciPy find_peaks function with set parameters for prominence (>0.2) and minimum peak separation (5 frames). Kinetic parameters analyzed included the number of peaks, amplitude, slope, and full width at half maximum (FWHM). Cells exhibiting two or more peaks were classified as active, thereby reducing false positives from single transient events likely arising from artifacts. Outlier removal for kinetic parameters was performed using the z-score method with a default threshold of 3.0. Statistical evaluations included normality assessments (Shapiro-Wilk tests) and subsequent pairwise comparisons (Mann-Whitney U or t-tests, depending on data distribution), facilitating rigorous determination of significant differences among experimental groups.

Flow Cytometry Analysis of Cell Death in Transfected AF22 Cells

To quantify cell death in AF22 cells post-transfection, flow cytometry was performed using the FACSCelesta Cell Analyzer (BD). Initially, both adherent and non-adherent cells were collected by combining the original culture medium with the cell suspension obtained after detachment using TrypleExpress. This mixture was centrifuged at 500 x g for 5 minutes to pellet the cells.

The supernatant was discarded, and the cell pellet was resuspended in 500 μL PBS. For staining dead cells, 5 μg of Propidium Iodide (Sigma-Aldrich) was added per 500 μL of

suspension and incubated in the dark for 15 minutes. The stained cells were then analyzed using the FACSCelesta to determine the proportion of cell death.

RNA Sequencing

RNA sequencing was performed to evaluate the off-target effects of small activating RNA (saRNA) transfections and to validate qPCR findings. The sequencing was conducted by USEQ using the Illumina NextSeq2000 platform. To ensure thorough analysis, sequences similar to the saRNA were identified using BLAST and checked for differential expression in the sequencing data.

Data was processed using the RNASeq-NF pipeline, available on the UMC Genetics Github repository (UMCUGenetics/RNASeq-NF: RNASeq-NF is an NGS analysis pipeline for RNA expression quantification. (github.com)). This pipeline facilitated RNA expression quantification. For comprehensive visualization, data from two independent RNA sequencing experiments were combined. The resultant plots are presented in the Appendix.

Further data analysis and plotting were conducted using a custom R script, details of which can be found on the project's GitHub repository (JoepVondervoort/RNA-sequencing-analysis: RNA seq analysis for two feature count tables combined (github.com)).

Statistical analysis

The saRNAs were transfected using Lipofectamine RNAiMAX into the respective cell lines, and the mRNA expression levels of *SCN1A* were assessed using qPCR. Two biological replicates were tested for the transfections, and each biological replicate was assessed in triplicate within the qPCR experiment.

Statistical analyses were performed on ΔCt values using GraphPad Prism (version 9). Data are presented as mean fold change in *SCN1A* mRNA expression relative to scrambled saRNA control ± standard deviation (SD), with individual data points shown. Each data point represents a single qPCR measurement normalized to the average scrambled control. Normality of ΔCt values was assessed using the Shapiro-Wilk test. For experiments comparing multiple saRNA treatments to scrambled control, statistical significance was determined using either one-way ANOVA followed by Dunnett's multiple comparisons test (for normally distributed data) or Kruskal-Wallis test followed by Dunn's multiple comparisons test (for non-normally distributed data). For pairwise comparisons, unpaired t-tests were used. All statistical tests were two-tailed. Significance levels are indicated as: ns (not significant), * ($p < 0.05$), ** ($p < 0.01$), and *** ($p < 0.001$).

Results

Design small activating RNAs

The small activating RNAs (saRNAs) utilized in this study were designed based on the methodology previously described by Wang et al. [17]. The screening panel reported here comprised ten candidates, SCN1A-6 to SCN1A-15, synthesized according to Wang's design rules to target the *SCN1A* promoters (Figure 1; Supplementary material 1). The most proximal and distal promoters were selected as target regions of the *SCN1A* gene. The same control saRNAs were used as in previous studies [9], [12], [17].

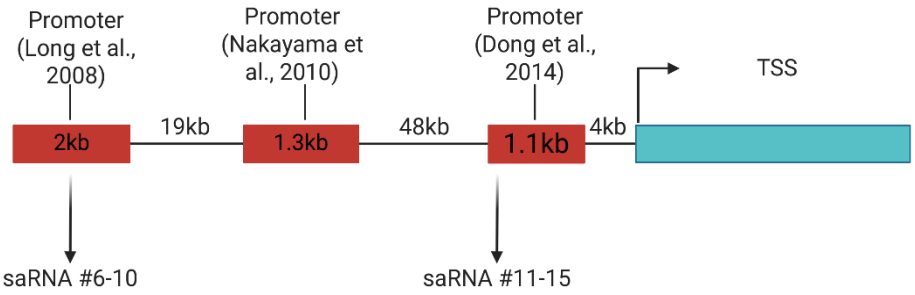


Figure 1: Localization of saRNAs Relative to *SCN1A* Promoters. *Figure 1 provides a visual representation of the localization of saRNAs designed to target the promoters of the SCN1A gene. The Transcription Start Site (TSS) is highlighted in blue, serving as the point of initiation for gene transcription. SCN1A features multiple promoters, as elucidated by earlier studies [18], [19], [20], depicted here in red.*

During the saRNA design process, adherence to specific criteria was crucial. Notably, the design rules prioritized the GC content of the target sequences, which were carefully maintained within a range of 40% to 60%. In addition, consecutive nucleotides in the saRNA targets were limited to a maximum of five in a row, ensuring sequence diversity. Furthermore, the presence of CpGs within the saRNA targets was restricted to a maximum of one.

Applying Wang's design principles, the synthesized saRNAs were tailored to target the selected promoter regions of *SCN1A* effectively.

Screening saRNAs targeting *SCN1A* promoters

The designed saRNAs were transfected into HeLa and AF22 cells using Lipofectamine RNAiMAX, and subsequent quantitative polymerase chain reactions (qPCRs) were performed to assess their potential to upregulate the expression of *SCN1A* mRNA (Figure 2). Notably, the experimental results revealed distinct patterns of saRNA efficacy in the two cell lines.

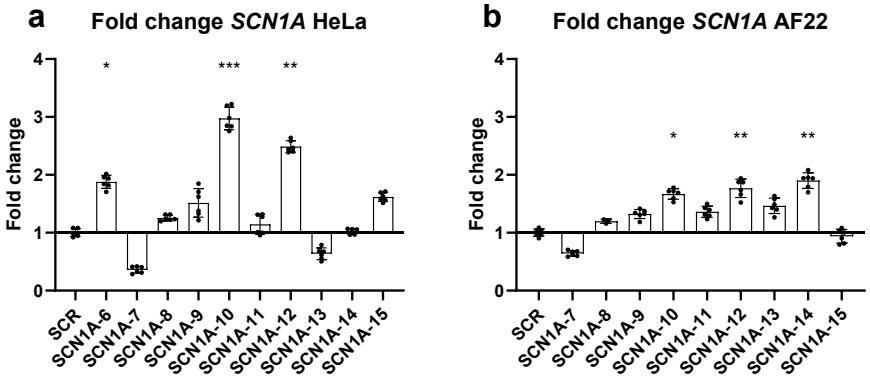


Figure 2: Screening RT-qPCR after transfection in HeLa (a) and AF22 (b). *The saRNAs were transfected using Lipofectamine RNAiMAX into the respective cell lines, and the mRNA expression levels of SCN1A were assessed using qPCR. The bars represent the mean fold change in SCN1A mRNA expression compared to the control group (scrambled saRNA), indicating the saRNAs' capacity to upregulate or downregulate the expression of SCN1A. Two biological replicates were tested for the transfections, and each biological replicate was assessed in triplicate within the qPCR experiment. Each dot represents an individual qPCR result normalized to the average scrambled control. Statistical differences in mean Delta CT values were evaluated using a Kruskal-Wallis test, with significance levels indicated as follows: ns (not significant), * ($p < 0.05$), ** ($p < 0.01$), and *** ($p < 0.001$).*

SCN1A-1 through *SCN1A*-5 were tested in HeLa cells, but they did not demonstrate consistent effectiveness, so these saRNAs were not included in this experiment. In the case of AF22 cells, *SCN1A*-1 through *SCN1A*-6 did not show promising results previously and were therefore excluded from this specific experiment. However, we do present data for *SCN1A*-6 in HeLa cells, where we observed upregulation.

In HeLa cells, transfection with *SCN1A*-10 exhibited the highest level of upregulation, significantly increasing the expression of *SCN1A* mRNA with an average fold change of 2,97 (p=0.0007). Conversely, in AF22 cells, *SCN1A*-14 displayed a greater efficacy in promoting upregulation with a fold change of 1.9 (in AF22, p=0.0004) compared to 1.02 (in HeLa). Interestingly, saRNAs *SCN1A*-10 and *SCN1A*-12 demonstrated effectiveness in both cell lines, indicating their potential as versatile candidates for upregulating *SCN1A* expression.

However, it is noteworthy that the transfection of *SCN1A*-7 led to a robust downregulation of *SCN1A* mRNA in both HeLa and AF22 cells (having a fold change of 0.36 and 0.64 respectively), presenting a contrasting outcome compared to the other saRNAs.

RNA sequencing corroborates qPCR findings

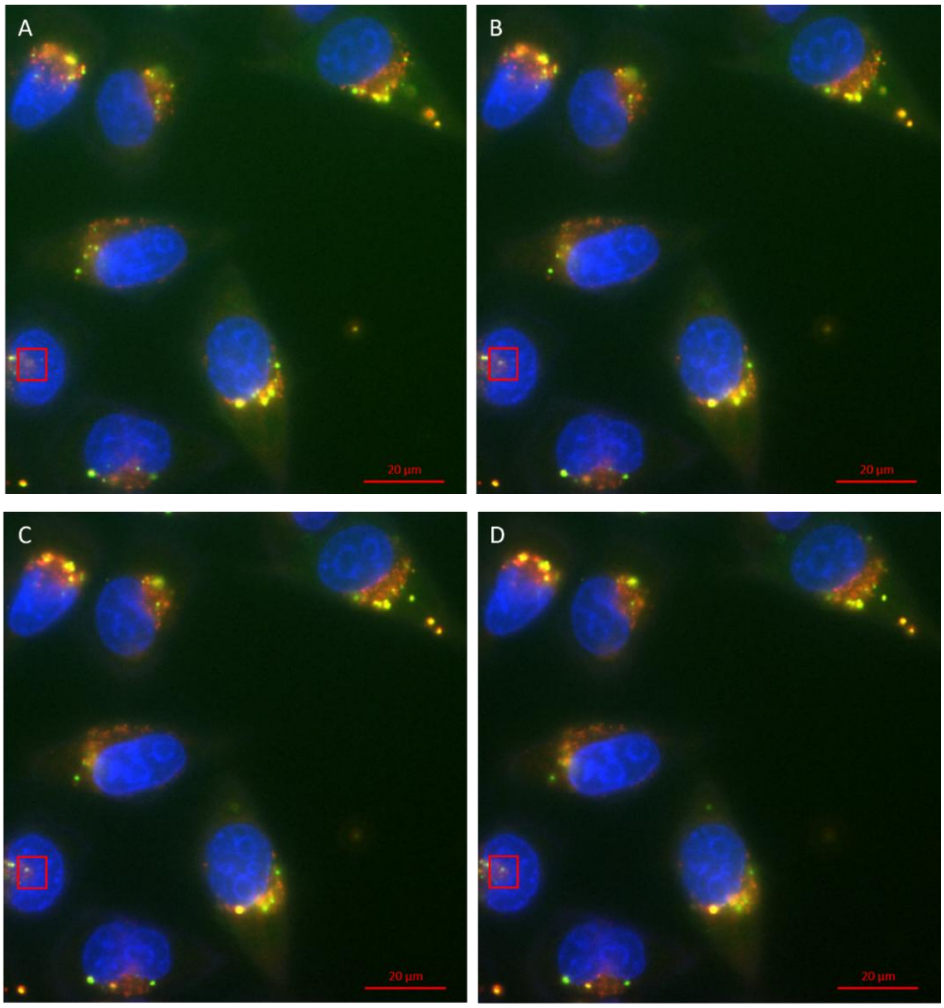
Complementing the qPCR data, RNA sequencing analysis of the same RNA samples provided independent validation of *SCN1A* modulation by the tested saRNAs in AF22 cells. As shown in Supplementary Figure 9B, normalized expression counts from RNA sequencing

confirmed that saRNA-10, saRNA-12, and saRNA-14 significantly upregulated *SCN1A* expression compared to scrambled control, with median expression levels approximately 2-fold higher than control. These results corroborate the 1.5 to 2.5-fold increases observed by qPCR for these same saRNAs.

Interestingly, saRNA-7 showed a contrasting effect, with RNA sequencing revealing downregulation of *SCN1A* expression relative to control, though this observation was based on only two biological replicates. The consistency between qPCR and RNA sequencing data across multiple saRNA treatments demonstrates the robustness and reliability of the observed *SCN1A* upregulation. This orthogonal validation using two independent molecular techniques strongly supports the effectiveness of saRNA-10, saRNA-12, and saRNA-14 in enhancing *SCN1A* expression in AF22 cells.

Transfection efficiency of RNAiMAX and localization of saRNAs in HeLa and AF22 cells

To evaluate the internalization efficiency of the small activating RNAs (saRNAs) targeting *SCN1A*, we employed fluorescent labeling on saRNA-10. Specifically, the sense strand was labeled with FAM, while the antisense strand was labeled with Cyanin5. This labeling strategy allowed us to visualize the saRNA within cells and assess its localization. We utilized the Zeiss cell imager for this purpose.


A
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C
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D
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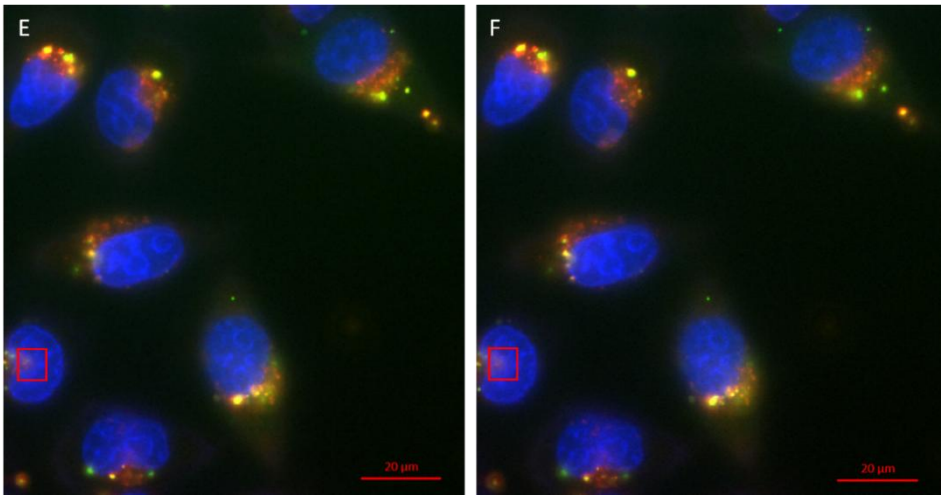

Figure 3: HeLa cells three days after transfection with fluorescently labeled *SCN1A*-10. *Nuclei are stained with Hoechst (Blue), the sense strand of SCN1A-10 is labeled with FAM (Green), and the antisense strand with Cy5 (Red). Figures A-F show a horizontal Z-stack through the cell with an interval of 0.5 µm. The red frame indicates a fluorescent signal localized in the nucleus.*

Upon examination of HeLa cells, we observed a remarkably high internalization efficacy of the saRNAs Figure 3. The majority of saRNAs were concentrated in the perinuclear space, indicating successful cellular uptake and distribution within the intracellular environment, most likely representing an accumulation in the endosomes. Notably, further scrutiny of the images revealed a subset of saRNAs exhibiting a distinct nuclear localization, as evidenced by the presence of saRNA signals within the nucleus (highlighted with a red square in the figure). Additionally, to confirm the efficiency of transfections with AF22-labeled saRNAs, we conducted similar visualization studies in AF22 cells. The results obtained from these experiments are presented in Figure 4 and show a similar intracellular localization. These imaging findings provide valuable insights into the intracellular behavior of saRNAs targeting *SCN1A*, demonstrating their successful internalization and distribution within the perinuclear space. The observation of saRNA localization within the nucleus suggests the possibility of nuclear-mediated gene regulation, highlighting the potential of saRNAs as a promising tool for modulating gene expression. Picture of differentiated AF22 wildtype stained with nucleus, astrocyte and neuronal markers is attached in the appendix 6.

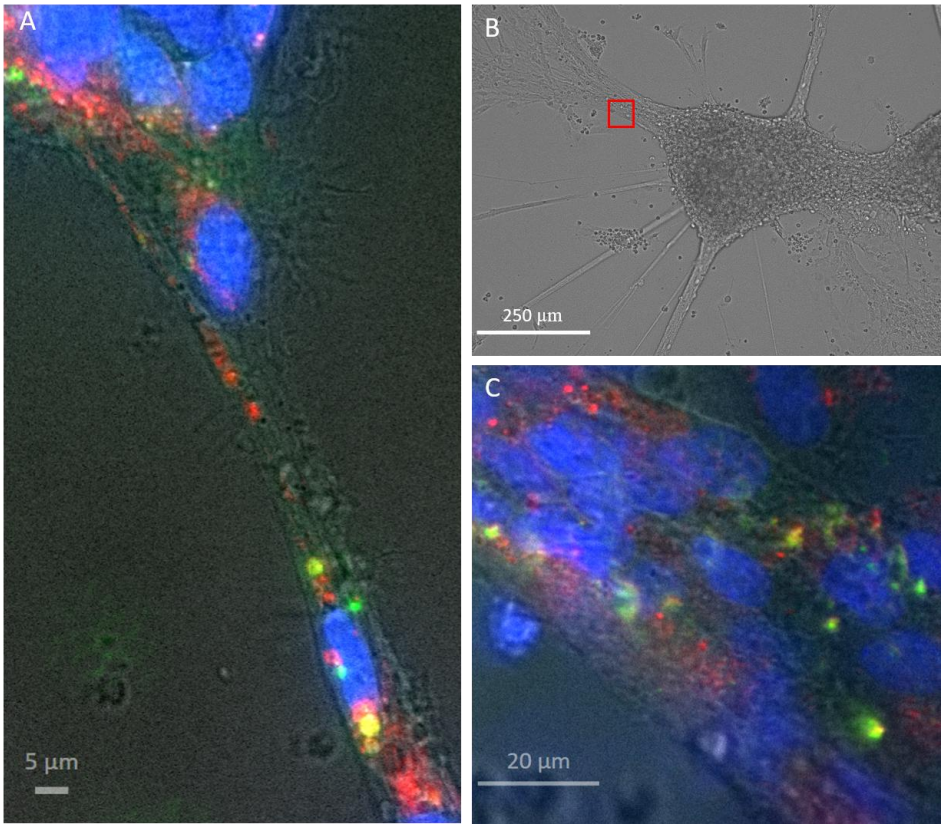

Figure 4: AF22 cells three days after transfection with fluorescently labeled *SCN1A*-10. *Nuclei are stained with Hoechst (Blue), the sense strand of SCN1A-10 is labeled with FAM (Green), and the antisense strand with Cy5 (Red). A) Undifferentiated AF22 cells captured with the Zeiss observer microscope. B) Differentiated AF22 cells, the red frame indicates a region representative for figure C. C) Differentiated AF22 captured with Zeiss observer microscope.*

$Na_v1.1$ protein expression in HeLa and undifferentiated AF22 cells after treatment with *SCN1A*-10

Following the observation of *SCN1A* mRNA upregulation in HeLa and AF22 cells, we further investigated the effect of saRNA transfection on the protein levels of $Na_v1.1$. Western blot experiments were conducted to assess the expression of $Na_v1.1$ protein levels (Figure 5). For this analysis, cells were transfected with *SCN1A*-10 saRNA and harvested 72 hours post-transfection.

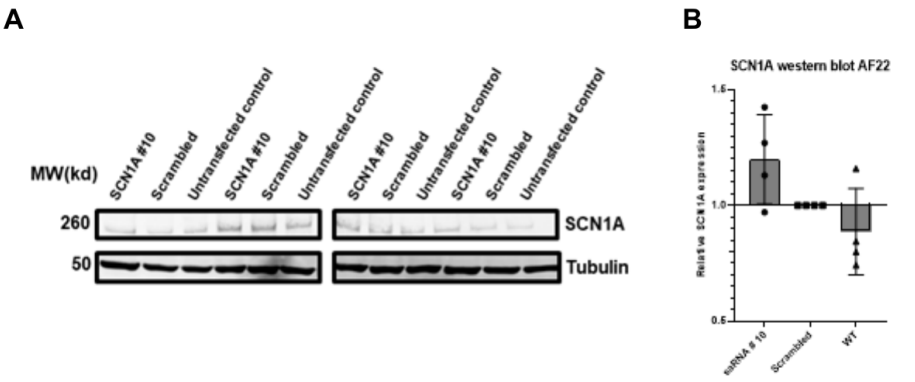

Figure 5: Western Blot Analysis of $Na_v1.1$ Protein Expression in AF22 Cells. *The cells were treated with saRNA SCN1A-10, scrambled, or left untreated. To normalize the data, α-Tubulin, a housekeeping protein, was used. Each lane in the blot corresponds to a separate sample from the same experimental set. The figure is composed of two distinct gels, which were run in parallel. B) Bar plot displays the normalized expression of* $Na_v1.1$ *protein relative to the scrambled condition. This quantification reveals that cells treated with saRNA-10 demonstrate a higher* $Na_v1.1$ *protein expression in comparison to those treated with the scrambled sequence. However, statistical analysis using one-way ANOVA indicated that this increase in expression was not statistically significant (p > 0.05).*

Conformingly, the western blot results demonstrated higher band intensity corresponding to $Na_v1.1$ protein levels following transfection with *SCN1A*-10 compared to the scrambled control groups in both HeLa (Supplementary material 3) and AF22 cells. This increase in band intensity suggests upregulation of $Na_v1.1$ protein expression induced by *SCN1A*-10 saRNA. To ensure the accuracy of the analysis, normalization was performed using the housekeeping protein α-Tubulin, which confirmed the observed upregulation of $Na_v1.1$.

SCN1A mRNA upregulation in differentiated AF22 cells after treatment with *SCN1A*-10

In addition to observing the upregulation of *SCN1A* in HeLa and AF22 NES cells, we further investigated the effects of saRNA transfection on differentiated AF22 cells, which serve as a more representative cellular model due to their inclusion of GABAergic, but also dopaminergic and excitatory neurons [15]. Differentiated AF22 cells were transfected with saRNA *SCN1A*-10, followed by qPCR analysis to evaluate the mRNA expression levels of *SCN1A*.

The qPCR results confirmed a significant upregulation of *SCN1A* mRNA expression in differentiated AF22 cells subsequent to transfection with *SCN1A*-10 saRNA (Figure 6). The fold change in mRNA expression compared to the control group exhibited a substantial increase in *SCN1A* expression (1.4 log2 Fold Change, p=0.01), providing further evidence of the effectiveness of *SCN1A*-10 in promoting upregulation.

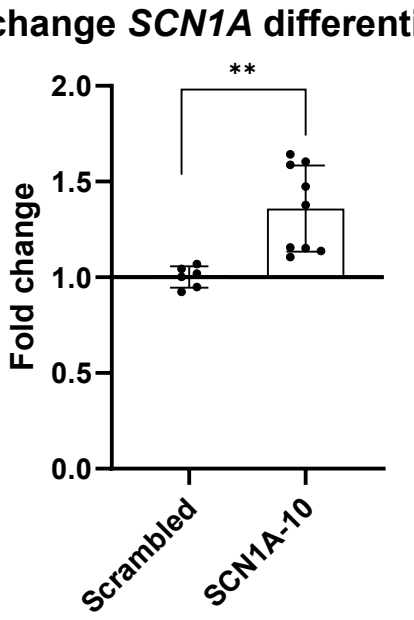

Figure 6: qPCR of *SCN1A* mRNA expression in differentiated AF22 cells transfected with *SCN1A*-10 saRNA compared to scrambled control. *Bar plot displays fold change in SCN1A mRNA expression (mean ± SD) from three biological replicates. Dots represent individual qPCR results, illustrating variability within the dataset. The figure shows the results of qPCR analysis, indicating a significant upregulation of SCN1A mRNA expression in differentiated AF22 cells transfected with SCN1A-10 saRNA compared to the scrambled control. Each bar represents the mean value of the technical replicates. The difference in mean Delta CT values among different saRNAs (SCN1A-10 vs scrambled) was tested using an unpaired t-test with a significance code indicating the level of significance: $p < 0.01$ (**).*

These findings underscore the therapeutic potential of *SCN1A*-10 saRNA for modulating *SCN1A* expression in differentiated AF22 cells, which closely resemble the cellular context of Dravet syndrome with the presence of GABAergic neurons.

No toxicity of transfection with RNAiMAX and saRNAs in AF22 cells

In addition to assessing the efficacy of small activating RNAs (saRNAs) targeting the *SCN1A* promoters on AF22 cells, we also evaluated their potential toxicity. To investigate the impact of saRNA transfection on cell viability, we conducted transfections when the cells reached approximately 70% confluency. After 72 hours of transfection, we employed flow cytometry to assess cell death quantitatively. To facilitate the quantification of dead cells, we utilized Propidium Iodide, which selectively stains only dead cells by binding to DNA but is unable to enter cells with intact plasma membranes.

For comparison and validation of our approach, we employed P21 saRNA as a positive control, known to upregulate the *CKDN1A* gene, leading to cell cycle arrest and cellular

response to DNA damage. As negative controls, we included untransfected cells (UTC), cells transfected with RNAiMAX alone, and cells transfected with RNAiMAX and a non-targeting saRNA (scrambled). Notably, we observed a higher percentage of cell death in the sample treated with the P21 saRNA compared to all other saRNAs (supplementary 4).

Our saRNA *SCN1A*-10 specifically targeted *SCN1A* and did not induce significant cell death, displaying similar percentages of viable cells compared to all our negative control conditions. This finding suggests that the saRNA-10 did not impose substantial cytotoxicity on the AF22 cells.

A comprehensive representation of the flow cytometry settings utilized to accurately distinguish viable and dead cells is shown in the supplementary material.

Modulation of Calcium Dynamics in AF22 Neurons Following *SCN1A*-10 saRNA Transfection

To investigate whether increased expression of *SCN1A* affects neuronal activity, calcium imaging experiments were performed on AF22 differentiated cells transfected with either scrambled control RNA, *SCN1A*-targeting small activating RNA (saRNA), or *SCN1A*-targeting small interfering RNA (siRNA) (Figure 7). These experiments aimed to determine whether modulation of *SCN1A* expression could lead to detectable changes in neuronal excitability and calcium kinetics. Outliers in these kinetic parameters were systematically removed using the z-score method with an adjustable threshold, set by default at 3.0.

Initially, we assessed the percentage of active cells, defined as cells exhibiting at least two calcium transient peaks. For the subsequent analysis of calcium kinetics, only active cells were considered. The results showed a clear decrease in active cells in the siRNA-treated group (8.6%) compared to scrambled control (25.4%) and saRNA-treated groups (30.3%). This indicates that reducing *SCN1A* expression significantly diminishes neuronal activity.

Detailed calcium kinetics analysis revealed subtle yet significant differences. Specifically, neurons treated with *SCN1A*-targeting siRNA exhibited significantly fewer calcium peaks compared to the scrambled control group ($p = 0.0001$). Interestingly, saRNA-treated cells also showed a mild yet statistically significant reduction in the number of peaks compared to the scrambled control ($p = 0.016$).

No significant differences were observed in amplitude across all groups. Regarding the slope, the siRNA treatment resulted in significantly altered calcium transient kinetics compared to scrambled control ($p < 0.001$), while the saRNA group did not show significant differences. Analysis of the full width at half maximum (FWHM) revealed a significantly narrower transient in the siRNA group compared to scrambled control ($p < 0.001$), indicating faster calcium transient dynamics, whereas no significant differences were found between saRNA and scrambled controls.

These findings suggest that alterations in *SCN1A* expression have a minor yet distinctly influence on neuronal calcium dynamics. Downregulation via siRNA consistently reduced neuronal excitability and calcium kinetics. In contrast, saRNA-mediated upregulation led to a subtle reduction in peak number without broadly altering other kinetic parameters, highlighting the nuanced relationship between *SCN1A* expression and neuronal excitability.

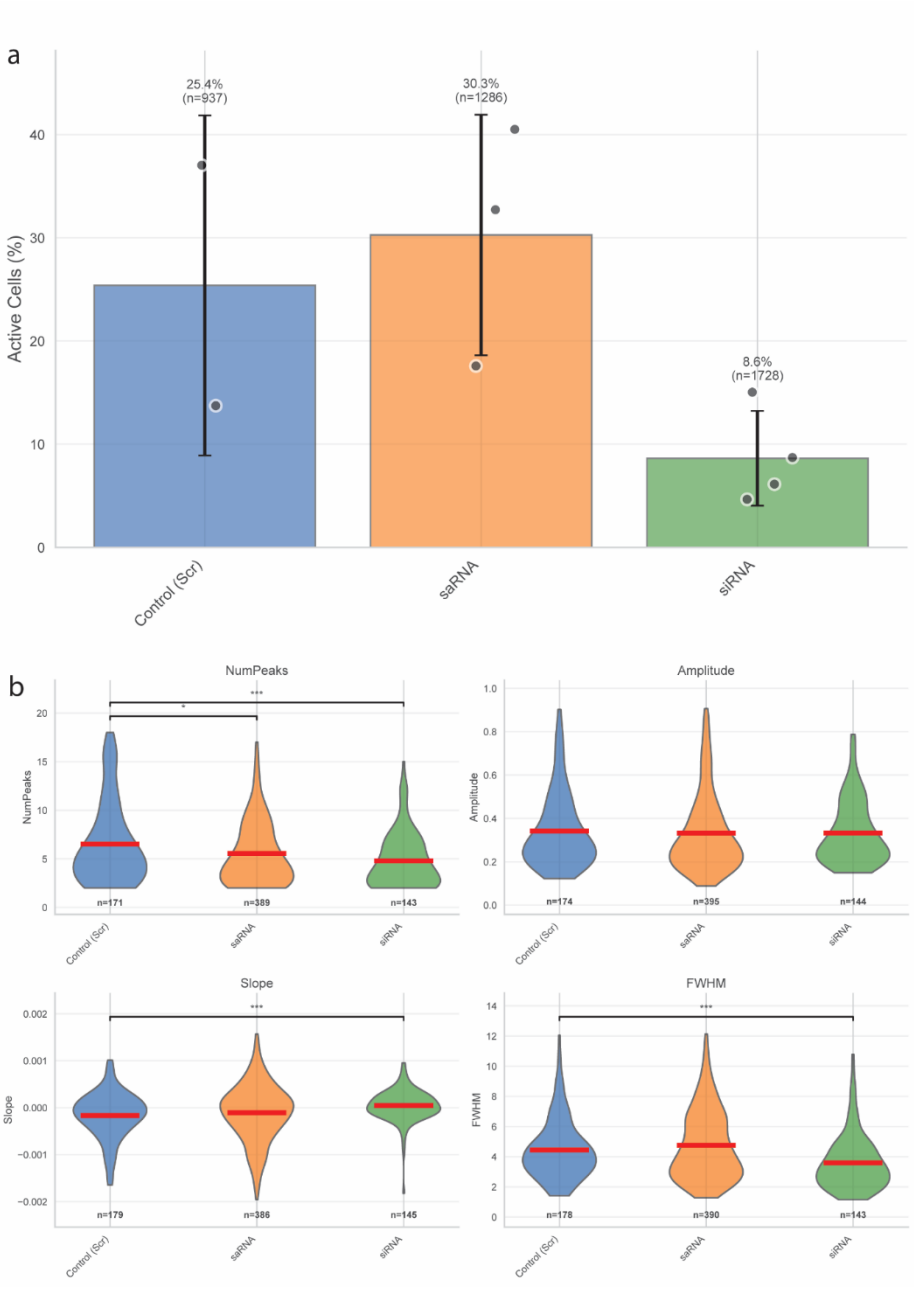
a
25.4%
(n=937)
30.3%
(n=1286)
8.6%
(n=1728)
Active Cells (%)
0
10
20
30
40
Control (Scr)
saRNA
siRNA
b
NumPeaks
Amplitude
Slope
FWHM
n=171
n=389
n=143
n=174
n=395
n=144
n=179
n=386
n=145
n=178
n=390
n=143

Figure 7. Calcium imaging analysis of AF22-derived neurons transfected with *SCN1A*-targeting small activating or silencing RNAs. *(A) Percentage of active cells, defined as cells exhibiting at least two calcium transients during the recording period. Differentiated AF22 neurons were transfected with either a non-targeting scrambled control (Scr), SCN1A-targeting saRNA (saRNA-10), or SCN1A-targeting siRNA. Transfection with saRNA-10 slightly increased the proportion of active cells (30.3%) compared to control (25.4%), while siRNA transfection markedly reduced activity (8.6%), consistent with effective SCN1A knockdown impairing network excitability. Data is from 2 biological replicates per condition and following z-score-based outlier removal (|z| > 3). (B–E) Quantification of calcium transient kinetics in active cells (≥2 peaks), following z-score-based outlier removal (|z| > 3). The Red horizontal line represents the mean. (B) NumPeaks: Number of calcium events per active cell. saRNA and control groups exhibited similar frequencies, while siRNA resulted in a significant reduction in transient counts. (C) Amplitude: Mean amplitude of calcium transients. saRNA-treated cells showed a modest but statistically significant increase compared to control, while siRNA-transfected cells displayed lower amplitudes. (D) Slope: Maximum rise slope of calcium transients. saRNA treatment significantly increased slope values relative to control, suggesting steeper activation kinetics. siRNA reduced slopes compared to both control and saRNA. (E) FWHM (Full Width at Half Maximum): Duration of calcium transients. saRNA significantly increased FWHM values, while siRNA-treated cells exhibited broader peaks compared to controls, albeit with a lower number of events. Statistical significance annotations: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns = not significant ($p > 0.05$). Statistical comparisons were performed against the Control (Scr) group using Mann-Whitney U test following Shapiro-Wilk normality testing. Sample sizes indicated on violin plots represent the number of active cells analyzed per condition after outlier removal.*

Next, we evaluated the effects of acute chemical stimulation using a selective $Na_v1.1$ activator (BI-7150), pentylenetetrazole (PTZ), and veratridine on differentiated AF22 cells (Figure 8). The rationale behind this experiment was to determine if specific activation of *SCN1A* channels (with BI-7150) could enhance inhibitory activity, if reducing inhibition (with PTZ that acts as a GABA receptor antagonist) would impact neuronal activity, and whether generalized neuronal activation with veratridine that acts on all relevant sodium channels could still be achieved in these cells. Veratridine acts on all sodium channels ($Na_v1.1$, Nav1.2, and Nav1.6) and extends their open state, leading to persistent sodium influx and membrane depolarization.

Chemicals were added to the medium immediately prior to measurement. The percentage of active cells was highest in veratridine-treated cells (33.6%) compared to wild-type (WT, 4.8%), $Na_v1.1$ activator (7.0%), and PTZ-treated cells (10.8%). Detailed calcium kinetics analysis revealed significant alterations specifically following veratridine treatment. $Na_v1.1$ activator-treated cells showed significantly less peaks compared to WT ($p = 0.026$). Veratridine treatment resulted in significantly higher amplitude values compared to all other groups ($p < 0.001$). Furthermore, veratridine exhibited significantly higher slope values compared to WT ($p = 0.005$). Full width at half maximum (FWHM) was significantly greater with veratridine treatment compared to both WT and $Na_v1.1$ activator ($p < 0.001$). Furthermore, the $Na_v1.1$ activator showed a significant reduction of the FWHM as compared to WT ($p<0.001$).

From these observations, we conclude that the $Na_v1.1$ activator also showed limited yet distinct effects on neuronal activity, suggesting limited efficacy in enhancing *SCN1A*-mediated inhibition in this model, similar to previous outcomes observed with *SCN1A*-targeting saRNA. The minimal impact of PTZ implies that inhibitory mechanisms in these cells might already be maximally engaged. However, veratridine treatment showed clear generalized neuronal activation.

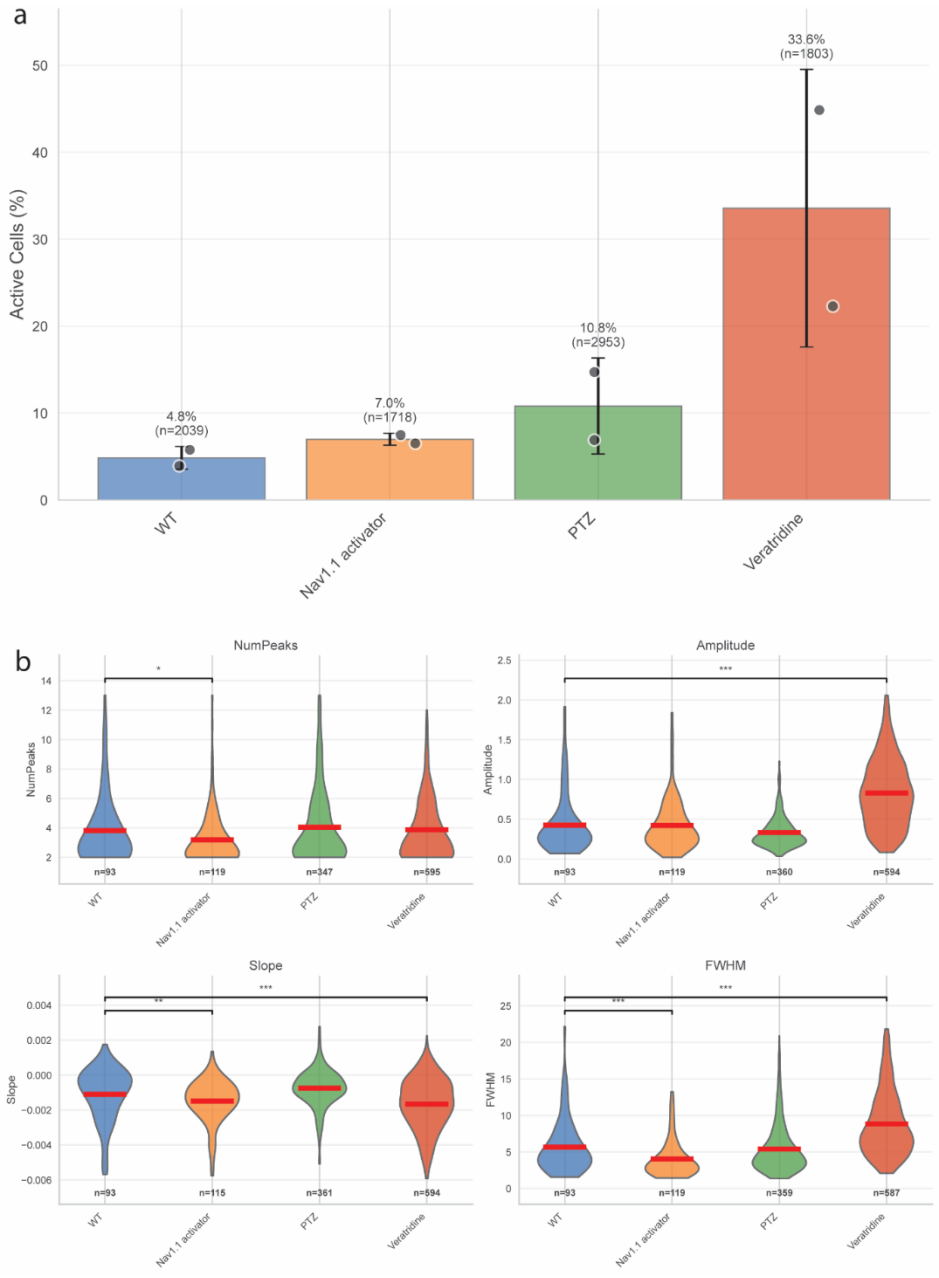
a
Active Cells (%)
4.8%
(n=2039)
7.0%
(n=1718)
10.8%
(n=2953)
33.6%
(n=1803)
WT
Nav1.1 activator
PTZ
Veratridine
b
NumPeaks
Amplitude
Slope
FWHM
n=93
n=119
n=347
n=595
n=360
n=594
n=115
n=361
n=359
n=587

Figure 8. Calcium imaging analysis of AF22-derived neurons treated with small molecules affecting neuronal excitability. (A) Percentage of active cells per condition, defined as cells exhibiting at least two calcium transients during the recording period. Treatments included wild-type untreated controls (WT), a $Na_v1.1$-selective activator, the GABA_A receptor antagonist *pentylenetetrazole* (PTZ), and the broad-spectrum sodium channel opener veratridine. Veratridine treatment led to a marked increase in the percentage of active cells (33.6%), while the $Na_v1.1$ activator and PTZ *produced moderate increases (7.0% and 10.8%, respectively) compared to WT (4.8%). Data is from 2 biological replicates per condition and following z-score-based outlier removal ($|z| > 3$). (B–E) Quantification of calcium transient kinetics for active cells only (defined as cells with ≥2 peaks) across conditions. Outliers were removed using a z-score threshold method ($|z| > 3$) applied per condition. The Red horizontal line represents the mean. (B) NumPeaks*: Number of calcium transients detected per active cell. PTZ and veratridine significantly increased transient frequency compared to WT and $Na_v1.1$ activator conditions. *(C) Amplitude: Mean transient amplitude per cell, with veratridine producing the most pronounced increase. (D) Slope: Maximum rise slope of calcium transients, reflecting activation kinetics. Veratridine and PTZ both resulted in steeper slopes than WT and* $Na_v1.1$ *activator conditions.* (E) FWHM (Full Width at Half Maximum): Duration of calcium transients at half-maximal amplitude. Veratridine and PTZ induced significantly broader peaks, while WT and $Na_v1.1$ activator conditions showed shorter transient durations. *Statistical significance annotations: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns = not significant ($p > 0.05$). Statistical comparisons were performed against the Control (Scr) group using Mann-Whitney U test following Shapiro-Wilk normality testing. Sample sizes indicated on violin plots represent the number of active cells analyzed per condition after outlier removal.*

Discussion

Restoring *SCN1A* expression represents a promising therapeutic strategy for treating *SCN1A* haploinsufficiency, in Dravet syndrome. Dravet syndrome arises from loss-of-function mutations in *SCN1A*, reducing $Na_v1.1$ channel expression and impairing inhibitory interneuron function, resulting in network hyperexcitability and seizures [1], [21]. Due to technical limitations of traditional gene replacement approaches, mainly the large size of *SCN1A*, epigenetic upregulation of the functional allele using small activating RNAs (saRNAs) presents an attractive alternative strategy.

In this study, we successfully demonstrated saRNA-mediated upregulation of *SCN1A* in human cell models. Robust transcriptional activation was observed in proliferative cells such as HeLa and AF22 neural progenitors (NES), whereas differentiated AF22 cells, comprising a mixture of inhibitory and excitatory neurons, exhibited more modest increases. This differential efficacy likely reflects cellular state differences; dividing cells with looser chromatin structures and enhanced nuclear accessibility respond better than post-mitotic neurons. Despite these limitations, significant induction of *SCN1A* transcripts and $Na_v1.1$ protein was confirmed by both qPCR and Western blot analyses, validating the functional relevance of the saRNA approach. RNA sequencing, performed mainly to confirm qPCR results, indeed showed clear upregulation of *SCN1A* with saRNAs 10, 12, and 14, and a surprising downregulation with saRNA 7 in AF22 NES cells. This highlights the complexity

of saRNA-promoter interactions and emphasizes the necessity for comprehensive functional screening.

Functional evaluation of increased $Na_v1.1$ expression using calcium imaging revealed mild alterations in spontaneous neuronal network activity in differentiated AF22 cultures (a slight reduction in calcium peak count, with other kinetic parameters unchanged). Conversely, siRNA-mediated *SCN1A* knockdown markedly reduced calcium transient frequency and amplitude, confirming assay sensitivity to *SCN1A* levels. Neurons with *SCN1A* knockdown exhibited fewer and smaller calcium transients, indicative of decreased excitability and impaired neuronal network functionality. This outcome aligns well with the established role of $Na_v1.1$ in inhibitory interneuron function, where reduced channel activity disrupts interneuronal firing and compromises inhibitory regulation. This condition closely mimics the underlying pathophysiology observed in Dravet syndrome, where *SCN1A* haploinsufficiency leads to hypoexcitable interneurons and insufficient inhibitory control, ultimately resulting in seizures and disturbed network dynamics [4]. The pronounced impact of *SCN1A* knockdown on neuronal excitability thus validates the sensitivity and biological relevance of our assay system to changes in $Na_v1.1$ expression.

Further pharmacological investigations supported these findings. The absence of a strong response to pentylenetetrazole (PTZ), a GABA_A receptor antagonist, suggests that baseline inhibitory tone in these cultures was already low or ineffective. *SCN1A* encodes only the α-subunit of the sodium channel, $Na_v1.1$, which alone may not substantially alter overall channel function unless the other subunits are simultaneously regulated. In our study, veratridine, which activates the entire sodium channel complex, induced clear increases in neuronal activity, confirming that sodium channel function was intact and capable of heightened activity when fully activated. Therefore, enhancing inhibitory capacity solely through upregulation of the α-subunit $Na_v1.1$ might not significantly alter firing patterns if excitatory neurons are already operating near their intrinsic firing limits, potentially due to inherent pacemaker-like properties. Such a scenario contrasts with in vivo conditions, where inhibitory interneurons strongly regulate cortical and hippocampal network excitability, making even slight modifications of $Na_v1.1$ levels consequential [22]. However, dissociated neuronal cultures, characterized by a heterogeneous mixture of inhibitory neurons, excitatory neurons and astrocytes [23] (supplementary 6), commonly exhibit dominant excitatory activity with reduced emergent inhibition, likely diminishing the functional effects of increasing $Na_v1.1$. Interestingly, the $Na_v1.1$-specific activator caused a mild reduction in calcium peak frequency, similar to the subtle decrease observed in the saRNA-treated group compared to scrambled controls. This may result from enhanced inhibitory interneuron firing reducing overall network activity, underscoring the complexity of modulating $Na_v1.1$ channels in heterogeneous neuronal cultures.

Importantly, these findings imply therapeutic saRNA-mediated *SCN1A* elevation would predominantly benefit haploinsufficient neurons, where restoring $Na_v1.1$ to physiological levels is critical. Indeed, prior animal studies support this therapeutic concept, demonstrating phenotype reversibility by reactivating the silenced *Scn1a* allele [6], [7]. However, the saRNA sequences identified in our study are not conserved in mice, precluding further in vivo

experimentation in rodent models. Although our wild-type cultures showed only a modest response to additional $Na_v1.1$, this also suggests minimal risk of pathological hyperactivity from overexpression, enhancing the safety profile for clinical translation. Future studies must evaluate saRNA effectiveness in disease-relevant human models such as patient-derived iPSC neurons or haploinsufficient brain organoids, where correcting inherent *SCN1A* deficits could directly validate therapeutic potential. Comprehensive genomic and safety assessments remain critical to mitigate risks of off-target effects or immune activation.

Delivery and stability of saRNAs in neurons pose considerable translational challenges, evident from reduced efficacy in differentiated neurons. Achieving effective in vivo brain delivery will likely require chemical modifications (e.g., 2′ O-methyl), targeted ligands, nanoparticle encapsulation, or intrathecal administration, mirroring advances seen with antisense oligonucleotides (ASOs) currently under clinical investigation, such as STK-001 [24]. Compared to ASOs, saRNAs offer potential advantages including DNA-level specificity, harnessing endogenous transcriptional regulation, and preserving normal RNA processing patterns. However, their stability and delivery hurdles necessitate substantial optimization.

In conclusion, this study establishes proof-of-concept that saRNAs effectively increase *SCN1A* expression at transcriptional and protein levels. Although saRNA-induced $Na_v1.1$ increases do not alter neuronal activity under baseline conditions, they hold significant therapeutic promise for correcting *SCN1A* haploinsufficiency, particularly in conditions such as Dravet syndrome. Addressing delivery, efficacy, and safety through further translational research is essential to realize saRNA therapy as a viable precision medicine for genetic epilepsies.

Data availability statement

The custom code used for data analysis, written in Python and R, is available in a publicly accessible GitHub repository shown in the methods section. The repository includes README files explaining the usage and requirements for each script.

Where applicable, supplementary information is provided within the article and its supplementary files. For further inquiries regarding the data or materials used in this study, please contact the corresponding author.

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Statement on AI use.

During the preparation of this work, the authors used ChatGPT in order to improve the readability and accessibility of the content of this article. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Author contributions

Joep van de Vondervoort, Dr. B.P.C. Koeleman, Nore van Loon and Martijn Koetsier designed the experimental framework. Joep van de Vondervoort, Nore van Loon and Martijn Koetsier performed the cellular experiments and analyzed the data. Serife Uysal contributed extensively to cell culture and experimental work. Joep van de Vondervoort performed the calcium-imaging analysis with advice from Wout Weuring and Dr. T.P. de Boer. Carolin Vetter contributed to RNA-sequencing and gene-expression analyses.

Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman provided scientific supervision and contributed to interpretation of the results. Joep van de Vondervoort and Dr. B.P.C. Koeleman drafted the manuscript. All authors reviewed and revised the final chapter.

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Declaration of interest statement

JvdV, WW, and BK have a joint patent application filed for the saRNA described in this study and have a collaboration agreement with Ractigen Therapeutics. No other author has competing interests.

Ethics, consent to participate, consent to publish and clinical trials number

Ethics approval: not applicable. This study involved no human participants, animal subjects, or the use of their biological material. Consent to participate: not applicable. Consent to publish: not applicable. Clinical trial number: not applicable.

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Supplementary material

Supplementary material 1

Sequences of saRNAs and siRNA

	RNA sequence strand 1 (5'-3')	RNA sequence strand 2 (5'-3')
#	*SCN1A* (Located in the promoter region described by Dong (2014))	
1	CCUGACUUCUUUCUGAUCA [dT][dT]	UGAUCAGAAAGAAGUCAGG [dT][dT]
2	CUCUCCUUCUUACUCACAU [dT][dT]	AUGUGAGUAAGAAGGAGAG [dT][dT]
3	UAAGGUUGGUCACUGCAAA [dT][dT]	UUUGCAGUGACCAACCUUA [dT][dT]
4	UUCCCCUCAUCCUUGUUAA [dT][dT]	UUAACAAGGAUGAGGGGAA [dT][dT]
5	CACAGUUCCUGUAUCAGUA [dT][dT]	UACUGAUACAGGAACUGUG [dT][dT]
	SCN1A (Located in the promoter region described by Long (2008))	
6	ACAACUCACACCCCCAAUA [dT][dT]	UAUUGGGGGUGUGAGUUGU [dT][dT]
7	CAAUCCUCUUGCUCCAUUA [dT][dT]	UAAUGGAGCAAGAGGAUUG [dT][dT]
8	GGUGACUGUCUUUGAGUUA [dT][dT]	UAACUCAAAGACAGUCACC [dT][dT]
9	UGGUCAUUCUUCCCACAAA [dT][dT]	UUUGUGGGAAGAAUGACCA [dT][dT]
10	CUUAACUCGCCUUCCAAAA [dT][dT]	UUUUGGAAGGCGAGUUAAG [dT][dT]
	SCN1A (Located in the promoter region by Dong (2014))	
11	GAGCUUUUCUCAAAUGCAG [dT][dT]	CUGCAUUUGAGAAAAGCUC [dT][dT]
12	AAGGAAGUACAGGGUUCAA [dT][dT]	UUGAACCCUGUACUUCCUU [dT][dT]
13	AGGGUUCAAUUCACUCCAA [dT][dT]	UUGGAGUGAAUUGAACCCU [dT][dT]
14	UGCCUCUCCCAACUUGGAA [dT][dT]	UUCCAAGUUGGGAGAGGCA [dT][dT]
15	CCCAACUUGGAACAGUCAA [dT][dT]	UUGACUGUUCCAAGUUGGG [dT][dT]
Scrambled control	ACUACUGAGUGACAGUAGA [dT][dT]	UCUACUGUCACUCAGUAGU [dT][dT]
siRNA *SCN1A. Silencer Predesigned* Lot#: ASO2NFWZ	GGGUUUUUUGGAUGCACUA[dT][dT]	UAGUGCAUCCAAAAAACCC[dT][dC]

Supplementary material 2

Validation of saRNA effects on *SCN1A* expression

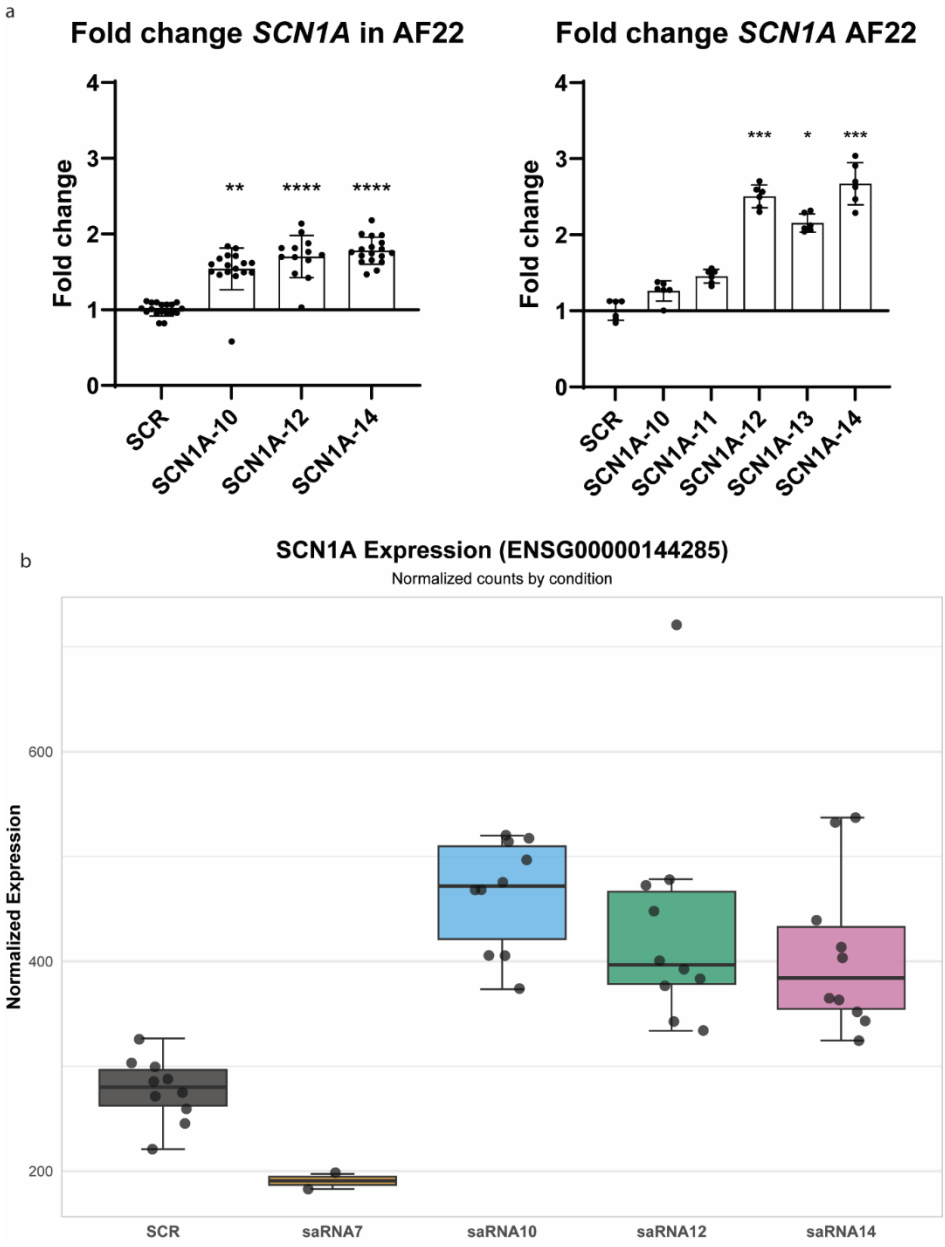


Figure 9: Validation of saRNA effects on *SCN1A* expression *This figure presents validation data demonstrating the effects of small activating RNAs (saRNAs) on SCN1A expression in AF22 NES cells through both qPCR and RNA sequencing analyses.*

(a) qPCR Analysis of SCN1A Expression *Quantitative PCR was performed to assess SCN1A mRNA levels following transfection with different saRNAs. The bar graphs show mean fold change in SCN1A expression relative to scrambled saRNA control. Statistical analysis was performed on ΔCt values using Kruskal-Wallis test followed by Dunn's multiple comparisons test. Each data point represents a technical replicate from the qPCR experiment. Results demonstrate significant upregulation of SCN1A with saRNA-10, saRNA-12, and saRNA-14 compared to scrambled control, with all three treatments showing approximately 1.5 to 2.5-fold increases in expression.*

(b) RNA Sequencing Confirmation *To independently validate the qPCR findings, RNA sequencing (Poly-A) was performed on the same RNA samples used for qPCR analysis, including samples from multiple independent transfections. The boxplot displays normalized SCN1A expression counts across different saRNA treatments. Each data point represents a biological replicate. RNA sequencing data confirms the upregulation of SCN1A in cells treated with saRNA-10, saRNA-12, and saRNA-14 compared to scrambled control. Notably, saRNA-7 (n=2) showed downregulation of SCN1A expression. These RNA sequencing results corroborate the qPCR findings and provide orthogonal validation of saRNA-mediated SCN1A upregulation.*

Supplementary material 3

Western blots showing upregulation of $Na_v1.1$ 72 hours post transfection of saRNA-10 in both HeLa and AF22 NES cells.

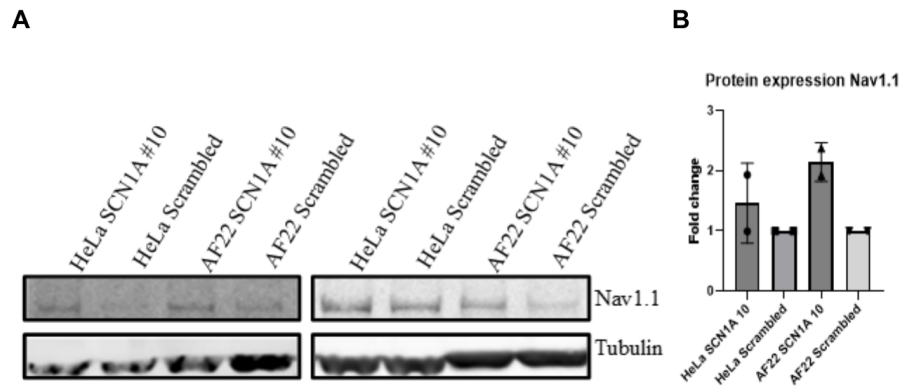


Figure 10: $Na_v1.1$ protein expression after treatment of HeLa and AF22 cells with *SCN1A*-10 *A) Western blots of HeLa and AF22 cells harvested 72h after transfection with SCN1A-10 or scrambled control, the upper part of the blot is stained with Anti-*$Na_v1.1$ *Antibody and the lower part with Anti-αTubulin. B) Fold changes + SD of* $Na_v1.1$ *protein expression normalized to scrambled treated samples of the same cell line. The intensity of bands in the western blots was quantified in ImageJ and normalized to αTubulin as a loading control.*

Supplementary material 4

AF22 cell viabilities after transfections

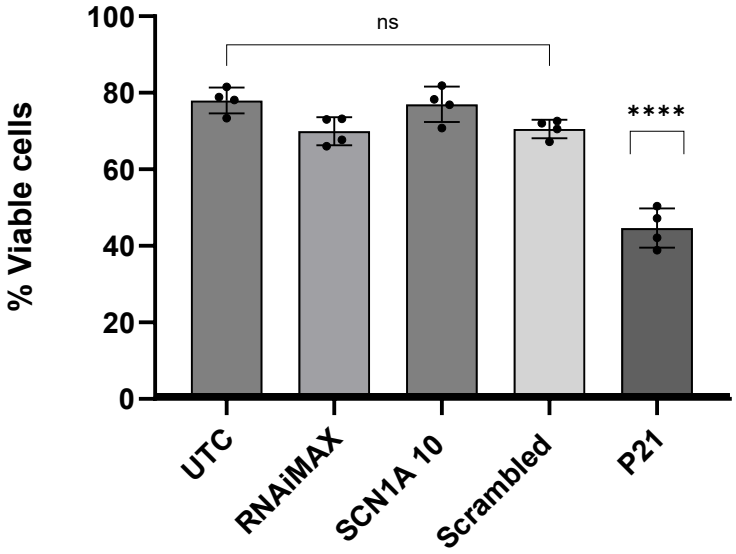


Figure 11: AF22 cell viabilities after transfections *Percentages viable AF22 cells 72h after transfection with RNAiMAX transfection reagent alone or in combination with saRNAs or scrambled control. Bars are mean + SD of four biological replicates. Differences in mean viable cell percentages were tested with One-way ANOVA and Tukey's multiple comparison test. The figure indicates the significance of comparisons with UTC of Tukey's multiple comparison tests. Significance codes: p >0.05 (ns), <0.0001 (****).*

Supplementary material 5

Dose-response experiments of *SCN1A*-10 in AF22 cells

We investigated the dose-response relationship to determine the optimal concentration for achieving the highest efficiency of transfections with saRNA targeting *SCN1A*. To this end, we performed transfections using multiple doses of saRNA in AF22 cells (Figure 12).

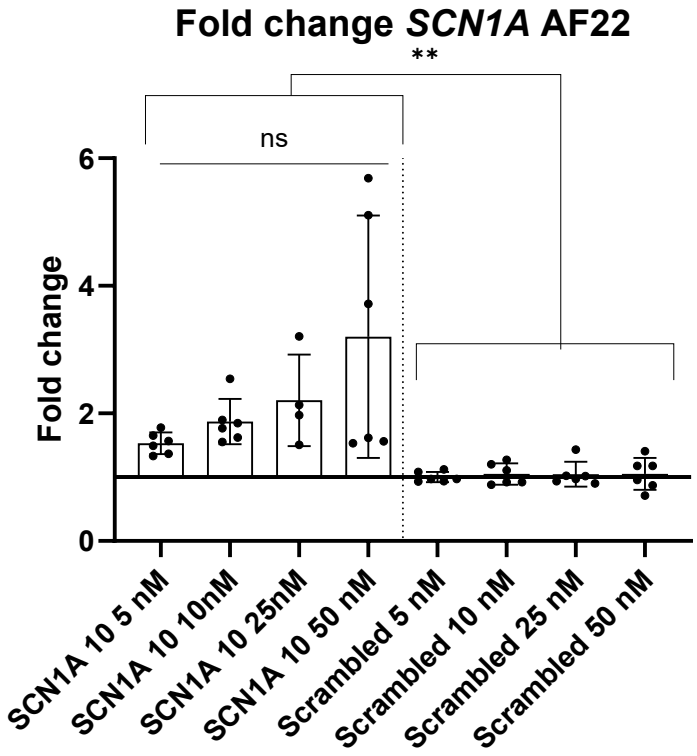


Figure 12: *SCN1A* mRNA expression after treatment of AF22 cells with *SCN1A*-10 *qPCR results of SCN1A mRNA expression in AF22 cells harvested three days after transfection with different concentrations of SCN1A-10 or scrambled control. The fold change in SCN1A mRNA expression was determined by normalizing to GAPDH and comparing it to Scrambled 5nM treated samples. The data are presented as fold change ± standard deviation (SD) on a logarithmic scale. Each dot in the figure represents an experiment with two biological replicates, and each replicate consisted of three qPCR plate replicates (except for SCN1A-10 25 nM, which had two plate replicates). The difference in mean Delta CT values between SCN1A-10 and scrambled was assessed using a two-sample t-test (**) to determine statistical significance. The difference in mean Delta CT values among different concentrations of SCN1A-10 was tested using a one-way ANOVA (ns) with significance codes indicating the level of significance: p < 0.05 (ns), 0.05 (*), 0.01 (**), 0.0001 (***), and <0.0001 (****).*

The tested concentrations of saRNA included 5, 10, 25, and 50 nM. Remarkably, all tested concentrations demonstrated a significant induction of *SCN1A* expression compared to the samples treated with scrambled saRNA, indicating the effectiveness of saRNA-mediated upregulation of *SCN1A*. The highest upregulation of *SCN1A* expression was achieved following treatment with 50 nM saRNA.

Based on these results, a concentration of 50 nM saRNA is most effective in inducing the desired upregulation of *SCN1A* expression and may be considered optimal for future studies involving saRNA-mediated modulation of *SCN1A* in AF22 cells.

Supplementary material 6

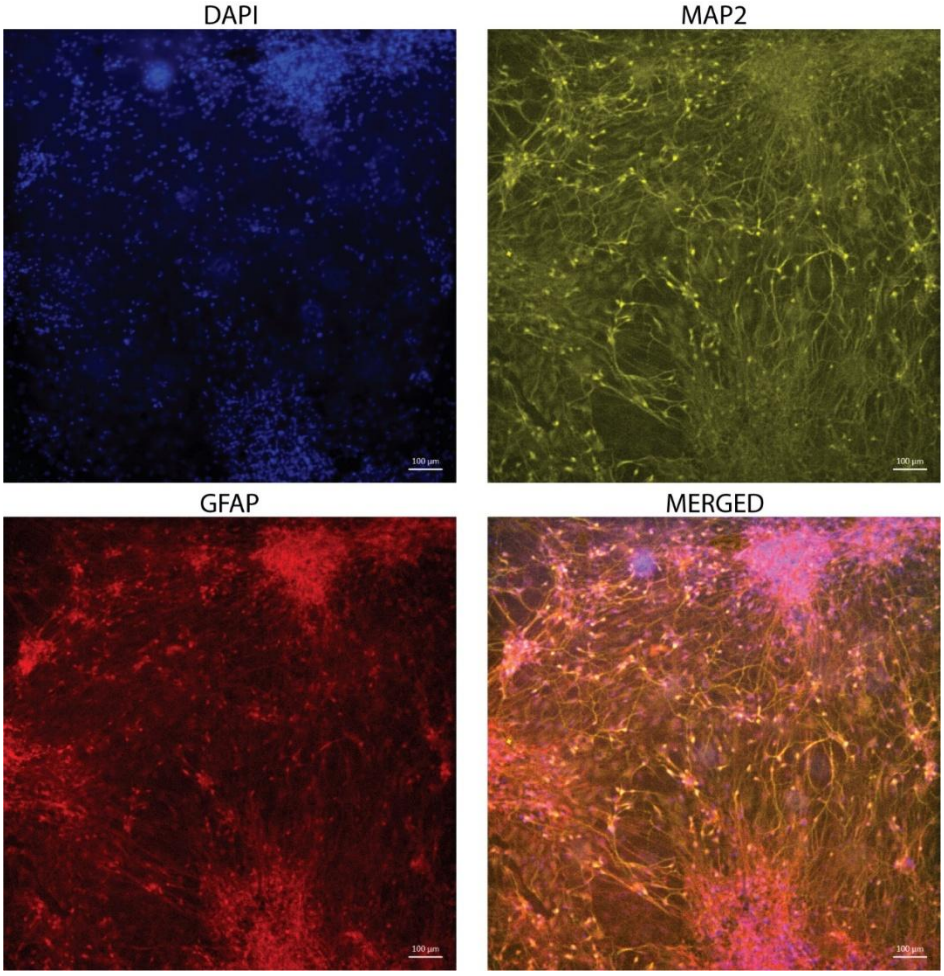


Figure 13. Immunofluorescence characterization of differentiated AF22 cells demonstrates mixed neuronal and glial populations. *AF22 cells were differentiated for 4 weeks and immunostained for cell-type specific markers. DAPI (blue) shows nuclear staining revealing high cell density with some clustering. MAP2 (yellow) labels mature neurons, showing extensive neuronal networks with elaborate dendritic arborizations throughout the culture. GFAP (red) identifies astrocytes, which appear as distinct populations interspersed among the neuronal networks. The merged image demonstrates the co-culture of neurons and astrocytes, typical of differentiated neural cultures, with areas of high neuronal density and supporting glial cells. Images were acquired using a Zeiss Cell Observer microscope.*

Supplementary material 7

Examples of P21 (positive control) saRNA transfections

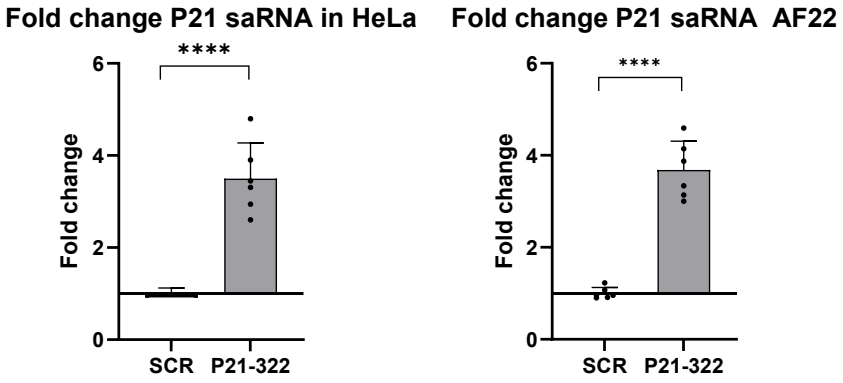
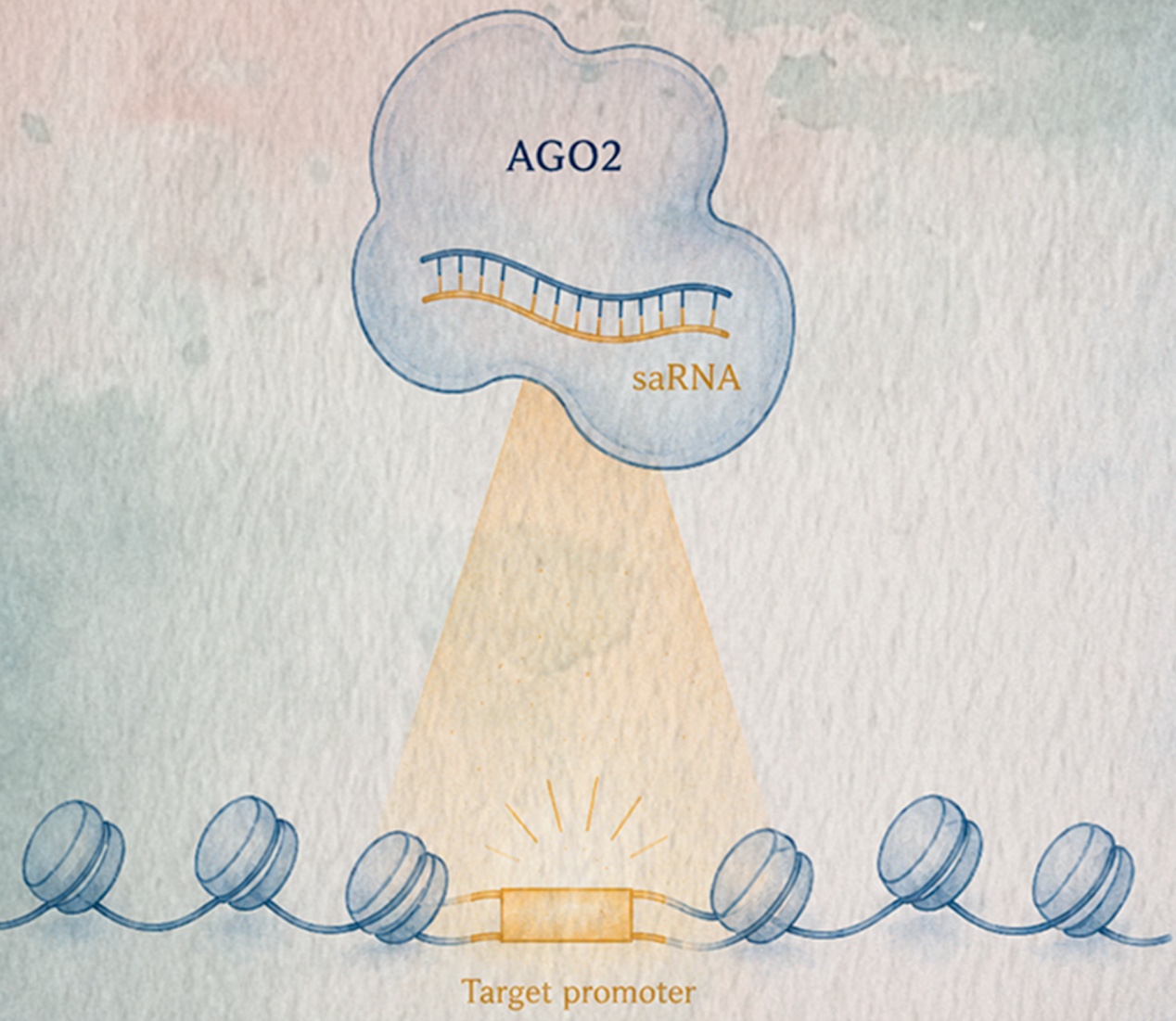


Figure 14: P21 mRNA expression following saRNA treatment in HeLa and AF22 cells

*qPCR results showing the fold change in P21 mRNA expression in HeLa and AF22 cells harvested three days post-transfection with 50 nM of P21-322 saRNA compared to scrambled (SCR) control. The data represent fold change ± standard deviation (SD), and each dot corresponds to a technical replicate. Statistical significance was determined using an unpaired two-sample t-test, with significance indicated as **** ($p < 0.0001$) on $\Delta\Delta Ct$ values. Both HeLa and AF22 cells showed highly significant upregulation of P21 expression following treatment with the P21-322 saRNA, confirming the efficacy of this saRNA for inducing gene expression in both cell types.*


AGO2
saRNA
Target promoter

Chapter 3

Mechanistic link to AGO2 in neural cells

Chapter 3

Mechanistic Insights into saRNA-Mediated Upregulation of *SCN1A*

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Abstract

This chapter investigates how small activating RNAs (saRNAs) upregulate *SCN1A* and to what extent this follows the canonical AGO2-dependent RNA activation (RNAa) pathway. Building on the identification of *SCN1A*-10 as a potent activator in neuronal cells, we combined AGO2 knockdown, chromatin immunoprecipitation (ChIP-qPCR), and RNA sequencing in HeLa and AF22 NES cells using a panel of saRNAs (*SCN1A*-7, -10, -12, -14).

siRNA-mediated AGO2 knockdown reduced AGO2 protein to ~50–75% of control and attenuated saRNA-induced upregulation of both *SCN1A*-10 and the *CDKN1A* (*P21*) positive-control saRNA, indicating that efficient RNAa requires intact AGO2 levels. In contrast, AGO2 ChIP-qPCR did not show detectable enrichment at *SCN1A* or *CDKN1A* promoter regions, despite robust saRNA-mediated transcriptional activation and clear H3K4me3 enrichment at *GAPDH* as a positive control. These data suggest that AGO2 involvement in *SCN1A* RNAa is either highly transient, mediated via RNA intermediates, or below the sensitivity of conventional ChIP-qPCR.

To assess a potential role for antisense long noncoding RNA, Ribo-Zero RNA-seq was performed in AF22 cells. While numerous lncRNAs were detected, *SCN1A-AS1* (previously implicated in *SCN1A* regulation) was undetectable in both scrambled and *SCN1A*-10 conditions, arguing against a *SCN1A-AS1*–dependent mechanism in this model.

Poly(A) RNA-seq confirmed robust *SCN1A* upregulation by *SCN1A*-10, -12, and -14 and a trend toward downregulation by *SCN1A*-7. Transcriptome-wide responses differed markedly between saRNAs, with *SCN1A*-10 showing the most favorable balance between target activation and global specificity. Gene Ontology analysis for *SCN1A*-10 revealed enrichment of neuronal and ion-channel-related components and functions among upregulated genes, and extracellular matrix terms among downregulated genes, indicating coordinated modulation of neuronal excitability pathways.

In summary, saRNAs can activate *SCN1A* in an AGO2-associated manner in the absence of detectable *SCN1A-AS1* expression, while also inducing broader transcriptional programs. *SCN1A*-10 emerges as a lead candidate with comparatively high specificity, but the precise molecular mode of promoter engagement remains unresolved and will require RNA-centric chromatin approaches to define.

Introduction

Small activating RNAs (saRNAs) represent a distinct class of small RNAs capable of inducing gene expression through RNA activation (RNAa), a nuclear process involving Argonaute 2 (AGO2)-mediated promoter targeting and chromatin remodeling [1,2]. Unlike RNA interference (RNAi), which primarily mediates gene silencing in the cytoplasm, RNAa activates transcription by assembling a nuclear RNA-induced transcriptional activation (RITA) complex composed minimally of AGO2, RNA helicase A (RHA/DHX9), and CTR9, a member of the PAF1 transcription elongation complex [1,3]. The RITA complex promotes an open chromatin state, characterized by enrichment of activating marks such as H3K4 trimethylation and H2B monoubiquitination, alongside a reduction in repressive histone modifications like H3K9me3 and H3K27me3 [1]. Importantly, the nuclear localization of AGO2, essential for RNAa, is context-dependent, influenced by cell density, mature miRNA availability, and interactions with cytoplasmic TNRC6 proteins [3].

However, the exact targeting mechanism utilized by saRNA-loaded AGO2 remains uncertain for most saRNAs, with clear mechanistic evidence available for only a limited number of targets. Two non-exclusive models are proposed: direct binding of AGO2–saRNA complexes to promoter DNA guided by seed-region complementarity, and indirect targeting through RNA–RNA interactions involving promoter-associated antisense transcripts (NATs) [4,5]. To date, direct DNA binding has been conclusively demonstrated for only a single saRNA, targeting the P21 promoter by Portnoy et al. (2016), where mutations in the saRNA seed sequence or target DNA completely abolished transcriptional activation [4]. Conversely, NATs have been implicated in RNAa through RNA–RNA interactions in several contexts, as illustrated by saRNAs targeting the progesterone receptor, where antisense transcripts were required to facilitate the recruitment of transcriptional machinery [5].

Antisense transcripts represent compelling targets for RNAa-based therapeutic strategies, given that approximately 40% of active promoters produce overlapping antisense transcripts [6]. The clinical candidate saRNA MTL-CEBPA targets an antisense transcript overlapping the CEBPA gene, successfully activating both sense and antisense transcription without RNA degradation, highlighting a distinct activation mechanism from traditional antisense oligonucleotide (ASO) strategies [7]. Similarly, ASO-mediated suppression of *SCN1A-AS1* significantly restored *SCN1A* expression in a Dravet syndrome mouse model, underscoring the therapeutic potential of antisense RNA modulation [8]. These findings highlight the complexity of RNAa mechanisms, suggesting that saRNAs could function via different pathways depending on promoter architecture and the presence of antisense transcripts.

Our previous work (Chapter 2) identified *SCN1A*-10, an saRNA capable of robustly increasing *SCN1A* expression in neuronal cells. The *SCN1A*-10 binding site overlaps *SCN1A-AS1*, raising critical mechanistic questions about whether its activity involves canonical AGO2-mediated promoter targeting or modulation of the antisense transcript. To explore these possibilities comprehensively, we also included saRNAs *SCN1A*-12 and *SCN1A*-14 targeting alternative promoter regions that do not overlap with *SCN1A-AS1*. These additional saRNAs provide

essential comparative controls, facilitating the dissection of promoter-specific versus antisense RNA-dependent mechanisms. Additionally, assessing the specificity and broader transcriptional effects of these saRNAs is essential, given potential off-target interactions and indirect transcriptional changes triggered by RNAa-based strategies [9,10].

This study aimed to elucidate the molecular mechanisms underlying *SCN1A* activation by *SCN1A*-targeting saRNAs through four main objectives: (1) determining the necessity of AGO2 in *SCN1A*-10-mediated activation via AGO2 knockdown experiments; (2) assessing AGO2 occupancy at the *SCN1A* promoter using chromatin immunoprecipitation followed by quantitative PCR (ChIP-qPCR); (3) investigating potential involvement of *SCN1A-AS1* via strand-specific RNA sequencing; and (4) characterizing transcriptome-wide effects of *SCN1A*-targeting saRNAs, providing insight into their specificity, broader biological impact, and mechanistic complexity.

Methods

Cell Culture – AF22 and HeLa Cells

AF22 neuroepithelial stem cells, derived from induced pluripotent stem cells (iPSCs), were cultured at 37°C and 5% CO_2 in NES medium, prepared freshly every seven days. The NES medium consisted of DMEM/F12 with 0.02% L-alanyl-L-glutamine, supplemented with 1% N-2, 1% penicillin/streptomycin (P/S), 0.1% B27, 0.05% basic fibroblast growth factor, and 0.01% epidermal growth factor. Culturing flasks were pre-coated with 0.1% gelatin in PBS for 30 minutes at 37°C. To split cells, medium was aspirated, cells were washed with dPBS and detached using TrypLE Express (37°C for 3 minutes). Cells were then resuspended in NES medium and seeded into new flasks. AF22 cells were split every 3-4 days upon reaching 80% confluency.

HeLa cells were cultured in high-glucose DMEM supplemented with 10% FBS and 1% P/S. For subculturing, cells were washed with dPBS, treated with Trypsin-EDTA solution (0.05%) for 3 minutes at 37°C, and then resuspended in fresh medium. Cells were split upon reaching 100% confluency once or twice weekly.

Transfection

One day before transfection, AF22 or HeLa cells were seeded to reach approximately 60-70% confluency in 12-well plates or 10 cm dishes. The next day, HeLa cells were switched to Opti-MEM four hours prior to transfection. Cells were transfected with saRNAs or siRNAs at final concentrations of 50 nM, using Lipofectamine RNAiMAX (Invitrogen) in Opti-MEM, according to the manufacturer's instructions. SaRNAs were designed as described previously (van de Vondervoort, chapter 2) and a scrambled RNA consisting of a nonhuman sequence of 19 nucleotides which is not complementary to anything in the genome and transcriptome was included as a control in all experiments. List of sequences of saRNAs and siRNAs can be

found in the supplementary 1. Cells were incubated at 37°C for 72 hours post-transfection and harvested for RNA isolation, Western blot, or chromatin immunoprecipitation (ChIP).

RNA Isolation and Sequencing

Cells were washed with dPBS and detached using TrypLE Express. Cell suspensions were centrifuged at 300 rcf for 3 minutes, after which cell pellets were snap-frozen and stored at -80°C until RNA extraction. RNA was isolated using the Nucleospin RNA Mini Kit (Macherey-Nagel) according to the manufacturer's instructions and eluted in 25 µL of RNase-free water. RNA concentrations were quantified using a NanoDrop 2000 spectrophotometer.

To investigate potential interactions between small activating RNA (saRNA) *SCN1A*-10 and overlapping long noncoding RNAs (lncRNAs) at the target genomic site, we conducted two RNA sequencing (RNA-seq) experiments. In experiment 1, AF22 NES cells were transfected with scrambled control (SCR), saRNA-10, saRNA-12, and saRNA-14. Experiment 2 included the same conditions as experiment 1, with the addition of saRNA-7, selected due to consistent downregulation observed in preliminary qPCR analyses.

RNA samples were sent to the Utrecht Sequencing Facility (USEQ) for library preparation and sequencing. Sequencing was performed using the Illumina NextSeq2000 platform. Two separate library preparations were utilized to comprehensively capture the transcriptome: poly(A)-selected libraries for mRNA sequencing and ribosomal RNA-depleted libraries for lncRNA sequencing. Raw data was aligned and quantified using the UMCUGenetics/RNASeq-NF pipeline v1.1 using default parameters (UMCUGenetics/RNASeq-NF: RNASeq-NF is an NGS analysis pipeline for RNA expression quantification).

Aligned reads were visualized and analyzed using Integrative Genomics Viewer (IGV) to identify lncRNAs overlapping the *SCN1A* saRNA binding region. Ribosomal RNA-depleted RNA-seq (Ribo-Zero) datasets allowed verification of lncRNA expression specifically at the genomic locus targeted by the saRNA.

Additionally, feature count data from the two Poly-A sequencing experiments were merged and analyzed with a custom R script (available via GitHub; https://github.com/JoepVondervoort/RNA-sequencing-analysis). Differential gene expression analysis used DESeq2 with condition as the sole design variable; Gene Ontology (GO) enrichment analysis used clusterProfiler and DOSE.

We acknowledge the Utrecht Sequencing Facility (USEQ) for providing sequencing service and data. USEQ is subsidized by the University Medical Center Utrecht and The Netherlands X-omics Initiative (NWO project 184.034.019).

RNA Isolation and cDNA Synthesis

Total RNA was extracted using the Machery Nagel RNA Mini Kit, with elution performed in 25 μL of nuclease-free water. cDNA synthesis was conducted using the iScript cDNA synthesis kit with an input of 400 ng total RNA in a 20 μL reaction volume. The reaction mix consisted of 1 μL iScript Reverse Transcriptase, 4 μL iScript Reaction Mix, RNA (400 ng), and RNase-free water to a final volume of 15 μL. Reverse transcription was performed under the following conditions: priming at 25°C for 5 minutes, reverse transcription at 46°C for 20 minutes, and RT inactivation at 95°C for 1 minute, followed by a hold at 4°C. After completion of the reaction, 100 μL RNase-free water was added.

Primer Design and Preparation

Primer sequences were designed to target *SCN1A*, *AGO2*, *CDKN1A*, and *C1ORF34* as a housekeeping gene. A primer mix was prepared by combining 10 μL forward primer (100 μM), 10 μL reverse primer (100 μM), and 80 μL nuclease-free water to reach a final volume of 100 μL at a 20 μM concentration.

qPCR reactions were prepared using a Micro-AMP Optical 384-well plate and Sybr Green assays. Each reaction contained 4 μL Sybr Green (2X), 1.84 μL nuclease-free water, 0.16 μL primer mix (20 μM), and 2 μL cDNA sample, resulting in a total reaction volume of 8 μL.

qPCR Cycling Conditions

qPCR reactions for cDNA analysis were conducted under the following cycling conditions: initial enzyme activation at 95°C for 2 minutes, followed by 40 cycles of denaturation at 95°C for 5 seconds and annealing/extension at 60°C for 30 seconds. The dissociation step was performed with an initial denaturation at 95°C for 15 seconds (ramp rate: 1.99°C/second), followed by a 60°C hold for 1 minute (ramp rate: 1.77°C/second), and a gradual increase to 95°C at 0.075°C/second.

Sonication Optimization

Sonication parameters were optimized to shear chromatin to approximately 300 bp (supplementary 2). Aliquots of chromatin from 10 million AF22 cells were sonicated using a Covaris Bioruptor Pico at various cycle counts (10, 15, 20, 24, 26 and 30 cycles). Chromatin fragments were analyzed on a 1% agarose gel stained with SYBR Gold to confirm fragment size. The optimal sonication was established at 24 cycles, with fragments around 300 bp.

Chromatin Immunoprecipitation (ChIP) assay

To investigate the interaction between small activating RNAs (saRNAs) and their target genomic regions, chromatin immunoprecipitation (ChIP) was performed using HeLa cells

harvested 48 hours post-transfection. Cells were first washed and counted, after which 10 million cells per experimental condition were crosslinked with 1% formaldehyde diluted in Dulbecco's phosphate-buffered saline (dPBS) for 10 minutes at room temperature. Crosslinking was quenched by the addition of glycine to a final concentration of 0.125 M, to terminate formaldehyde fixation and prevent excessive crosslinking. Cells were then centrifuged, washed with cold dPBS, and pellets were stored at -70°C to preserve sample integrity.

Prior to chromatin isolation, cell pellets were thawed on ice and sequentially lysed in three separate buffers to facilitate efficient nuclear isolation and chromatin release. Initially, pellets were incubated in Lysis Buffer A for 10 minutes at 4°C, followed by a brief incubation in Lysis Buffer B for 5 minutes on ice, and subsequently in Lysis Buffer C for 10 minutes on ice (complete buffer compositions are provided in supplementary 5). Chromatin was fragmented by sonication into approximately 200-500 bp fragments to ensure effective antibody accessibility and resolution.

Sonicated chromatin was pre-cleared with Protein G magnetic beads to reduce non-specific background binding. Immunoprecipitation was performed overnight at 4°C using Protein G magnetic beads conjugated to either anti-AGO2 (FUJIFILM Wako, 20 μg per sample (shown in figure)), and another AGO2 antibody [EPR10411] (data not shown) to detect RNA-induced transcriptional complexes, or anti-H3K4me3 (Abcam AB213224, 5 μg per sample) as a positive control indicative of active transcriptional marks. Following incubation, beads were subjected to stringent washing twice each with RIPA, RIPA-500 (high-salt wash), and LiCl buffers, concluding with a single wash in TE buffer to reduce non-specific interactions and improve signal specificity. Chromatin was subsequently eluted from the beads using Elution Buffer, and crosslinking was reversed by overnight incubation at 65°C.

Post-reversal of crosslinks, contaminating RNA and proteins were enzymatically digested using RNase A (1 hour at 37°C) and Proteinase K (2 hours at 55°C), respectively. DNA was purified using Ampure beads coupled with isopropanol precipitation to efficiently recover fragmented DNA. Final purified DNA was eluted in 40 μL of nuclease-free water, quantified using a NanoDrop spectrophotometer, and stored at -20°C until further analysis.

Quantitative PCR (qPCR) analysis of the ChIP-enriched DNA was conducted to assess enrichment at specific regions of the *SCN1A* promoter targeted by the saRNAs. Primers were designed to amplify the saRNA binding site within the *SCN1A* promoter, as well as regions located 3000 bp upstream, 300 bp upstream, 300 bp downstream, and 3000 bp downstream of the binding site. Additional primers were included targeting the *CDKN1A* (P21) saRNA binding region and the *GAPDH* promoter as positive controls for H3K4me3 enrichment.

Each 8 μL qPCR reaction contained 4 μL SYBR Green master mix, 2 μL purified DNA, 0.16 μL each of forward and reverse primers, and 1.84 μL nuclease-free water. Cycling conditions for ChIP-qPCR consisted of an initial enzyme activation at 95°C for 2 minutes, followed by 40 cycles of denaturation at 95°C for 15 seconds and annealing/extension at 60°C for 60 seconds. Melt curve analysis (dissociation curve) was performed with an initial denaturation at 95°C for 15 seconds (ramp rate: 1.6°C/second), followed by a 60°C hold for 1 minute (ramp rate: 1.6°C/second), and a gradual increase to 95°C at 0.075°C/second to verify product specificity.

Threshold cycle (Ct) values obtained from ChIP samples were normalized against input DNA and a negative control genomic region ("desert region") lacking known regulatory elements. Fold enrichment was calculated using the ΔΔCt method (2^- ΔΔCt). Statistical analyses were performed in GraphPad Prism, and ΔΔCt analyses were conducted using Excel to determine relative enrichment levels.

Western Blot for AGO2

Western blotting was employed to confirm the efficiency of AGO2 knockdown mediated by siRNA treatment. Protein extracts were prepared from cultured cells using RIPA buffer supplemented with protease inhibitors. Lysates were sonicated at 4°C using a Bioruptor Pico device under optimized conditions: three cycles of 30 seconds on and 30 seconds off at a low-frequency setting. Protein samples were stored at -20°C until further analysis.

Protein samples were prepared for electrophoresis by mixing 10 µL lysate with 5 µL NuPAGE LDS Sample Buffer, 2 µL NuPAGE Reducing Agent, and 3 µL Milli-Q water to a final volume of 20 µL, followed by heating at 80°C for 10 minutes to denature proteins. Subsequently, 15 µL of each prepared sample was loaded onto a 4-12% Bis-Tris gradient gel for SDS-PAGE, alongside a PageRuler prestained protein ladder to determine molecular weights. Electrophoresis was conducted in MES SDS Running Buffer (prepared by diluting 20 µL of 20X MES buffer in 380 µL distilled water) with the addition of 200 µL antioxidant, at 150 V for 105 minutes to ensure adequate protein separation.

Proteins were transferred from the gel onto a 0.2 µm nitrocellulose membrane using the Bio-Rad Turbo-Blot System set to the "1.5 mm Gel" transfer program (1.3 A, 25 V, 10 minutes). To minimize nonspecific antibody binding, membranes were blocked in PBS containing 0.1% Tween-20 and 5% non-fat milk powder for at least 30 minutes at room temperature.

Following blocking, membranes were cut between 55 and 97 kDa to separately detect target proteins and incubated overnight at 4°C with primary antibodies diluted in blocking buffer. AGO2 was targeted using a rabbit monoclonal anti-AGO2 antibody (AB57113, Abcam) at a dilution of 1:100, and α-tubulin, used as a loading control, was detected using a mouse monoclonal antibody at 1:1500 dilution. After incubation, membranes were washed three times for 5 minutes each with PBS containing 0.1% Tween-20.

Secondary antibody incubation was performed using goat anti-mouse or anti-rabbit, 680 or 800CW respectively (LiCor, IRDye®) secondary antibodies conjugated to fluorescent dyes at a dilution of 1:7500 in blocking buffer for 1 hour at room temperature in the dark. Membranes were then washed again three times for 5 minutes each with PBS-0.1% Tween-20. Protein bands were visualized and quantified using an Amersham Typhoon Scanner and analyzed using FIJI software (ImageJ) to determine knockdown efficiency. Tubulin was used as a loading control to normalize protein levels across conditions. Relative AGO2 expression was quantified by calculating the ratio of AGO2 band intensity to tubulin band intensity, and results were expressed as fold change compared to scrambled control.

Primer Efficiency Testing for ChIP-qPCR

New primers were designed (appendix 1) to flank potential AGO2 binding sites on the *SCN1A* promoter, validated for efficiency by qPCR in HeLa DNA at 100, 10, and 1 ng/μL. Each reaction contained 2 μL DNA, 3.5 μL water, 5 μL SYBR green mix, and 0.5 μL primer mix. Efficiency was confirmed if ΔCt values across dilutions fell within 3.1 to 3.6.

Statistical Analysis

Statistical analysis was conducted using GraphPad Prism 9 and R. Normality was assessed with Shapiro-Wilk tests and Q-Q plots. For qPCR analysis, statistical tests were performed on ΔCt values to ensure appropriate statistical assumptions. One-way ANOVA followed by Tukey's multiple comparisons test was used for normally distributed data, while Kruskal-Wallis test followed by Dunn's multiple comparisons test was applied for non-normally distributed data.

Volcano plots were generated using custom Python scripts to visualize differential gene expression results. The analysis was performed using Python 3.10.9 with the following libraries: pandas (v1.5.3), numpy (v1.23.5), matplotlib (v3.7.0), seaborn (v0.13.2), and adjustText (v1.3.0). ChIP-qPCR results were visualized using the same Python environment and core visualization libraries as described above. Bar plots with individual data points were generated using matplotlib (v3.7.0) and seaborn (v0.13.2). Data from three independent experiments were presented as mean ± standard error of the mean (SEM). Statistical analyses were performed using scipy (v1.10.0) for calculating standard errors and statistical significance.

Results

AGO2 is associated with saRNA efficiency

To investigate the importance of AGO2 for small activating RNA (saRNA) function, we tested whether its knockdown would affect saRNA efficacy. HeLa cells were first transfected with a AGO2-targeting siRNA 72 hours prior to harvesting. After 48 hours, cells were subsequently transfected with either *SCN1A*-10 saRNA, *CDKN1A* (P21) saRNA, or a scrambled control. Twenty-four hours later, cultures were split for AGO2 protein analysis and target-gene mRNA quantification.

The western blot analysis confirmed successful AGO2 knockdown in AGO2 siRNA treated samples (Figure 1), with AGO2 protein levels reduced to 0.50-0.75 of scrambled control.

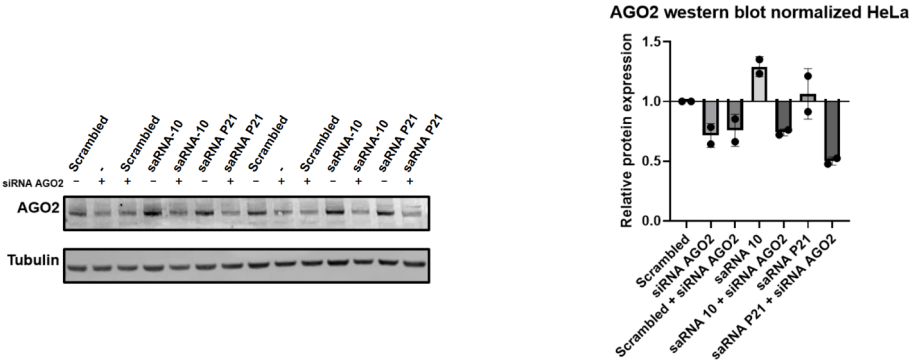

Figure 1: AGO2 protein levels are effectively reduced by siRNA knockdown in HeLa cells. *HeLa cells were transfected with siRNA targeting AGO2 72 hours prior to harvesting, followed by transfection with either SCN1A-10 saRNA, P21 saRNA, or scrambled control 24 hours later. Protein lysates were collected 48 hours post-saRNA transfection and analyzed by Western blot. Left panel: Representative Western blot showing AGO2 protein levels across all experimental conditions. Tubulin serves as a loading control to ensure equal protein loading. The "+" and "-" symbols indicate the presence or absence of siRNA AGO2 treatment, respectively. Right panel: Quantification of AGO2 protein expression normalized to tubulin and presented relative to scrambled control. Each data point represents an individual biological replicate (N=2 per condition), with error bars representing standard error of the mean.*

We next performed quantitative PCR (qPCR) analysis to measure mRNA levels of *AGO2*, *SCN1A*, and *CDKN1A* (Figure 2). *SCN1A*-10 saRNA increased *SCN1A* mRNA 2.9-fold ($p = 0.023$) in control cells, but this effect was diminished upon AGO2 knockdown. Similarly, *CDKN1A* (P21) saRNA induced *CDKN1A* 3.6-fold ($p = 0.078$), and this activation was impaired by AGO2 knockdown. AGO2 mRNA itself was reduced to ~0.50–0.62-fold by siAGO2, confirming knockdown at the transcript level. Notably, P21 saRNA also increased AGO2 mRNA, an effect not seen with *SCN1A*-10 saRNA and not predicted from promoter complementarity, suggesting an indirect or feedback response.

Overall, these results indicate that AGO2 plays a critical role in the activation of gene expression by saRNAs, supporting the hypothesis that saRNA function is AGO2-dependent.

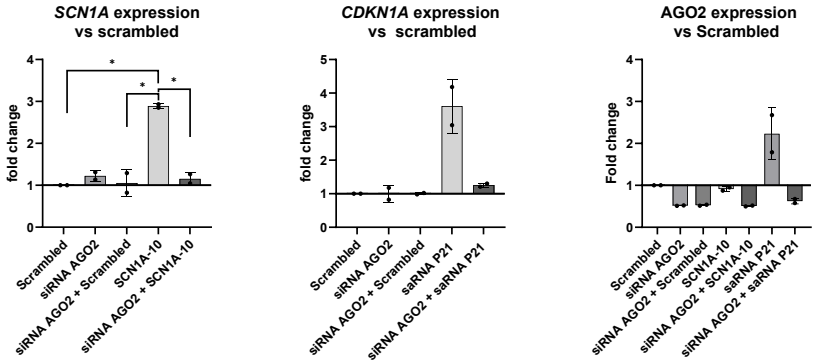

Figure 2: AGO2 knockdown impairs saRNA-mediated gene activation in HeLa cells. *HeLa cells were transfected with siRNA targeting AGO2 72 hours prior to harvesting, followed by transfection with either SCN1A-10 saRNA, P21 saRNA, or scrambled control 48 hours later. RNA was isolated 24 hours post-saRNA transfection and analyzed by qPCR. Data are presented as fold change relative to scrambled control, with each data point representing an individual biological replicate (N=2 per condition). Error bars represent standard error of the mean. Left panel (SCN1A expression): SCN1A-10 saRNA significantly upregulated SCN1A mRNA levels compared to scrambled control (p=0.023). However, this saRNA-mediated activation was significantly reduced when AGO2 was knocked down (SCN1A-10 vs. siAGO2+SCN1A-10, p=0.040), demonstrating AGO2-dependent gene activation. Middle panel (CDKN1A expression): P21 saRNA showed a trend toward increased CDKN1A expression with 3,6 increased fold change, though this did not reach statistical significance. AGO2 knockdown appeared to further diminish this response. Right panel (AGO2 expression): siRNA treatment successfully reduced AGO2 mRNA levels across all conditions containing siAGO2, confirming effective knockdown. Statistical analysis: Data were analyzed using ΔCt values to ensure appropriate statistical assumptions. For SCN1A expression, one-way ANOVA followed by Tukey's multiple comparisons test was performed. For CDKN1A and AGO2 expression, non-parametric Kruskal-Wallis test followed by Dunn's multiple comparisons test was used due to data distribution characteristics. Statistical significance was set at *p<0.05.*

ChIP-qPCR Analysis of AGO2 Binding at saRNA Target Sites

We aimed to verify the presumed mechanism of action of saRNAs, which involves AGO2-mediated binding to the specific promoter sequence for which it is specifically designed. Therefore, we performed a Chip-qPCR analysis for the specific region in the *SCN1A* promoter for which *SCN1A*-10 has been designed. First, we confirmed successful transfection and saRNA-induced gene activation prior to chromatin immunoprecipitation (ChIP). To do so, we transfected HeLa cells with *SCN1A*-10 and the P21-saRNA. At 48 hours post-transfection, we harvested 10% of the total cell lysate for RNA extraction and confirmed upregulation of both target genes, with an 2,8-fold increase of *SCN1A* (p=0.070) and ~5-fold increase of *CDKN1A* (p=0.267), confirming successful saRNA-mediated gene activation (Figure 3).

Following RNA extraction, the remaining 90% of the cell lysate was subjected to ChIP-qPCR as previously described for the P21-saRNA ([1]). In brief, formaldehyde crosslinking was used to stabilize protein-DNA interactions. Crosslinked chromatin was fragmented by sonication,

and fragments bound by AGO2 protein were selectively immunoprecipitated using an AGO2-specific antibody. We tested two different antibodies as described in methods. Both antibodies were confirmed to bind specifically to AGO2 using Western blot (data not shown).

ChIP-qPCR was carried out using primer pairs designed around the saRNA target sites within the promoters of *SCN1A* and *CDKN1A*, as well as distal control regions located approximately 3000 bp upstream and downstream from these sites.

ChIP-qPCR signals were normalized against input chromatin (total chromatin prior to immunoprecipitation) and further corrected using amplification data from a genomic "desert" region on chromosome 12 as an additional negative control. However, no detectable AGO2 enrichment was observed at any of the tested regions, including the predicted saRNA binding sites, in either saRNA-treated or control samples (Figure 4). All enrichment values remained at or near baseline levels (fold change ≈ 1) across the tested primer positions. While no AGO2 association was detected, ChIP with an H3K4me3 antibody yielded robust enrichment at the *GAPDH* promoter, confirming chromatin integrity and assay functionality, even though this control does not validate sensitivity for transiently bound factors such as AGO2. These results suggest that either AGO2 binding occurs below the detection threshold of our ChIP-qPCR assay or that saRNA-mediated gene activation may involve alternative mechanisms independent of stable AGO2-chromatin interactions.

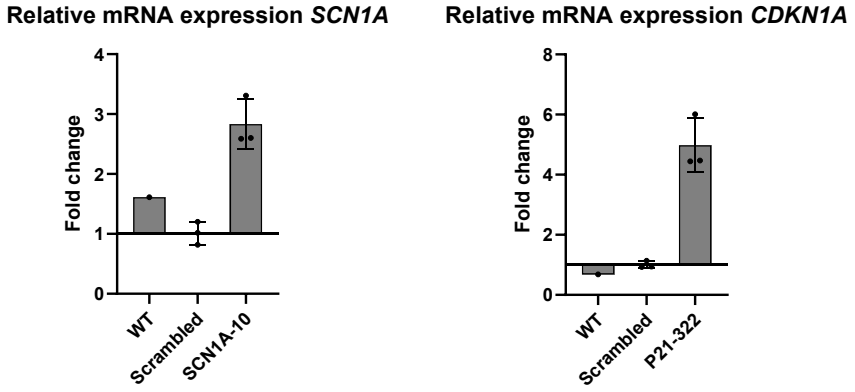

Figure 3: saRNA transfection successfully induces target gene expression in HeLa cells that were further analyzed for ChIP-qPCR. *HeLa cells were transfected with either SCN1A-10 saRNA, P21-322 saRNA, or scrambled control. RNA was isolated 24 hours post-transfection and analyzed by qPCR to confirm successful saRNA-mediated gene activation prior to ChIP-qPCR analysis. Left panel (SCN1A expression): SCN1A-10 saRNA treatment resulted in approximately 3-fold upregulation of SCN1A mRNA compared to both wild-type (WT) untreated cells and scrambled control, demonstrating effective target gene activation. Right panel (CDKN1A expression): P21-322 saRNA treatment induced approximately 5-fold upregulation of CDKN1A mRNA relative to control conditions, confirming successful saRNA-mediated gene activation. Data are presented as fold change relative to wild-type control, with each bar representing the mean of biological replicates. Error bars represent standard error of the mean. Individual data points are overlaid to show the distribution of values. Statistical analysis: Data were analyzed using ΔCt values with Kruskal-Wallis test followed by Dunn's multiple comparisons test. While the observed upregulation showed strong trends (SCN1A: p=0.070; CDKN1A: p=0.135 and p=0.267), statistical significance was not reached, likely due to limited sample size.*

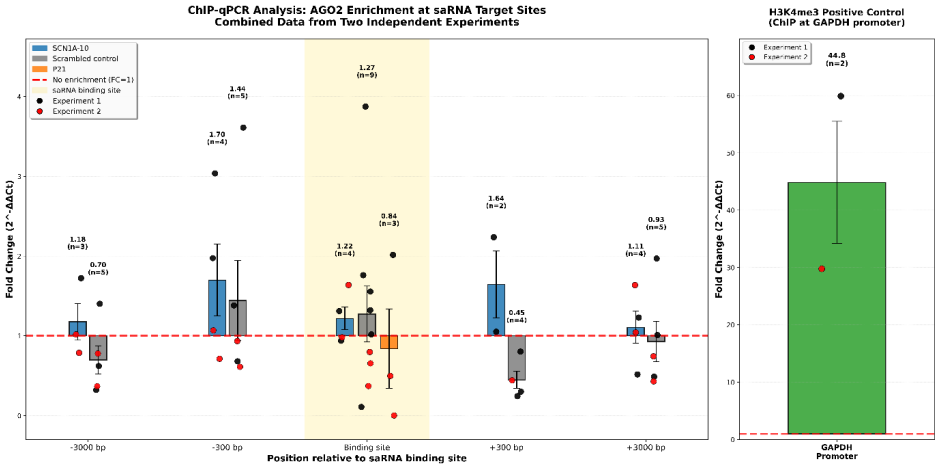

Figure 4: ChIP-qPCR analysis shows no detectable AGO2 enrichment at the *SCN1A* promoter following saRNA transfection in two independent experiments. *ChIP-qPCR was performed to assess AGO2 occupancy at the SCN1A promoter region in cells transfected with SCN1A-10 saRNA, scrambled control saRNA, or P21 saRNA. (Left panel) AGO2 enrichment at various positions relative to the SCN1A-10 saRNA binding site (highlighted in yellow). Positions examined include -3000 bp, -300 bp, the binding site (0 bp), +300 bp, and +3000 bp. Cells were transfected with SCN1A-10 (blue bars), scrambled control (gray bars), or P21 saRNA (orange bars, binding site only). Data represent fold change (2^-ΔΔCt) normalized to input DNA and a gene desert region on chromosome 12, with the baseline set at 1.0 (red dashed line) indicating no enrichment. Individual data points from two independent experiments are shown as black dots (Experiment 1) or red dots (Experiment 2). Bars represent mean ± SEM with sample sizes indicated above each bar. (Right panel) H3K4me3 positive control showing robust enrichment at the GAPDH promoter in wild-type cells (44.8-fold, n=2), confirming ChIP efficiency.*

Undetectable *SCN1A-AS1* expression suggests absence of lncRNA-mediated interference

Given the genomic proximity of the antisense long noncoding RNA *SCN1A-AS1* to our saRNA target site, we also investigated whether this lncRNA can be involved in the mechanism of action of our saRNA. Ribo-Zero RNA sequencing analysis of both scrambled control and saRNA10-treated samples revealed complete absence of detectable *SCN1A-AS1* expression in AF22 NES cells (Figure 5). Despite robust expression of neighboring genes *SCN1A* and *SCN9A*, and other lncRNAs (appendix 3) no transcriptional activity was observed across any of the annotated *SCN1A-AS1* isoforms in either treatment condition. This finding makes potential lncRNA-mediated competition for saRNA binding unlikely and confirmed that the observed *SCN1A* upregulation resulted directly from saRNA10 activity rather than indirect effects through antisense RNA modulation.

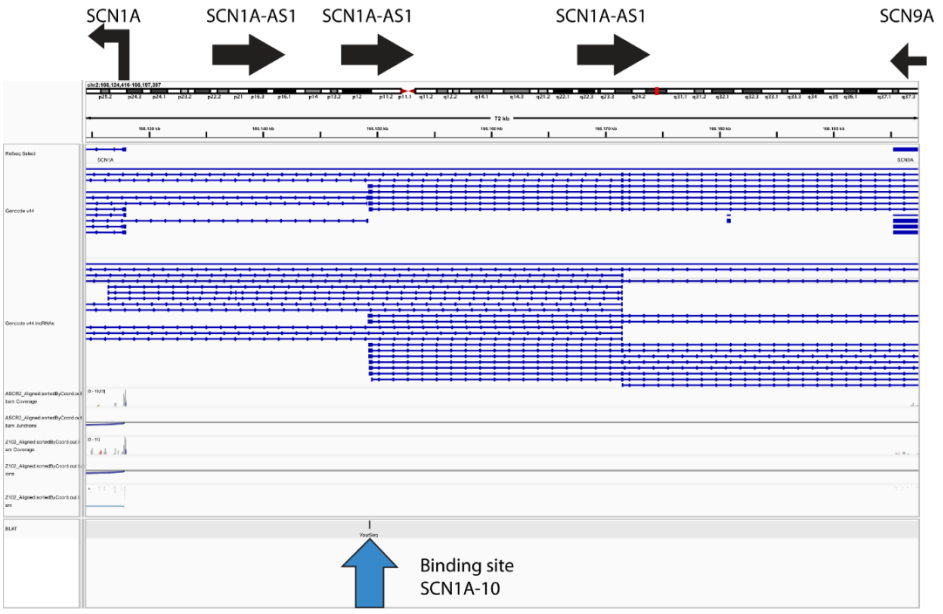

Figure 5: Ribo-Zero RNA sequencing analysis reveals absence of SCN1A-AS1 expression in AF22 NES cells. *Integrative Genomics Viewer (IGV) visualization of Ribo-Zero RNA sequencing data aligned to the SCN1A genomic region. The display shows the genomic locus spanning SCN1A, multiple copies of the SCN1A-AS1 long noncoding RNA, and SCN9A genes (indicated by black arrows showing gene orientation). RNA-seq tracks from scrambled control and SCN1A-10 saRNA-treated AF22 NES cells demonstrate robust expression of SCN1A and SCN9A genes (blue read coverage), while no detectable expression of SCN1A-AS1 is observed across any of the annotated SCN1A-AS1 transcripts in either condition. The blue arrow at the bottom indicates the binding site for saRNA SCN1A-10, designed to target this genomic region*

SCN1A upregulation is robust across distinct saRNAs, while genome-wide transcriptomic effects diverge

To confirm upregulation of *SCN1A* mRNA after transfection with our saRNAs and investigate potential off-target effects we performed RNA sequencing and transcriptome analysis of AF22 NES cells treated with four different saRNA constructs (saRNA7, saRNA10, saRNA12, and saRNA14) compared to scrambled control. These saRNA showed robust downregulation (saRNA7) and upregulation (saRNA10, -12, and 14). We hypothesized that effective saRNA candidates would demonstrate robust *SCN1A* upregulation with minimal global transcriptomic perturbation, indicating high specificity for the target gene.

Poly-A RNA sequencing analysis confirmed the upregulation of *SCN1A* mRNA as measured in 10 biological replicates of AF22 NES cells transfected with saRNA10, saRNA12, and saRNA14 (log2FC = 0.74, 0.65, and 0.55, respectively; all padj < 1×10^{-5}) compared to scrambled control (n=10). Also, downregulation of *SCN1A* after saRNA7 transfection (n=2)

showed a trend toward *SCN1A* downregulation (log2FC = -0.54, padj = 0.056), but without statistical significance, potentially due to the limited sample size.

We observed considerable differences in the genome-wide transcriptomic changes after transfection between saRNAs, which may correlate to the different promoter regions that are targeted (Figure 6). At padj < 0.01 and |log2FC| > 0.5, saRNA12 showed the most differentially expressed genes (3,353), followed by saRNA14 (1,149), saRNA7 (585), and saRNA10 (214). Eight genes were significantly upregulated across saRNA10, saRNA12, and saRNA14 (Table 1 and Figure 7). *SCN1A* ranked eighth by mean log2FC (0.649). Additional co-upregulated genes included *CACNA1E*, *VGF*, and *ABCA12*, suggesting coordinated activation of neuronal development and ion channel pathways. Similarly, six genes were consistently downregulated across all three effective treatments, which could indicate reproducible off-target effects that warrant further investigation, or represent downstream effects induced by transfection, or *SCN1A* upregulation.

Table 1. Genes consistently regulated across multiple saRNA treatments. *Tables showing genes that are significantly upregulated (top) or downregulated (bottom) across saRNA10, saRNA12, and saRNA14 treatments compared to scrambled control (padj < 0.01, |log2FC| > 0.5).*

Up-regulated Genes in All Three Conditions

Analysis Date: 2025-06-17 | **Significance Threshold:** padj < 0.01, |log2FC| > 0.5 | **Total Genes:** 8

Rank	Gene Symbol	Ensembl ID	saRNA10		saRNA12		saRNA14		Mean log₂FC
			log₂FC	padj	log₂FC	padj	log₂FC	padj	
1	ENSG00000229081	ENSG00000229081	3.015	1.33e-03	2.421	3.41e-03	2.311	9.92e-03	2.582
2	CACNA1E	ENSG00000198216	1.748	1.17e-07	2.006	1.40e-11	1.347	3.23e-05	1.700
3	VGF	ENSG00000128564	0.701	2.65e-03	1.112	2.11e-09	1.540	7.00e-17	1.118
4	ABCA12	ENSG00000144452	0.715	1.40e-03	1.606	1.46e-20	0.759	1.11e-04	1.027
5	CLDN11	ENSG00000013297	1.119	1.17e-05	0.643	9.58e-03	0.956	8.46e-05	0.906
6	POLR3G	ENSG00000113356	0.711	1.17e-07	1.186	1.39e-23	0.682	7.87e-08	0.860
7	KDR	ENSG00000128052	0.764	1.81e-03	0.882	1.36e-05	0.800	2.10e-04	0.815
8	SCN1A	ENSG00000144285	0.743	2.79e-09	0.649	2.56e-08	0.555	6.68e-06	0.649

Down-regulated Genes in All Three Conditions

Analysis Date: 2025-06-17 | **Significance Threshold:** padj < 0.01, |log2FC| > 0.5 | **Total Genes:** 6

Rank	Gene Symbol	Ensembl ID	saRNA10		saRNA12		saRNA14		Mean log₂FC
			log₂FC	padj	log₂FC	padj	log₂FC	padj	
1	ZNF385B	ENSG00000144331	-0.741	2.94e-04	-0.857	9.16e-07	-1.242	1.24e-12	-0.947
2	ENSG00000279177	ENSG00000279177	-0.671	5.47e-04	-1.020	4.87e-10	-1.069	2.19e-10	-0.920
3	ENSG00000280330	ENSG00000280330	-0.668	5.62e-04	-1.020	4.42e-10	-1.069	1.99e-10	-0.919
4	RAB11FIP1	ENSG00000156675	-0.794	1.61e-04	-0.751	4.07e-05	-1.078	6.67e-09	-0.874
5	NAA50	ENSG00000121579	-0.531	1.33e-40	-0.511	1.41e-39	-0.981	5.78e-142	-0.674
6	FBN3	ENSG00000142449	-0.680	3.55e-07	-0.738	9.37e-10	-0.543	2.38e-05	-0.654

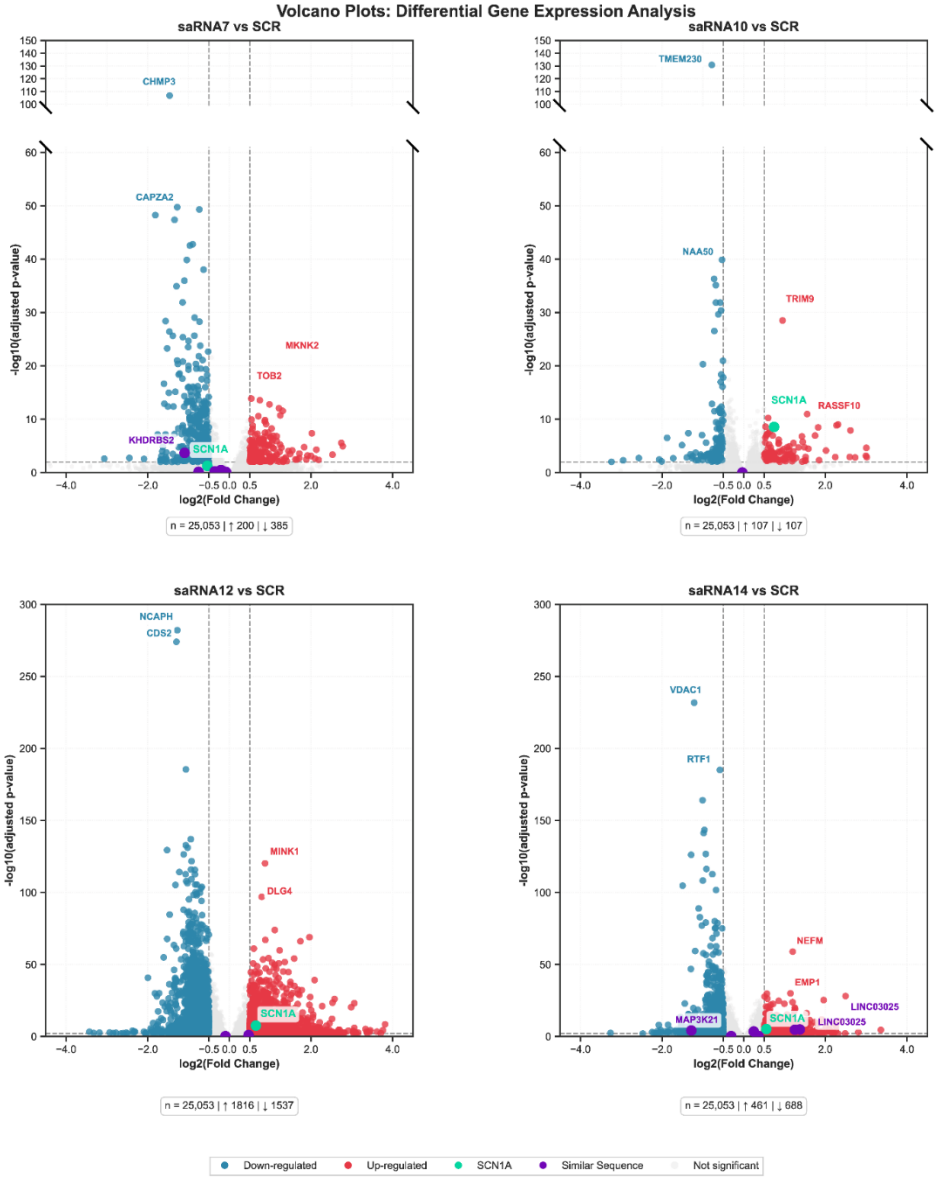
Volcano Plots: Differential Gene Expression Analysis
saRNA7 vs SCR
CHMP3
CAPZA2
MKNK2
TOB2
KHDRBS2
SCN1A
-log10(adjusted p-value)
log2(Fold Change)
n = 25,053 | ↑ 200 | ↓ 385
saRNA10 vs SCR
TMEM230
NAA50
TRIM9
SCN1A
RASSF10
n = 25,053 | ↑ 107 | ↓ 107
saRNA12 vs SCR
NCAPH
CDS2
MINK1
DLG4
SCN1A
n = 25,053 | ↑ 1816 | ↓ 1537
saRNA14 vs SCR
VDAC1
RTF1
NEFM
EMP1
LINC03025
MAP3K21
SCN1A
LINC03025
n = 25,053 | ↑ 461 | ↓ 688
Down-regulated
Up-regulated
SCN1A
Similar Sequence
Not significant
Significance thresholds: adjusted p-value < 0.01, |log2(FC)| > 0.5
Y-axis breaks (60-100) applied to saRNA7 and saRNA10. Genes in break region shown at edges with triangular markers.

Figure 6: Differential gene expression analysis of saRNA-treated AF22 NES cells reveals SCN1A upregulation. *Volcano plots showing differential gene expression between saRNA-treated conditions (saRNA7, saRNA10, saRNA12, saRNA14) and scrambled control (SCR) samples from poly-A RNA sequencing analysis. Each plot displays log2 fold change (x-axis) versus -log10 adjusted p-value (y-axis) for all analyzed genes. To accommodate the wide range of significance values, saRNA7 and saRNA10 plots feature y-axis breaks between 60 and 100, with the y-axis extending to 150, while saRNA12 and saRNA14 plots use a continuous scale extending to 300. Genes falling within the break region are represented at the axis edges with triangular markers. Red dots represent significantly upregulated genes (padj < 0.01, log2FC > 0.5), blue dots represent significantly downregulated genes (padj < 0.01, log2FC < -0.5), and gray dots represent genes not meeting both thresholds. SCN1A is highlighted in teal green and shows significant upregulation in saRNA10, saRNA12, and saRNA14 conditions but not in saRNA7 treatment. Purple dots indicate genes with sequence similarity to the target region, representing potential off-target candidates. Vertical dashed lines indicate the log2FC threshold (±0.5), while the horizontal dashed line represents the adjusted p-value threshold (0.01). The top two most significantly up- and down-regulated genes are labeled for each condition, with labels positioned above and to the right for upregulated genes and above and to the left for downregulated genes. Gene counts below each plot indicate the total number of analyzed genes and the numbers of significantly upregulated (↑) and downregulated (↓) genes, demonstrating condition-specific transcriptional responses.*

To understand the biological significance of saRNA-induced transcriptional changes, we performed Gene Ontology enrichment analysis focusing on saRNA10 treatment due to its optimal balance of efficacy and specificity (Figure 7a, b). Upregulated genes were significantly enriched in neuronal and synaptic cellular components, including plasma membrane protein complexes, neuronal cell bodies, and voltage-gated potassium channel complexes, suggesting that saRNA10-induced *SCN1A* activation may be accompanied by coordinated upregulation of genes involved in neuronal excitability. Conversely, downregulated genes showed enrichment primarily in extracellular matrix components, suggesting coordinated modulation of cellular architecture alongside ion channel expression.

Molecular function analysis further confirmed this pattern, with voltage-gated monoatomic ion channel activity representing the most significantly enriched functional category among upregulated genes, and network analysis revealing coordinated activation of functionally related ion channel genes including *SCN1A*, *CACNA1E*, and *KCNJ16* (Supplementary Figure 4).

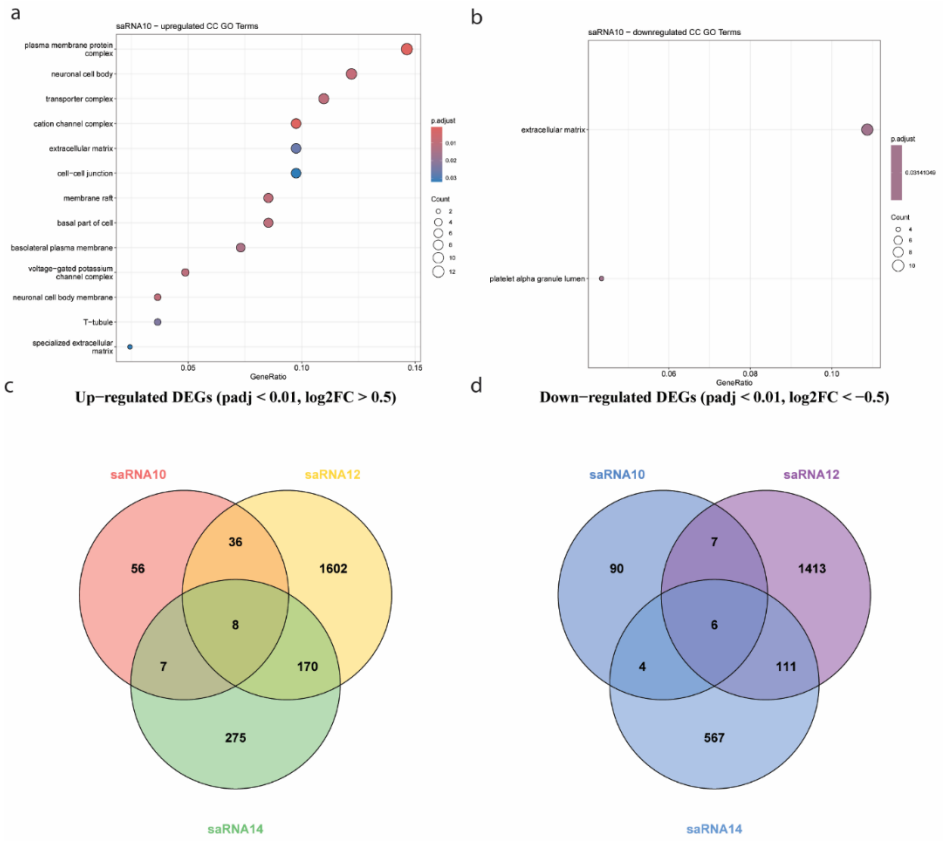

Figure 7: Gene Ontology enrichment analysis and overlap of differentially expressed genes across saRNA treatments. *(a,b) Dot plots showing significantly enriched Gene Ontology Cellular Component (CC) terms for upregulated (a) and downregulated (b) genes following saRNA10 treatment compared to scrambled control. Dot size represents gene count, while color intensity indicates adjusted p-value significance. Upregulated genes are enriched in neuronal and synaptic components including plasma membrane protein complex, neuronal cell body, and voltage-gated potassium channel complex. Downregulated genes show enrichment primarily in extracellular matrix components. (c,d) Venn diagrams illustrating the overlap of significantly differentially expressed genes (padj < 0.01, |log2FC| > 0.5) between saRNA10, saRNA12, and saRNA14 treatments. (c) Upregulated genes show substantial overlap between treatments, with 8 genes commonly upregulated across all three conditions and 170 genes shared between saRNA12 and saRNA14. saRNA12 shows the largest number of unique upregulated genes (1602). (d) Downregulated genes demonstrate similar patterns of overlap, with 6 genes commonly downregulated across all treatments and varying degrees of pairwise overlap. The analysis reveals both treatment-specific and shared transcriptional responses, with saRNA12 producing the most extensive changes in gene expression in both directions.*

Discussion

The primary aim of this study was to elucidate the molecular mechanism underlying saRNA-mediated activation of *SCN1A*, with particular focus on the role of AGO2 and potential involvement of antisense transcripts. We employed a multi-faceted approach combining functional knockdown experiments, chromatin immunoprecipitation, and comprehensive transcriptomic analysis to investigate how *SCN1A*-10 upregulates *SCN1A* expression. Our analysis of multiple saRNAs targeting the *SCN1A* promoters enabled us to assess both mechanism and specificity, addressing critical questions about the therapeutic potential of RNA activation for treating *SCN1A* haploinsufficiency disorders such as Dravet syndrome.

AGO2 Association with saRNA-Mediated *SCN1A* Activation

The involvement of AGO2 has only been demonstrated for the P21 saRNA [1]. Here we are also able to show dependence on AGO2 for the action of our *SCN1A* saRNA. The siRNA-mediated knockdown experiments revealed a clear association between AGO2 levels and saRNA efficacy, where reduction of AGO2 protein correlated with diminished *SCN1A* and *CDKN1A* (P21) upregulation by their respective saRNAs. While this correlation suggests AGO2 involvement, we acknowledge that knockdown experiments alone cannot definitively establish direct mechanistic dependence. The parallel reduction in activity for both *SCN1A*-10 and the well-characterized P21 saRNA upon AGO2 depletion suggests these saRNAs may share common cellular machinery, likely the conserved RNAa pathway described by Portnoy et al. (2016) [1], though alternative interpretations remain possible.

The inability to detect AGO2 enrichment at the *SCN1A* promoter by ChIP-qPCR, despite four independent attempts using two different antibodies, contrast our functional observations. This lack of detectable AGO2 occupancy was consistent across all tested conditions, including experiments with our positive control saRNA previously demonstrated to recruit AGO2 in Portnoy et al. (2016) in PC3 (human prostate cancer cell line) cells. This apparent discrepancy between functional association and physical detection has been reported in multiple RNAa studies and may reflect several possibilities. First, AGO2 might not directly bind promoter DNA but instead associate with promoters through RNA intermediates such as nascent transcripts or antisense RNAs [1], [2]. Standard ChIP protocols may fail to preserve these protein-RNA-DNA complexes due to RNA degradation during processing or insufficient crosslinking efficiency for RNA-protein interactions. Second, the interaction between AGO2 and target promoters during RNAa might be highly transient, occurring only briefly during specific phases of transcriptional activation, thus falling below the detection threshold of ChIP-qPCR. Third, the RITA complex components, including AGO2, RNA helicase A, and CTR9, are thought to associate dynamically with chromatin through mechanisms that differ fundamentally from the stable binding patterns of classical transcription factors, potentially requiring alternative detection methodologies beyond conventional ChIP approaches [1].

Several technical limitations may have contributed to our inability to detect AGO2 by ChIP-qPCR. The timing of our analysis at 48 hours post-transfection represents a single snapshot that may not capture the optimal window for AGO2 detection. Previous studies have shown variable kinetics of AGO2 recruitment, with some reporting maximal occupancy within 24-36 hours [1]. Additionally, our positive control using H3K4me3 antibody confirmed successful chromatin immunoprecipitation, suggesting the negative AGO2 result was not due to general technical failure. The success of Portnoy et al. using the same AGO2 antibody indicates that detection is possible under certain conditions, though differences in cell type (HeLa cells versus HEK293T), chromatin state, or the specific promoter context could account for our different results.

Potential Mechanisms and the Question of Long Non-Coding RNA Involvement

The absence of detectable *SCN1A-AS1* expression in our ribosomal RNA-depleted RNA sequencing data, despite successful detection of other lncRNAs, raises important questions about potential mechanisms of *SCN1A*-10 action. Several interpretations merit consideration: (1) *SCN1A-AS1* expression may be cell-type specific or developmentally regulated, potentially absent in our neuronal progenitor model system, (2) the transcript might be expressed at levels below our detection threshold or with unusual processing that escaped our sequencing approach, (3) *SCN1A*-10 might function through alternative mechanisms not involving *SCN1A-AS1*, or (4) technical limitations in our RNA-seq library preparation might have missed this specific transcript.

The work of Hsiao et al. (2016), demonstrating *SCN1A* upregulation through antisense oligonucleotide targeting of *SCN1A-AS1*, establishes that antisense transcripts can regulate *SCN1A* expression. However, it is important to note that AntagoNATs operate through a fundamentally different mechanism, depleting inhibitory lncRNAs via RNase H-mediated degradation, whereas saRNAs are proposed to recruit transcriptional activation machinery. The possibility remains that *SCN1A*-10 could target an as-yet-uncharacterized promoter-associated transcript, as genome-wide studies have revealed extensive divergent transcription at human promoters [3], [4].

Current models of RNAa suggest multiple potential mechanisms that are not mutually exclusive. Direct promoter targeting through RNA-DNA interactions has been demonstrated for some saRNAs, with seed region complementarity determining specificity [5]. Alternatively, saRNAs might bind to nascent transcripts or promoter-associated RNAs, creating scaffolds for transcriptional machinery recruitment. Our data cannot distinguish between these possibilities, and it remains possible that *SCN1A*-10 functions through a mechanism distinct from those previously characterized.

Transcriptome Analysis Reveals Specific and Coordinated Gene Expression Changes

In the combined RNA-seq analysis, saRNA10 showed fewer differentially expressed genes than saRNA12: 214 versus 3,353 (padj < 0.01, |log2FC| > 0.5). This marked difference in transcriptome-wide effects guided our selection of saRNA10 as the lead candidate and suggests that saRNA design significantly impacts selectivity. The variable off-target profiles among saRNAs targeting the same promoter likely reflect differences in seed sequences and their complementarity to unintended transcripts throughout the genome.

The identification of eight genes commonly upregulated across treatments that successfully activated *SCN1A* (saRNA10, 12, and 14) may provide insights into coordinated transcriptional responses, though we cannot exclude that these represent secondary effects rather than direct saRNA targets. *CACNA1E*, encoding the R-type calcium channel CaV2.3, showed consistent upregulation and is particularly noteworthy given that mutations in this gene cause developmental and epileptic encephalopathy [6]. Whether the coordinate upregulation of *CACNA1E* and *SCN1A* reflects shared regulatory mechanisms, secondary compensatory responses, or coincidental off-target effects requires further investigation.

The upregulation of *VGF*, *CLDN11*, and *POLR3G* across effective saRNAs might indicate activation of broader cellular programs. VGF functions as a neuronal activity-regulated neuropeptide precursor, *CLDN11* maintains myelin integrity, and *POLR3G* participates in RNA polymerase III transcription. These changes could represent adaptive responses to altered neuronal excitability, compensatory mechanisms, or unrelated off-target effects. The enrichment of extracellular matrix components among downregulated genes further suggests complex cellular responses to saRNA treatment that extend beyond simple target gene activation.

Methodological Considerations for Studying RNA-Mediated Transcriptional Regulation

Our experience highlights important technical challenges in studying RNA-mediated transcriptional activation. The inability of standard ChIP-qPCR to detect enrichment at the saRNA binding site, despite functional experiments with siRNA against AGO2 strongly supporting its involvement, underscores the need for specialized approaches. Chromatin Isolation by RNA Purification (ChIRP) offers an alternative that captures RNA-dependent interactions without relying on protein antibodies [7]. The use of biotinylated saRNAs, as demonstrated by Portnoy [1], could enable direct tracking of saRNA-containing complexes through affinity purification approaches.

Recent technological advances including GRID-seq and RADICL-seq provide genome-wide mapping of RNA-chromatin interactions and might reveal both direct saRNA binding sites and indirect effects [8], [9]. These unbiased approaches could potentially identify the primary molecular targets of saRNAs without requiring prior knowledge of specific transcripts.

Implementation of such techniques might resolve current uncertainties about whether *SCN1A*-10 functions through direct promoter interaction, antisense transcript modulation, or alternative mechanisms.

Implications and Future Directions

Our findings suggest that *SCN1A* can be transcriptionally activated by saRNAs in an AGO2-associated manner, though the precise molecular mechanism requires further elucidation. The correlation between AGO2 knockdown and loss of saRNA activity provides strong indication of AGO2 involvement, consistent with the RNAa literature, but does not establish whether this requirement is direct or indirect. The superior specificity profile of saRNA10 demonstrates that careful design can minimize off-target effects, an essential consideration for therapeutic development.

Future studies should employ modified detection methods to clarify the molecular interactions underlying *SCN1A* activation. The absence of detectable *SCN1A-AS1* in our system, combined with the inability to detect AGO2 at the promoter, suggests our understanding of the *SCN1A* regulatory landscape remains incomplete. Systematic investigation using RNA-centric chromatin capture methods, and time-course analyses approaches may be necessary to resolve these mechanistic questions.

The coordinate changes in ion channel and synaptic gene expression observed with *SCN1A* activation warrant careful evaluation in disease-relevant models. While some of these changes might be beneficial, others could represent unintended consequences requiring optimization of saRNA design or delivery methods. Understanding the full spectrum of transcriptional changes induced by saRNA treatment will be crucial for predicting therapeutic outcomes and potential side effects.

In conclusion, our study provides evidence for an association between AGO2 and saRNA-mediated *SCN1A* activation, though the precise mechanism remains to be fully elucidated. The transcriptional consequences extend beyond *SCN1A* itself, suggesting complex cellular responses that may include both beneficial adaptations and off-target effects. These findings advance our understanding of saRNA technology while highlighting important technical and mechanistic questions that must be addressed to realize the therapeutic potential of transcriptional activation approaches for neurological disorders.

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Author contributions

Joep van de Vondervoort designed the study with Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman, performed and analyzed the mechanistic experiments, and drafted the chapter. Martijn Koetsier provided technical support and contributed to experimental work. Serife Uysal contributed extensively to cell culture and experimental work. Carolin Vetter contributed particularly to optimization, execution and analysis of the ChIP experiments. Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman supervised the study and contributed to interpretation and revision. All authors reviewed the final chapter.

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Appendix

Supplementary 1

Sequences saRNAs, and siRNA.

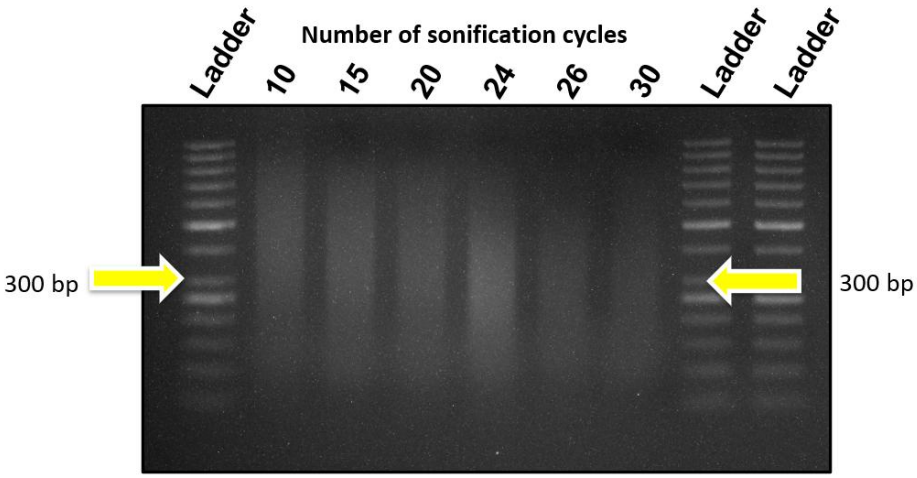
	RNA sequence strand 1 (5'-3')	RNA sequence strand 2 (5'-3')
#	*SCN1A* (Located in the promoter region described by Long (2008))	
7	CAAUCCUCUUGCUCCAUUA [dT][dT]	UAAUGGAGCAAGAGGAUUG [dT][dT]
10	CUUAACUCGCCUUCCAAAA [dT][dT]	UUUUGGAAGGCGAGUUAAG [dT][dT]
#	*SCN1A* (Located in the promoter region by Dong (2014))	
12	AAGGAAGUACAGGGUUCAA [dT][dT]	UUGAACCCUGUACUUCCUU [dT][dT]
14	UGCCUCUCCCAACUUGGAA [dT][dT]	UUCCAAGUUGGGAGAGGCA [dT][dT]
Scrambled control	ACUACUGAGUGACAGUAGA [dT][dT]	UCUACUGUCACUCAGUAGU [dT][dT]
P21 saRNA [10]	CCAACUCAUUCUCCA AGUA[dT][dT]	UACUUGGAGAAUGAGUUGG[dT][dT]
siRNA *SCN1A. Silencer Predesigned* Lot#: ASO2NFWZ	GGGUUUUUUGGAUGCACUA[dT][dT]	UAGUGCAUCCAAAAAACCC[dT][dC]
siRNA AGO2 [11]	AAGGATATGCCTTCAAGCCTCUU	GAGGCTTGAAGGCATATCCTTUU

Sequences primers

Primer name	Sequence	Application	binding
3000bfsa10_fw	GCTATTCCAGCCTGCCCATA	Chip-qPCR	Primer binding to 3000 bp before *SCN1A*-10 binding site, for use in ChIP-qPCR
3000bfsa10_rv	GCTACACTGAATAGCATGTGGC	Chip-qPCR	Primer binding to 3000 bp before *SCN1A*-10 binding site, for use in ChIP-qPCR
300bfsa10_fw	CGTTCTGAATTTGATATCCTATTACCT	Chip-qPCR	Primer binding to 300 bp before *SCN1A*-10 binding site, for use in ChIP-qPCR
300bfsa10_rv	ACTCAAAGACAGTCACCGAGG	Chip-qPCR	Primer binding to 300 bp before *SCN1A*-10 binding site, for use in ChIP-qPCR
300afsa10_fw	GCTGGAGGAAGCTGAGCT	Chip-qPCR	Primer binding to 300 bp after *SCN1A*-10 binding site, for use in ChIP-qPCR
300afsa10_rv	GCATGCTGACAGAACTGTAATCT	Chip-qPCR	Primer binding to 300 bp after *SCN1A*-10 binding site, for use in ChIP-qPCR
3000afsa10_fw	AGCTCCGCTATGAAAAGTTCC	Chip-qPCR	Primer binding to 3000 bp after *SCN1A*-10 binding site, for use in ChIP-qPCR

3000afsa10_rv	TCAAGGGTGCAAACAGGTCC	Chip-qPCR	Primer binding to 3000 bp after *SCN1A*-10 binding site, for use in ChIP-qPCR
hSCN1A-10_fw	TGCTTCTGGTCATTCTTCCCA	Chip-qPCR	Primer binding to *SCN1A*-10 binding site, for use in ChIP-qPCR
hSCN1A-10_rv	GCTGGGGAGCTCTCTGAATG	Chip-qPCR	Primer binding to *SCN1A*-10 binding site, for use in ChIP-qPCR
SCN1A-TSS_fw	AGGAATTTCATATGCAGAATAAATGGT	Chip-qPCR	Primer binding to *SCN1A* transcription start site, for use in ChIP-qPCR
SCN1A-TSS_rv	TGAAGAAGTTGAAGCTGTCAGGT	Chip-qPCR	Primer binding to *SCN1A* transcription start site, for use in ChIP-qPCR
SCN1A-intron_fw	ATGCTGATGAAGGAGATCCATGAG	Chip-qPCR	Primer binding to *SCN1A* intronic region, for use in ChIP-qPCR
SCN1A-intron_rv	TCATCTCTGACCAAGGAGAAGC	Chip-qPCR	Primer binding to *SCN1A* intronic region, for use in ChIP-qPCR
hp21-322_fw1	AATCCTTGCCTGCCAGAGTG	Chip-qPCR	Primer binding to P21-322 binding site, for use in ChIP-qPCR
hp21-322_rv1	CTGGACACATTTCCCCACGA	Chip-qPCR	Primer binding to P21-322 binding site, for use in ChIP-qPCR
H3-*GAPDH*_fw4	ACTAGGCGCTCACTGTTCTC	Chip-qPCR	Primer binding to H3 binding site in *GAPDH*, for use in ChIP-qPCR
H3-*GAPDH*_rv4	GCCCCCATACGACTGCAAA	Chip-qPCR	Primer binding to H3 binding site in *GAPDH*, for use in ChIP-qPCR
hGAPDH_fw2	CCATGTTCGTCATGGGTGTG	qPCR	Primer binding to exon-exon junction in human *GAPDH* to detect mRNA expression
hGAPDH_rv2	GGTGCTAAGCAGTTGGTGGTG	qPCR	Primer binding to exon-exon junction in human *GAPDH* to detect mRNA expression
hCDKN1A_fw	TGCCGAAGTCAGTTCCTTGT	qPCR	Primer binding to exon-exon junction in human *CDKN1A* to detect mRNA expression
hCDKN1A_rv	GTTCTGACATGGCGCCTCC	qPCR	Primer binding to exon-exon junction in human *CDKN1A* to detect mRNA expression
hSCN1A3-fw	GCTTTGATACCTTCAGTTGGGC	qPCR	Primer binding to exon-exon junction in human *SCN1A* to detect mRNA expression
hSCN1A3-rv	TGTACGTTTTCCCAGCAGCA	qPCR	Primer binding to exon-exon junction in human *SCN1A* to detect mRNA expression
qPCR-hAGO2_fw	GATATGCCTTCAAGCCTCCA	qPCR	Primer binding to exon-exon junction in human AGO2 to detect mRNA expression
qPCR-hAGO2_rv	AACTCTCCTCGGGCACTTCT	qPCR	Primer binding to exon-exon junction in human AGO2 to detect mRNA expression

Supplementary 2



Supplementary Figure 2. Optimization of sonication conditions for chromatin fragmentation in HeLa cells.

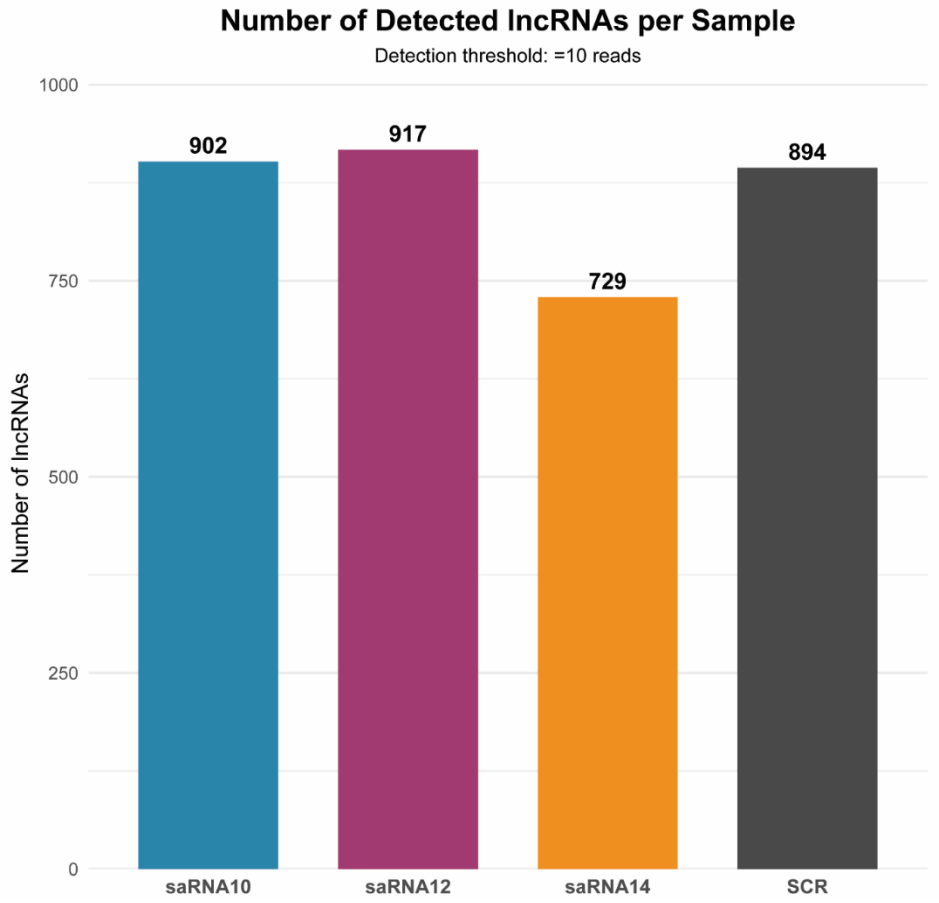
Agarose gel electrophoresis analysis of chromatin DNA fragments following different sonication cycles to determine optimal conditions for ChIP-qPCR. HeLa cells were crosslinked with formaldehyde and subjected to varying numbers of sonication cycles (10, 15, 20, 24, 26, and 30 cycles). Following sonication, chromatin was reverse-crosslinked, and DNA was purified and analyzed on a 1.5% agarose gel.

Molecular weight ladders are shown in the flanking lanes for size reference, with the 300 bp band indicated by yellow arrows. The gel demonstrates progressive fragmentation of chromatin DNA with increasing sonication cycles. At lower cycle numbers (10-15), DNA fragments remain relatively large, while higher cycle numbers (26-30) result in extensive fragmentation with smaller DNA pieces.

Based on this optimization, 24 sonication cycles were selected for subsequent ChIP experiments, as this condition produced the desired chromatin fragmentation with DNA fragments predominantly in the 200-500 bp range, centered around 300 bp. This fragment size range is optimal for ChIP-qPCR analysis as it provides sufficient resolution for mapping protein-DNA interactions while maintaining adequate DNA yield for downstream qPCR amplification.

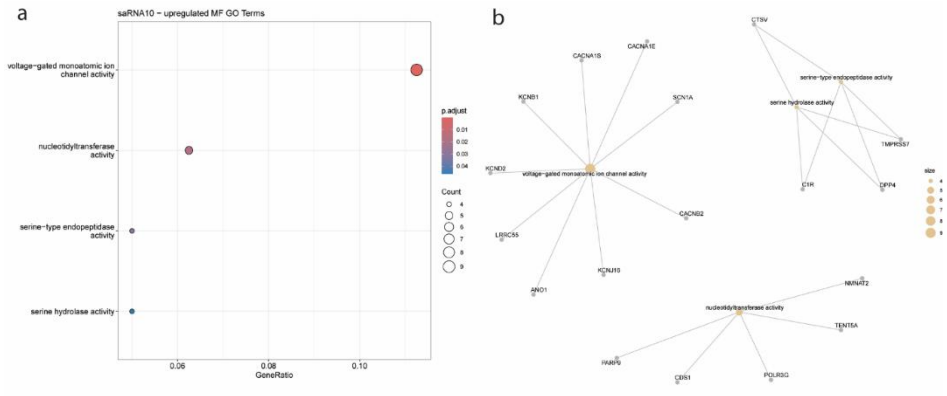
Proper chromatin fragmentation is critical for ChIP success, as it ensures that immunoprecipitated DNA fragments represent specific protein binding sites rather than large, non-specific chromatin regions. The selected sonication conditions were used consistently across all ChIP experiments in this study.

Supplementary 3



Supplementary Figure 3. Number of detected lncRNAs across experimental conditions in Ribo-Zero samples. *Bar chart showing the total number of lncRNAs detected in each sample condition using a detection threshold of ≥10 reads. The saRNA12 condition yielded the highest number of detected lncRNAs (917), followed by saRNA10 (902) and SCR control (894). The saRNA14 condition showed notably fewer detected lncRNAs (729). This shows that we could detect lncRNAs in our sample. SCN1A-AS1 did not have any detectable reads in the saRNA binding position.*

Supplementary 4



Supplementary Figure 4. Gene Ontology molecular function enrichment and network analysis of saRNA10 upregulated genes. *(a) Dot plot showing significantly enriched Gene Ontology Molecular Function (MF) terms for genes upregulated following saRNA10 treatment compared to scrambled control. Dot size represents gene count, while color intensity indicates adjusted p-value significance. The most significantly enriched term is voltage-gated monoatomic ion channel activity, followed by nucleotidyltransferase activity, serine-type endopeptidase activity, and serine hydrolase activity, highlighting the functional coherence of saRNA10-induced gene expression changes. (b) Network diagram illustrating the functional relationships between genes within the enriched GO molecular function categories. Nodes represent individual genes, with node size proportional to connectivity and color indicating different functional clusters. The central hub around "voltage-gated monoatomic ion channel activity" demonstrates the coordinated upregulation of ion channel-related genes, including SCN1A, CACNA1E, KCNJ16, and other channel subunits. Additional functional clusters include serine-type peptidase activity and nucleotidyltransferase activity, revealing the broader molecular pathways activated by saRNA10 treatment. The network topology suggests that saRNA10 triggers coordinated activation of functionally related gene sets rather than random transcriptional changes.*

Supplementary 5

ChIP Buffer Recipes

Lysis Buffer A

Component	Final Conc.
HEPES, pH 8.0	50 mM
NaCl	140 mM
EDTA	1 mM
Glycerol	10%
NP-40	0.5%
Triton X-100	0.25%

Lysis Buffer B

Component	Final Conc.
Tris, pH 8.0	10 mM
EDTA	1 mM
EGTA	0.5 mM
NaCl	200 mM

Lysis Buffer C

Component	Final Conc.
Tris, pH 8.0	50 mM
EDTA	5 mM
SDS	1%
NaCl	100 mM

10x ChIP Buffer

Component	Final Conc.
SDS	0.1%
Triton X-100	10%
EDTA	12 mM
Tris-HCl, pH 8.0	167 mM
NaCl	1.67 M

1x TE Buffer

Component	Final Conc.
Tris-HCl, pH 8.0	10 mM
EDTA, pH 8.0	1 mM

5% DOC

Component	Final Conc.
Na-Deoxycholate	5% (w/v)

2.5M Glycine

Component	Final Conc.
Glycine	2.5 M

Heat to 65°C, filter

RIPA Buffer

Component	Final Concentration
Tris-HCl, pH 8.0	10 mM
EDTA, pH 8.0	1 mM
NaCl	140 mM
Triton X-100	1%
SDS	0.1%
DOC	0.1%

RIPA-500 Buffer

Component	Final Concentration
Tris-HCl, pH 8.0	10 mM
EDTA, pH 8.0	1 mM
NaCl	500 mM
Triton X-100	1%
SDS	0.1%
DOC	0.1%

LiCl Wash Buffer

Component	Final Concentration
Tris-HCl, pH 8.0	10 mM
EDTA, pH 8.0	1 mM
LiCl	250 mM
NP-40	0.5%
DOC	0.5%

Elution Buffer

Component	Final Concentration
Tris-HCl, pH 8.0	10 mM
EDTA, pH 8.0	5 mM
NaCl	150 mM
SDS	1%

Binding/Blocking Buffer

Component	Final Concentration
BSA	0.5%
Tween	0.5%
PBS	1x

Chapter 4

Extending saRNA activation to additional epilepsy genes

Chapter 4

Evaluating saRNA-Mediated Promoter Activation Beyond *SCN1A*: Challenges and Insights

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Abstract

This chapter extends promoter-targeted RNA activation (RNAa) beyond *SCN1A* by evaluating small activating RNAs (saRNAs) designed for six additional genes implicated in haploinsufficiency-driven epilepsies: *DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, *TSC2* (core repressors of mTORC1 signaling), and the chromatin remodeler *CHD2*. Because these disorders arise from insufficient dosage of a single functional allele, transcriptional upregulation represents a compelling therapeutic strategy. Using the established design framework of Wang et al., we generated ten promoter-targeting saRNAs per gene and systematically screened them in HeLa cells, known to be permissive for RNAa, and in AF22 neural stem cells, a more physiologically relevant neuronal model.

Across all loci, only a minority of saRNAs produced reproducible activation in HeLa cells, with modest but structurally consistent increases observed for subsets of *CHD2*, *DEPDC5*, *NPRL3*, and *TSC2* candidates. *NPRL2* and *TSC1* yielded no reliable activators. In striking contrast, none of the saRNAs produced robust activation in AF22 cells, and several instead induced mild repression. Even the canonical positive control, *P21*, showed dampened and variable activation in AF22, highlighting substantial cell-type differences in RNAa responsiveness. The reduced permissiveness of neural progenitors likely reflects more restrictive chromatin states and lower availability of RNAa cofactors such as AGO2, underscoring that promoter accessibility and cellular context are major determinants of saRNA function.

To understand why only a few structurally compliant saRNAs were active, we compiled a curated dataset of 14 validated functional sequences and 111 nonfunctional designs and performed exploratory sequence analyses. Principal component analysis showed no global compositional pattern distinguishing the two groups, indicating that saRNA efficacy depends on fine-grained positional features rather than overall sequence content. Position-specific motif profiling revealed strong enrichment of distinct nucleotides at a small set of "hotspot" positions, most prominently G at positions 10 and 16, U at position 15, and C depletion at position 9. Dinucleotide and trinucleotide analyses reinforced these findings, identifying specific motifs such as UG (pos 15), AC (pos 9), and AAA (pos 17) that were characteristic of functional sequences. A complementary stability-based analysis revealed enrichment of strong (G/C) bases in the central region of the guide and weak (A/U) bases near the 3′ end, suggesting that a local stability gradient may contribute to promoter hybridization, strand selection, or AGO2 loading.

Together, these results illustrate both the promise and the challenges of broadening RNAa to additional haploinsufficient epilepsy genes. Functional saRNAs can be identified, but success depends not only on sequence design rules but also on the chromatin environment of the target locus and the cellular context in which activation is attempted. The positional features identified here provide a foundation for refining saRNA design algorithms. Future integration of these sequence signatures with chromatin accessibility maps and neuron-specific promoter architecture will be essential for realizing predictable, locus-specific gene activation suitable for therapeutic development.

Introduction

Pathogenic variants in genes upstream of the mTOR (mechanistic target of rapamycin) signaling pathway are an important cause of developmental and epileptic encephalopathies (DEEs) and cortical malformations. Haploinsufficiency of *TSC1* or *TSC2* leads to tuberous sclerosis complex, a disorder marked by cortical tubers, intellectual disability, and often drug-resistant epilepsy [1]. Likewise, loss-of-function mutations in *DEPDC5*, *NPRL2*, and *NPRL3*, which encode the GATOR1 complex, a negative regulator of mTORC1 signaling, are among the most common genetic causes of focal epilepsies and cortical malformations [2]. In each of these disorders, insufficient expression of a single functional allele drives disease through mTOR hyperactivation and abnormal cellular signaling. Upregulating the remaining allele represents a rational therapeutic strategy that could help restore more physiological signaling and potentially reduce seizure burden [3], [4].

The *CHD2* gene is another example of the many recently characterized monogenic causes of DEE. *CHD2* encodes a chromodomain helicase DNA-binding protein involved in chromatin remodeling and neuronal differentiation. Haploinsufficiency of *CHD2* is linked to myoclonic epilepsies and neurodevelopmental disorders [5], [6], making it another important candidate for upregulation strategies. Unlike the GATOR1/TSC genes, *CHD2* does not act through mTOR signaling, allowing us to test whether promoter-targeted upregulation can be extended beyond a single pathway.

In previous chapters (2 and 3) of this thesis, we established and characterized small activating RNAs (saRNAs) targeting the *SCN1A* promoter, demonstrating robust and reproducible upregulation in multiple cell models. This work provided a proof-of-principle for saRNA-mediated promoter activation in genes implicated in haploinsufficiency-driven epilepsies. Here, our goal was twofold; first, to expand saRNA development to additional genes relevant to monogenic haploinsufficient epilepsy and second, to explore whether common sequence features could explain why some saRNAs function as expected, while others do not.

To this end, we designed ten candidate saRNAs per gene targeting *DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, *TSC2*, and *CHD2* according to the sequence guidelines of Wang et al. (2015) [7]. These candidates were screened in two cellular contexts: HeLa cells, which are known to support robust RNAa responses, and AF22 human neural stem cells, a disease-relevant neuronal model. This two-system approach allows us to evaluate both sequence-dependent effects and cell-type-dependent variability.

Because only a minority of saRNAs are typically functional, understanding what distinguishes successful activators is crucial. We therefore systematically analyzed positional nucleotide composition, short motif patterns (1–3 mers), and base-pairing stability features (G/C vs A/U) in functional versus non-functional saRNAs. This exploratory motif analysis aims to identify recurrent sequence characteristics that might inform improved design strategies in future work.

In summary, this chapter describes the development and screening of saRNAs for six epilepsy-associated genes and the identification of positional sequence and stability

preferences linked to saRNA activity. By contrasting working and non-working sequences, we aim to better understand the design features that contribute to effective promoter-targeted activation, with the ultimate goal of improving the reliability of saRNA-based strategies for haploinsufficiency-driven epilepsies.

Material and methods

Cell Culture – AF22 and HeLa Cells

AF22 neuroepithelial stem (NES) cells, derived from induced pluripotent stem cells (iPSCs) [8], were cultured at 37°C and 5% CO_2 in NES medium, prepared freshly every seven days. The NES medium consisted of DMEM/F12 (Thermo Fisher Scientific) supplemented with 0.02% L-alanyl-L-glutamine (Sigma-Aldrich), 1% N-2 supplement (Thermo Fisher Scientific), 0.1% B27 supplement (Thermo Fisher Scientific), 1% penicillin-streptomycin (P/S, Thermo Fisher Scientific), 20 ng/mL basic fibroblast growth factor (bFGF, Peprotech), and 20 ng/mL epidermal growth factor (EGF, Peprotech). Culture flasks were pre-coated with 0.1% gelatin (Sigma-Aldrich) in PBS at 37°C for 30 minutes. Cells were passaged every 3-4 days upon reaching 80-90% confluency using TrypLE Express (Thermo Fisher Scientific) and reseeded at a density of 1×10^5 cells/cm².HeLa cells were cultured in high-glucose DMEM supplemented with 10% fetal bovine serum (FBS) and 1% P/S. Cells were passaged upon reaching confluency using Trypsin-EDTA (0.05%, Thermo Fisher Scientific).

saRNA Design

Ten small activating RNAs (saRNAs) were designed to target genes implicated in epileptogenesis, including repressors of the mTOR pathway (*DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, and *TSC2*) as well as the chromatin remodeler *CHD2*. saRNAs were designed according to previously established guidelines (Wang, 2015) [7]. For each gene, target sites were selected within promoter regions defined as the genomic interval spanning −2000 to +500 bp relative to the transcription start site (TSS) of the MANE Select transcript. This region was chosen to include both proximal promoter elements and potential regulatory sequences involved in transcriptional activation. Candidate saRNA sequences were evaluated for GC content, absence of repetitive elements, and minimal off-target complementarity using NCBI BLAST against the human genome.

A scrambled control RNA lacking significant complementarity to any human transcript (verified by BLAST analysis) was used as a negative control, and the P21 (*CDKN1A*) saRNA originally described by Li et al. (2006) [9] served as a positive control to confirm successful saRNA transfection and activation. Complete saRNA sequences are listed in Supplementary Table S1 and S2.

Transfections, RNA Isolation and Quantitative PCR (qPCR)

AF22 or HeLa cells were seeded at ~70% confluency in 12-well plates. Cells were transfected the following day with 50 nM saRNA using Lipofectamine RNAiMAX (Invitrogen) according to the manufacturer's protocol. After transfection, cells were incubated at 37°C for 72 hours and then harvested for RNA isolation.

Cells were washed with dPBS and lysed using the Nucleospin RNA Mini Kit (Macherey-Nagel). RNA was eluted in 25 µL RNase-free water and quantified using a NanoDrop spectrophotometer. cDNA was synthesized from 400 ng RNA using the iScript cDNA synthesis kit (Bio-Rad). qPCR reactions were prepared with SYBR Green master mix (Bio-Rad) and gene-specific primers (Table S3). qPCR was performed on a Bio-Rad CFX384 Real-Time PCR System. *GAPDH* and *GUSB* were used as housekeeping genes for normalization. Data were analyzed using the ΔΔCt method.

Bioinformatic Analysis

Compilation of Functional and Control saRNA Datasets

A comprehensive dataset of functional saRNA sequences was compiled for this study. This included the positive saRNAs identified and validated in the current experiments, as well as our *SCN1A*-saRNA and six additional positive controls derived from the literature, resulting in a total of 14 positive control sequences in the dataset.

For benchmarking, these positive controls were contrasted with 111 negative control sequences. Negative controls were thoroughly tested but never exhibited structural upregulation in either HeLa or AF22 cell lines. Not all data are presented for every target gene; the full list of sequences and corresponding data can be found in the Supplementary Material.

Statistical Analysis

Statistical analyses were performed using GraphPad Prism 9.0 and Python (scipy.stats). Normality of data distribution was assessed using the Shapiro-Wilk test. For normally distributed data, differences between groups were evaluated using one-way ANOVA followed by Dunnett's post-hoc test for multiple comparisons against control. Non-normally distributed data were analyzed using the Kruskal-Wallis test with Dunn's post-hoc correction. Pairwise comparisons used unpaired t-tests (parametric) or Mann-Whitney U tests (non-parametric) as appropriate.

Results are presented as mean ± standard error of the mean (SD) unless otherwise indicated. Statistical significance was defined as: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. All computational analysis scripts and processed datasets are available at: JoepVondervoort/qPCR-data-analysis.

Results

Design of Small Activating RNAs

Small activating RNAs (saRNAs) were designed to upregulate key repressors of mTOR signaling, *DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, and *TSC2*, and, in parallel, the epilepsy-associated chromatin remodeler *CHD2* (non-mTOR), to assess the generalizability of promoter-targeted RNA activation. Following the design principles of Wang et al., ten saRNAs per gene were synthesized by Sigma and positioned across proximal and distal promoter regions (Supplementary 1 and 2). Sequences were constrained to a GC content of 40–60%, no more than five identical nucleotides in a row, and a maximum of one CpG dinucleotide per target sequence, to balance stability, specificity, and synthesis robustness. This standardized framework yielded 10 saRNAs per target gene for comparative screening across cell models.

Screening of saRNAs in HeLa

In HeLa cells, a subset of saRNAs produced structurally consistent up-regulation at or above a fold change of 1.5 (relative to SCR), whereas most were neutral or variable. Clear, repeatable, yet nonsignificant, activations were observed for *CHD2* (saRNA8, mean fold change of 1.98 ± 0.13, $p = 0.333$; saRNA10, mean fold change of 1.78 ± 0.18, $p = 0.333$), *DEPDC5* (saRNA4, mean fold change of 1.55 ± 0.06, $p = 0.667$; saRNA9, mean fold change of 1.54 ± 0.50, $p = 0.667$), *NPRL3* (saRNA5, mean fold change of 1.83 ± 0.32, $p = 0.333$; saRNA6, mean fold change of 2.29 ± 1.06, $p = 0.333$), and *TSC2* (saRNA8, mean fold change of 1.93 ± 0.35, $p = 0.0173$). By contrast, *NPRL2* showed no saRNA with a mean at or above a fold change of 1.5, and *TSC1* exhibited high between-replicate variance without a stable activator at or above a fold change of 1.5 (*e.g.*, saRNA2/saRNA3/saRNA8 displayed elevated means but large dispersion; all $p \geq 0.200$). Additional near-threshold trends (between a fold change of 1.3 and 1.5 with limited spread) were noted for *CHD2* (saRNA6, mean fold change of 1.32 ± 0.14; saRNA7, mean fold change of 1.35 ± 0.02; saRNA9, mean fold change of 1.33 ± 0.04; each $p = 0.333$), *DEPDC5* (saRNA1, mean fold change of 1.38 ± 0.14; saRNA2, mean fold change of 1.39 ± 0.37; saRNA5, mean fold change of 1.46 ± 0.11; saRNA10, mean fold change of 1.32 ± 0.07; each $p = 0.667$), and *TSC2* (saRNA10, mean fold change of 1.50 ± 0.40, $p = 0.400$). *P*-values are unadjusted two-sided tests versus the per-gene SCR distribution; given limited replication for some panels and multiple testing, these comparisons are best interpreted as screening evidence rather than confirmatory statistics.

Mean baseline Ct values for all genes ranged around 25, indicating robust endogenous transcription well above the detection threshold and supporting the feasibility of promoter activation in HeLa cells. None of the genes showed Ct values exceeding 35, except *TSC1*, which displayed greater replicate variability and occasional late amplification cycles,

consistent with its lower baseline expression and contributing to the higher dispersion observed in its saRNA panel.

The plate-wide positive control P21 verified assay responsiveness and, in several plates, exceeded the y-axis range of the plotted summary: on the *CHD2* plate, P21 showed a median fold change of 1.37 (n = 5); on *DEPDC5*, a fold change of 2.24 (n = 2); on *NPRL2*, fold changes of 5.88 and 5.78 across two runs (n = 4 per run); on *NPRL3*, a fold change of 4.63 (n = 4); on *TSC1*, a fold change of 1.54 with a range up to a fold change of 8.86 (n = 3); and on *TSC2*, fold changes of 14.62 in both runs with maxima of 23.09 (n = 5). These controls confirm successful saRNA transfection and sufficient dynamic range to detect multi-fold activation, even where the figure truncates the upper tail.

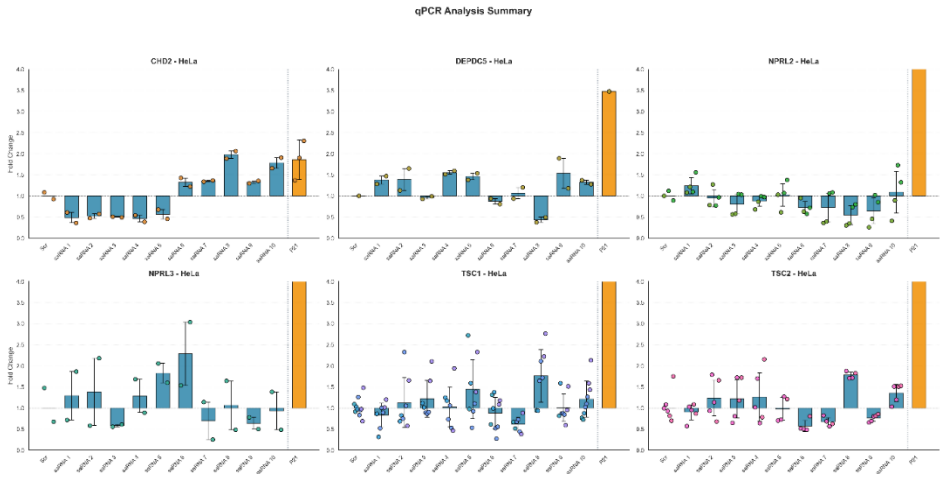

Figure 1. HeLa qPCR screen of saRNAs for *CHD2*, *DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, and *TSC2*. *Bars show fold change relative to scrambled (SCR) for each saRNA; dots represent biological replicates. Color differences indicate independent experiments. Prior to calculating fold change, all samples were normalized to housekeeping genes; P21 (positive control) is normalized to its own SCR on the same plate. Bars depict mean ± SD. In some panels, P21 exceeds the plotted y-range.*

Screening of saRNAs in AF22 NES

After the HeLa screen, we tested the same saRNAs in the more disease-relevant AF22 NES model. Across *CHD2*, *DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, and *TSC2*, responses were predominantly neutral, clustering around a fold change of ~0.8–1.2 relative to scrambled. A borderline increase was observed for *TSC1* (saRNA4, mean fold change of 1.55 ± 0.90, p = 0.686), but the replicate dispersion was substantial and therefore not comparable with the clearer down-regulatory behavior seen for *CHD2*. Consistent repression was detected for *CHD2* (saRNA4, fold change of 0.75 ± 0.13, p = 0.024; saRNA9, 0.80 ± 0.13, p = 0.035), *TSC1* (saRNA7, 0.68 ± 0.09, p = 0.028; saRNA9, 0.75 ± 0.07, p = 0.029), and *TSC2* (saRNA5, 0.81 ± 0.07,

p = 0.029). No saRNA for *DEPDC5* or *NPRL2* exceeded a mean fold change of 1.2. Taken together, AF22 cells did not exhibit structural saRNA-mediated activation for the tested loci.

The positive control P21 was variable in AF22 and generally showed lower upregulation than in HeLa (*e.g.*, *CHD2* plate, fold change of 2.08; *DEPDC5*, 1.16; *NPRL2*, 1.83 and 2.08 across two runs; *NPRL3*, median 2.80 with a range up to 3.65; *TSC1*, 1.04; *TSC2*, 1.14), indicating a reduced dynamic range in this cell model.

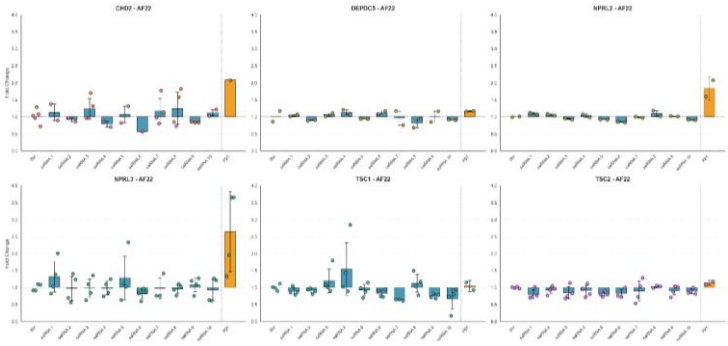

Figure 2. AF22 qPCR screen of promoter-targeting saRNAs for CHD2, DEPDC5, NPRL2, NPRL3, TSC1, and TSC2*. Bars show fold change relative to scrambled (SCR) for each saRNA; dots represent biological replicates. Color differences denote independent experiments. Prior to fold-change calculation, all samples were normalized to housekeeping genes. The positive control P21 is normalized to its own specific SCR. Bars depict mean ± SD. The dashed vertical line separates test saRNAs from the P21 control.*

Identifying Sequence Determinants of saRNA Functionality

While structural design rules for saRNAs have been established by Wang et al. (2015), these guidelines have proven insufficient for reliable prediction of functional sequences. In practice, the majority of saRNA designs that satisfy Wang's structural criteria fail to activate target genes, highlighting the need for additional sequence-level predictive features. Inspired by Wang's focus on sequence composition and positional effects, we hypothesized that specific nucleotide patterns and positional preferences, beyond basic structural requirements, distinguish effective saRNAs from inactive sequences. To identify these predictive features, we analyzed a dataset of 125 nineteen-nucleotide sequences comprising 14 experimentally validated functional saRNAs (positive controls) and 111 non-functional test sequences. All sequences were designed according to Wang's structural framework, yet only the positive controls demonstrated robust activity. The positive control sequences were derived from published literature and our own experimental studies in HeLa cells (including *SCN1A* saRNAs described in Chapter 2), selected based on stringent functional criteria: ≥1.5-fold gene activation and high reproducibility. The test sequences represent structurally compliant

designs that failed to show activity despite meeting Wang's criteria (all sequences provided in Supplementary Table S1 and S2). Our analytical strategy progressed from global sequence analysis using principal component analysis to fine-scale positional motif profiling, allowing us to identify position-specific nucleotide preferences that may serve as predictive markers for functional saRNA design.

Principal Component Analysis Reveals High-Dimensional Sequence Complexity

We first examined whether saRNA sequences cluster based on functionality using principal component analysis (PCA) on 125 sequences comprising 14 functional positive controls and 111 test sequences. The analysis revealed substantial overlap between functional and non-functional groups (Figure 3a), indicating that saRNA activity is not determined by simple, overall sequence composition. The first two principal components captured only 5.8% and 5.2% of total variance respectively, with cumulative analysis showing that approximately 70% of sequence variation required the first 20 components (Figure 3b). This gradual accumulation of explained variance across many dimensions demonstrates the high-dimensional nature of saRNA sequence space, where functionality likely depends on subtle, position-specific features rather than global sequence patterns. The lack of clear separation in reduced dimensionality space, combined with the low variance explained by initial components, justified a more granular analytical approach focused on individual nucleotide positions and local sequence motifs.

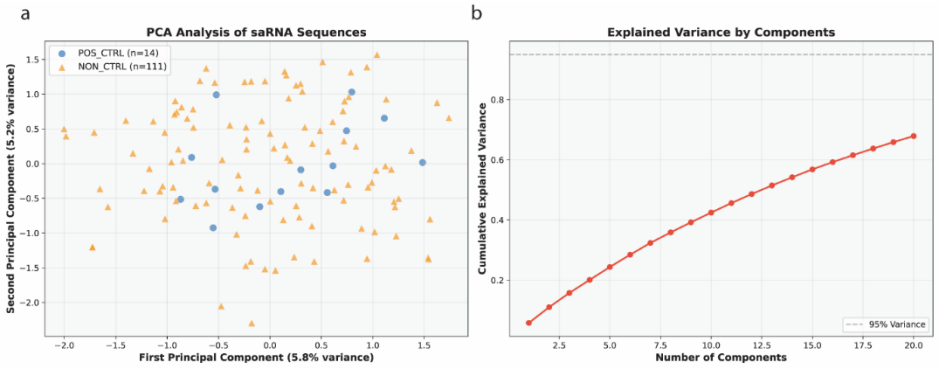

Figure 3. Principal component analysis of saRNA sequence composition. (a) *PCA scatter plot showing the distribution of 125 saRNA sequences in the space defined by the first two principal components. Blue circles represent functional positive control sequences (n=14), and orange triangles represent test sequences (n=111). The x-axis shows the first principal component (PC1), accounting for 5.8% of total variance, and the y-axis shows the second principal component (PC2), accounting for 5.2% of variance.* ***(b) Scree plot showing*** *cumulative explained variance as a function of the number of principal components. The red line shows the cumulative percentage of total variance explained by successive principal components (up to 20 components). The dashed gray line indicates the 95% variance threshold.*

Position-Specific Motif Analysis Reveals Nucleotide Preferences

Given the high-dimensional sequence variation revealed by PCA, we next explored whether specific positional sequence patterns could help distinguish functional from non-functional saRNAs. To this end, we calculated frequency differences between positive control sequences (n = 14) and non-functional test sequences (n = 111) for 1-mer, 2-mer, and 3-mer motifs along the 19-nucleotide guide strand. Because the number of functional sequences was limited, this analysis was exploratory and aimed at identifying trends rather than performing formal statistical testing.

Single-nucleotide analysis (Figure 4a) revealed clear position-specific preferences, with the largest differences occurring around positions 9–10 and 15–16. At position 10, guanosine was enriched by +24.8 pp in functional sequences, while at position 15 uracil was enriched (+23.0 pp) and guanosine depleted (−22.6 pp). Position 16 showed the opposite pattern (G +22.1 pp; U −17.2 pp), and position 9 also displayed strong effects (C −21.7 pp; G +20.3 pp). Additional smaller shifts were found at positions 7, 12, 14, and 17. Altogether, these positional differences highlight a small number of "hot spots" in the guide sequence where base identity appears to correlate with functionality.

Dinucleotide motif analysis (Figure 4b) revealed a similar pattern but with sharper contrasts at key sites. The strongest enrichments were observed for UG at position 15 (+24.1 pp), GG at position 12 (+20.5 pp), AC at position 9 (+19.6 pp), AA at position 17 (+18.6 pp), and CA at position 14 (+16.0 pp). These motifs are located in the same regions as the dominant 1-mer effects, suggesting that short local sequence context, not just single bases, may contribute to saRNA activity.

Trinucleotide motif analysis (Figure 4c) showed more diffuse patterns, as expected from increased motif complexity and lower individual frequencies. Even so, certain motifs stood out. In particular, AAA starting at position 17 was enriched by over 20 pp in functional sequences, suggesting that sequence composition at the 3' end of the guide strand may also influence functionality.

Taken together, these analyses indicate that a subset of positions, particularly around positions 9–10 and 15–17, carry disproportionately strong sequence signals associated with saRNA activity. While these are observational findings, they provide useful candidate features to guide improved saRNA design and future experimental validation.

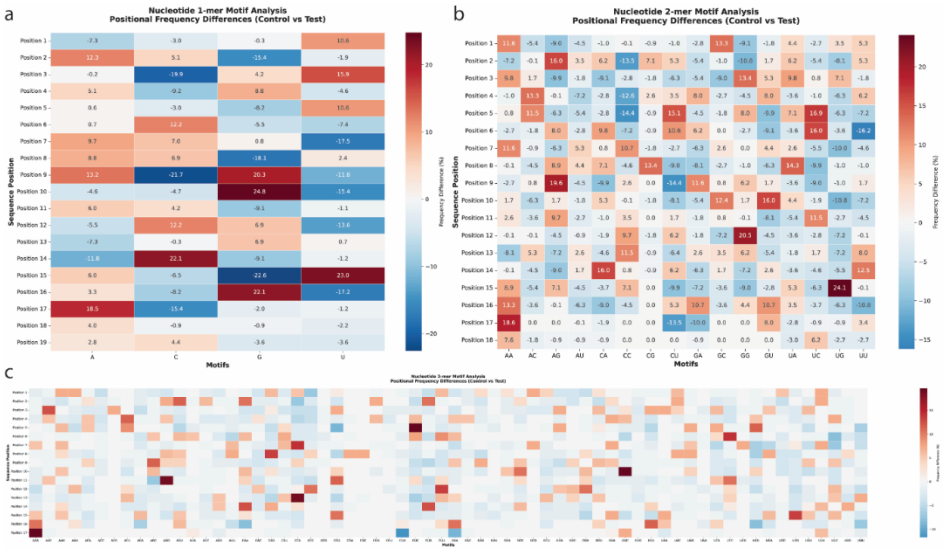

Figure 4. Positional motif frequency analysis of saRNA sequences. *Heatmaps showing the difference in motif frequencies between positive control sequences (n=14) and test sequences (n=111) at each sequence position. Values represent the percentage point difference (Control frequency - Test frequency), with red indicating higher frequency in controls and blue indicating higher frequency in test sequences.* ***(a)*** *Single nucleotide (1-mer) motif analysis across all 19 positions. Each cell shows the frequency difference for nucleotides A, C, G, and U at positions 1-19.* ***(b)*** *Dinucleotide (2-mer) motif analysis across 18 positions. Each cell shows the frequency difference for all 16 possible dinucleotide combinations (AA through UU) at positions 1-18.* ***(c)*** *Trinucleotide (3-mer) motif analysis across 17 positions. Each cell shows the frequency difference for all 64 possible trinucleotide combinations at positions 1-17.*

Stability Motif Analysis Reveals Base-Pairing Preferences

To examine sequence patterns from a structural perspective, we classified nucleotides as strong (S: G/C) or weak (W: A/U) and analyzed stability motif frequencies at each position (Figure 5). This binary classification highlights potential preferences in hydrogen-bonding strength that are independent of specific nucleotide identity and can reveal complementary patterns to the nucleotide-level motif analysis. This assessment was purely observational and descriptive; no statistical tests were applied, and the patterns shown should be interpreted as exploratory rather than inferential.

Single-position analysis (Figure 5a) revealed several prominent positional preferences. The most striking signal occurred at position 15, where W bases were enriched by +29.1 pp, with a corresponding depletion of S bases. This aligns with the U enrichment and G depletion observed in the nucleotide-level analysis (Figure 4). Additional strong effects were observed at position 10 (S +20.1 pp) and position 12 (S +19.2 pp), while position 3 showed a clear W enrichment (+15.7 pp). These results indicate that strong bases tend to be favored at positions 10 and 12, whereas weak bases dominate at position 15.

Dinucleotide stability motif analysis (Figure 5b) reinforced these observations. S-containing combinations were enriched at several key positions, including SS at position 12 (+21.2 pp) and position 13 (+20.3 pp), and SW at position 14 (+18.5 pp). Conversely, SW and SS motifs were depleted at position 15 (−23.6 pp for SW), consistent with the strong W preference seen in the 1-mer analysis. At the same time, WW motifs showed moderate enrichment at positions 7, 8, and 17, suggesting that clusters of weak bases occur more frequently at the 3′ end in functional sequences.

Trinucleotide stability motif analysis (Figure 5c) showed a more distributed pattern but still revealed a few notable trends. Strong base triplets (SSS) were enriched around positions 12–13, whereas WWW motifs were enriched near position 17, and depleted at some internal sites. This mirrors the single-nucleotide pattern of W enrichment at the 3′ end of the guide.

Taken together, the stability-based motif analysis offers an orthogonal view to the nucleotide-specific results in Figure 4. While G/C identity alone cannot explain activity, clusters of strong bases near the middle region (positions 10–13) and weak bases near position 15 and toward the 3′ end appear to be distinguishing features of functional saRNAs. These patterns may reflect subtle effects on guide–promoter hybrid stability, strand selection or Ago2 recruitment, and represent promising features for mechanistic follow-up and improved design strategies.

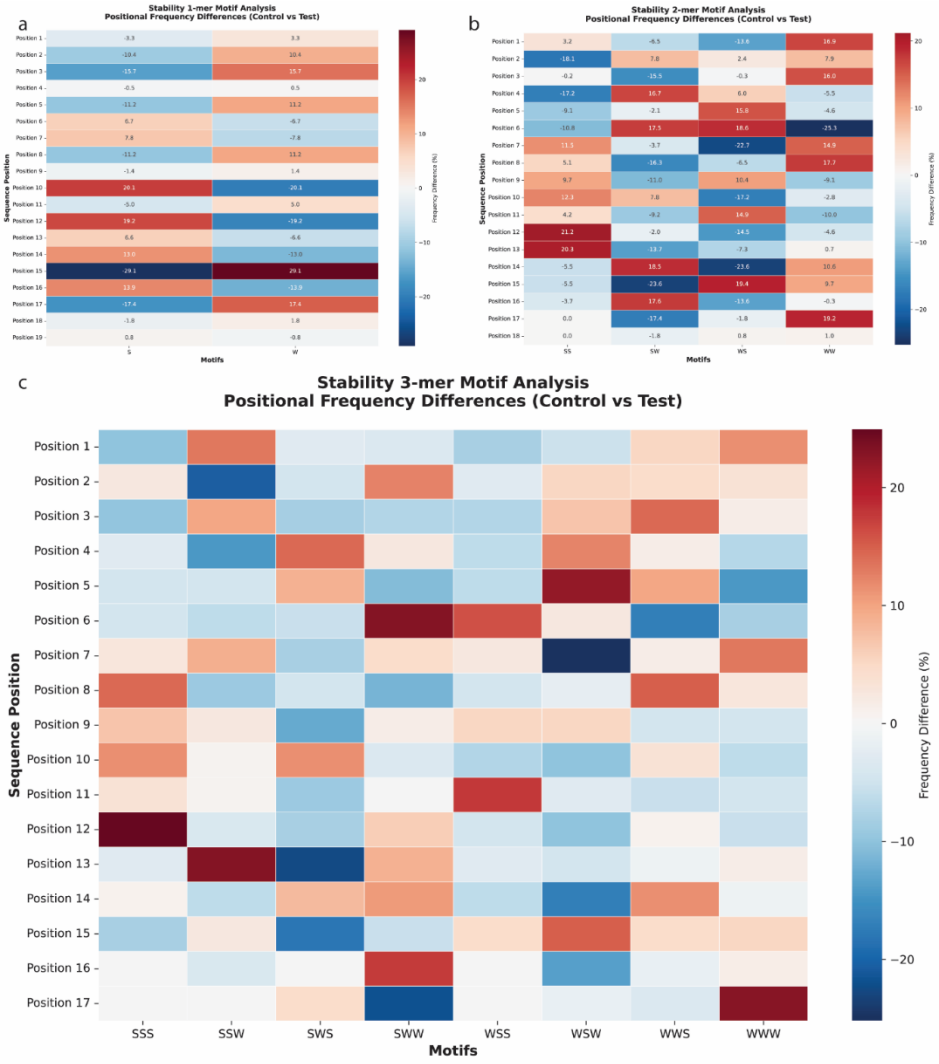

Figure 5. Stability-based motif frequency analysis of saRNA sequences. *Heatmaps showing frequency differences between positive control sequences (n=14) and test (non-successful saRNA) sequences (n=111) using stability classification, where nucleotides are mapped to strong (S: G, C) or weak (W: A, U) categories. Values represent percentage point differences (Control frequency - Test frequency), with red indicating higher frequency in controls and blue indicating higher frequency in test sequences. (a) Single-position stability (1-mer) analysis across all 19 positions, showing frequency differences for S and W classifications. (b) Dinucleotide stability (2-mer) analysis across 18 positions, showing frequency differences for all four possible two-position stability combinations (SS, SW, WS, WW). (c) Trinucleotide stability (3-mer) analysis across 17 positions, showing frequency differences for all eight possible three-position stability combinations (SSS, SSW, SWS, SWW, WSS, WSW, WWS, WWW).*

Discussion

In this chapter, we set out to expand the application of promoter-targeted RNA activation (RNAa) beyond *SCN1A* by designing and testing saRNAs against several additional epilepsy-associated genes, including repressors of the mTOR pathway (*DEPDC5*, *NPRL2*, *NPRL3*, *TSC1*, *TSC2*) and the chromatin remodeler *CHD2*. These genes are frequently implicated in developmental and epileptic encephalopathies through haploinsufficiency, making them attractive candidates for therapeutic upregulation [10], [11], [12]. Our aim was not only to identify additional functional saRNAs but also to explore whether common sequence or structural features might help explain their efficacy. Ultimately, the clinical value of saRNA functionality lies in whether such molecules can rescue disease phenotypes in relevant cellular models, not just increase transcript levels.

Our results confirm the intrinsic difficulty of identifying potent saRNAs across different loci. Despite designing ten candidates per gene following the well-established Wang *et al.* (2015) structural criteria, only a small subset induced ≥ 1.5-fold activation in HeLa cells [7]. This modest hit rate aligns with prior studies reporting that only about 15–20 % of structurally compliant saRNAs produce measurable activation. In contrast, previously characterized saRNAs targeting *SCN1A* (Chapter 1) showed more consistent effects, underscoring that some loci are more permissive to RNAa than others.

Several saRNAs even exhibited mild repression, consistent with previous reports that promoter-targeted RNAs can trigger transcriptional silencing when bound to promoter regions with repressive chromatin features, such as dense DNA methylation, low H3K4me3, or enrichment of H3K27me3, that hinder transcriptional initiation [13]. It is also important to note that saRNAs do not "switch on" completely silent genes; rather, they enhance transcription from loci that already exhibit a detectable baseline level of expression.

A striking feature of this study is the cell type–specific difference in RNAa responsiveness. In HeLa cells, both the *P21* positive control and several candidate saRNAs produced clear activation, whereas in AF22 neural progenitor cells responses were weaker and more variable, with P21 itself showing only modest upregulation. This suggests that AF22 cells are less permissive to RNAa, possibly due to lower transfection efficiency, reduced availability of AGO2 or other cofactors, or more restrictive chromatin states that limit promoter accessibility.

Unlike HeLa cells, which are cancer-derived and maintain a globally open and transcriptionally active chromatin landscape due to continuous proliferation, AF22 neural progenitors divide less frequently and retain tighter epigenetic control over lineage-specific genes. Their chromatin is comparatively compact, with fewer active histone marks (*e.g.*, H3K4me3, H3K27ac) at many promoters, reducing the likelihood that saRNAs can recruit activating complexes to initiate transcription. This highlights an important translational consideration: successful gene activation is not determined solely by saRNA sequence, but also by the cellular and epigenetic environment of the target locus.

To better understand why only a subset of saRNAs were active, we conducted exploratory positional motif analyses. This revealed pronounced sequence biases at specific positions

within the 19-nt guide strand. Functional saRNAs were characterized by G enrichment at positions 10 and 16, U enrichment at position 15, and C depletion at position 9, with additional subtler effects at positions 7, 12, 14, and 17. Dinucleotide and trinucleotide analysis reinforced these findings, with UG (pos 15), GG (pos 12), AC (pos 9), CA (pos 14), and AAA (pos 17) among the strongest enriched motifs. These "hotspots" around positions 9–10 and 14–17 suggest that local sequence context, not just overall GC content, plays a role in determining whether a saRNA is functional, potentially by modulating strand selection or RNA–DNA hybrid stability at the promoter.

In parallel, stability motif analysis offered a complementary perspective by collapsing sequence identity into strong (G/C) and weak (A/U) base categories. Functional sequences showed S enrichment at positions 10 and 12 and W enrichment at position 15, mirroring the nucleotide-level data. Dinucleotide analysis further revealed SS enrichment around positions 12–13 and SW depletion at 15, while trinucleotide analysis showed SSS enrichment in the central region and WWW enrichment near the 3′ end. This pattern suggests that a stable central core flanked by weaker bases at the 3′ end may favor functional activation, possibly by fine-tuning guide strand structure, hybrid stability, or AGO2 loading dynamics.

Taken together, these findings illustrate both the potential and the limitations of current saRNA design strategies. Potent activation, as seen for *SCN1A*, is achievable, but generalizing success to other loci requires deeper understanding of sequence and context dependencies. Our analyses suggest that incorporating positional nucleotide and stability preferences could enhance future saRNA design frameworks beyond the current Wang rules. At the same time, the cell-type variability observed between HeLa and AF22 underscores that epigenetic context, cofactor availability, and promoter accessibility must be considered in parallel.

A key next step will be to combine improved design rules with more physiologically relevant disease models, for example, patient-derived or iPSC-derived neurons, to assess whether saRNA-induced transcriptional activation can indeed rescue functional phenotypes caused by haploinsufficiency. By integrating sequence-level insights with cellular context, we can move closer to rational and predictive saRNA design strategies with therapeutic potential.

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Author contributions

Joep van de Vondervoort designed the study with Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman, coordinated the experiments, analyzed the data, and drafted the chapter. Martijn Koetsier provided technical support and contributed to experimental execution. Serife Uysal contributed extensively to cell culture and experimental work. Gina Rondei, Ruth van Brussel, and Carlijne Hagemans contributed to saRNA screening and validation experiments. Laia Avilés Domínguez contributed to experimental work and interpretation in the later stages of the study. Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman supervised the study and contributed to interpretation and revision. All authors reviewed the final chapter.

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Appendix

Table with positive control saRNA sequences

saRNA	Sequence	Reference
VEGF	GCAACUCCAGUCCCAAAUA	[14]
P53	UUACGGAAAGCCUUCCUAA	[15]
P21-322	CCAACUCAUUCUCCAAGUA	[9]
VHL-821	GGUACAUAGUAGGGCCUUA	[16]
SCN1A-10	CUUAACUCGCCUUCCAAAA	Chapter 2 and 3
SCN1A-12	AAGGAAGUACAGGGUUCAA	Chapter 2 and 3
SCN1A-14	UGCCUCUCCCAACUUGGAA	Chapter 2 and 3
CEBPA	GCGGUCAUUGUCACUGGUC	[17]
DEPDC5-4	AAUCACUUGAGCCCAGAAA	This chapter
CHD2-8	UACAGGUGGGCAAUGUAAA	This chapter
CHD2-10	GAGUUCUUAGGGGCUGAAA	This chapter
TSC2-8	CCUGACAAGGUCCAUGUUA	This chapter
NPRL3-5	CAGUCUCCAGCCUUUAAAA	This chapter
KLF4	GCUGUAGCGAAGGAAGUUA	[18]

Test saRNA sequences

saRNA	Sequence
SCN1A-1	CCUGACUUCUUUCUGAUCA
SCN1A-2	CUCUCCUUCUUACUCACAU
SCN1A-3	UAAGGUUGGUCACUGCAAA
SCN1A-4	UUCCCCUCAUCCUUGUUAA
SCN1A-5	CACAGUUCCUGUAUCAGUA
SCN1A-6	ACAACUCACACCCCCAAUA
SCN1A-7	CAAUCCUCUUGCUCCAUUA
SCN1A-8	GGUGACUGUCUUUGAGUUA
SCN1A-9	UGGUCAUUCUUCCCACAAA
SCN1A-11	GAGCUUUUCUCAAAUGCAG
SCN1A-13	AGGGUUCAAUUCACUCCAA
SCN1A-15	CCCAACUUGGAACAGUCAA
CHD2-1	CUGUCCAGGGAGGAAGUUA
CHD2-2	GGAGAAUUUGCUUUGUGAC

CHD2-3	GGAGUUUCUUGGGUUACUG
CHD2-4	GCCUGAAUCAGUAUGGGAA
CHD2-5	CUUCCAGUGGAGAUGGAAA
CHD2-6	GCAAUGUUGGCUGUGUUAA
CHD2-7	GAUGCCUAUGGUUUCCUAA
CHD2-9	AGCAAUGUUGGCUGUGUUA
KCNB1-1	AUGCUCCAGUGUUUGGGAA
KCNB1-2	CGGAAUUCCCCCAAAUCUU
KCNB1-3	CCCACCCCCAAAAUCUUGA
KCNB1-4	UCUCGAGAAGGGGAAGGAA
KCNB1-5	AGAGAGUUUCAGCACUCUA
KCNB1-6	CUCAUGAAAGCAAGGGCUA
KCNB1-7	CCUUCCUCCUGUUUCAUUA
KCNB1-8	CCCAUCUGUGGCAUCCUUA
KCNB1-9	GGUACCUCCUCCAGUUUUC
KCNB1-10	GCACAUAGGAAGCUGUCAA
CHD8-1	GAGUCAGACUCUGUCUCAA
CHD8-2	GCAAAGUGGACACUGAUUA
CHD8-3	CCCCAGGAAUUACUGUCUA
CHD8-4	GAAGGGUCACAGUGUCAAA
CHD8-5	GAAGAUUUGGCCCCUUUUA
CHD8-6	GGGAAACUCUGUCUCAAAA
CHD8-7	GGCUACUGCUGCCAUUAUA
CHD8-8	GGUACCUACACUGAUCUUA
CHD8-9	GAAACCCCGUCUCUACUAA
CHD8-10	AAACCCCGUCUCUACUAAA
GAD1-1	GCUGGAUGAGGAAACUGUA
GAD1-2	GCAGAAUUCUUCGUAGGAA
GAD1-3	CUGCCUAUUUCCAAAGGAA
GAD1-4	AAACCGUGAGCUGGAUUUA
GAD1-5	GAGAGGGAACUAGCGAGAA
GAD1-6	CCUCUUUCUUUCCAGGGAA
GAD2-1	GGAGCCAGACCUCAAACAA
GAD2-2	CCAAUCGAUGCACACAGAA
GAD2-3	CCUCCCUCUCUCGUGUUUU
GAD2-4	GCUCCCUUUCCCUCAAAUG

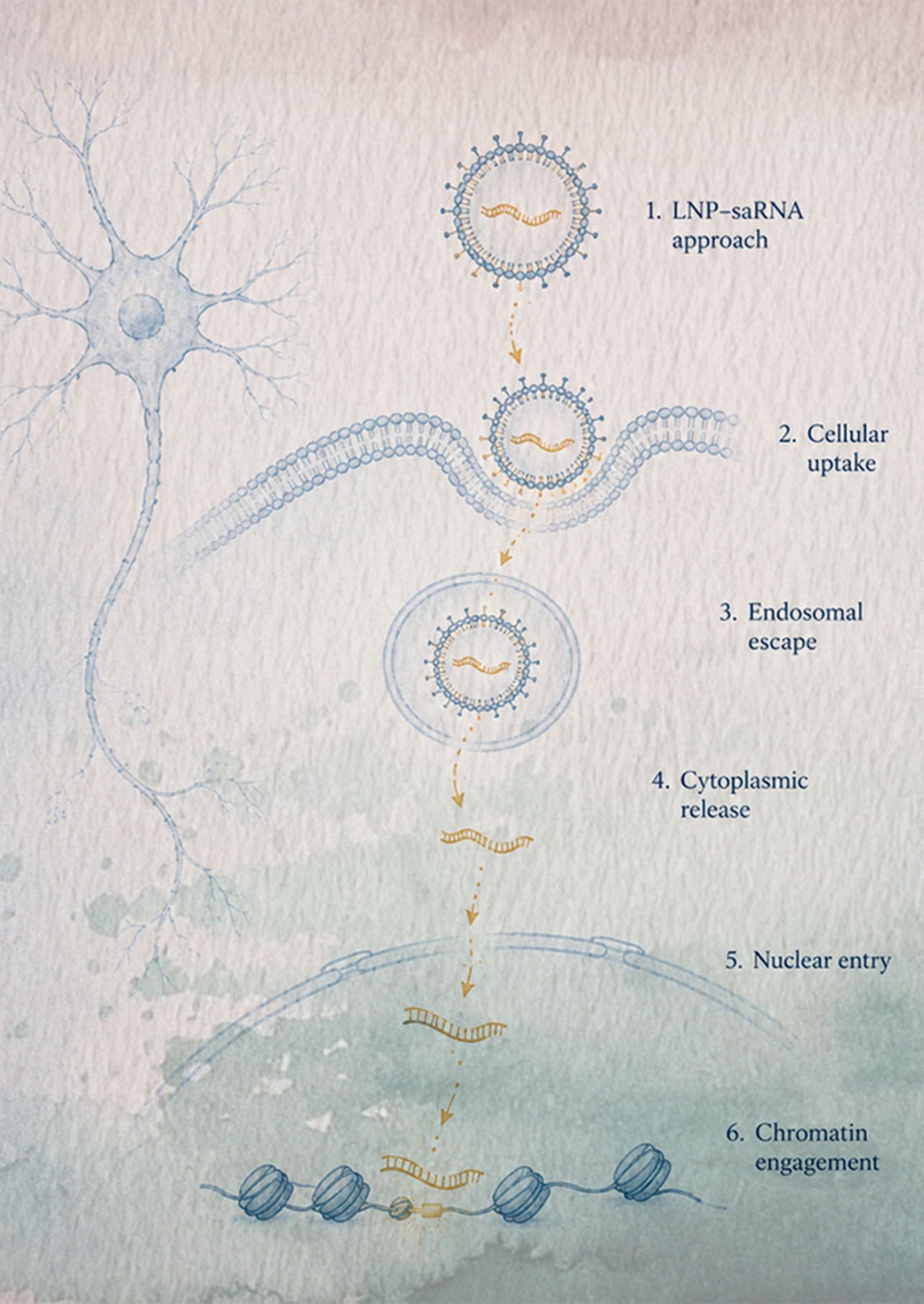
GAD2-5	AGGUAGUCCCGGUCUCUUU
GAD2-6	ACCUGCUCCAGUCUCCAAA
NPRL3-1	CAUCAGUCUCCAGCCUUUA
NPRL3-2	UCAGUCUCCAGCCUUUAAA
NPRL3-3	GUGCGUGCCCAUAAUCCUA
NPRL3-4	AUCAGUCUCCAGCCUUUAA
NPRL3-6	AGGUCUUGGGACAAAGCUA
NPRL3-7	AGGCGUUUUGCUCUCCAAA
NPRL3-8	GGCGUUUUGCUCUCCAAAA
NPRL3-9	GCAUUGUUCACCACACCAA
NPRL3-10	CAUUGUUCACCACACCAAA
NPRL2-1	AGCCGUGCUAGUGGUUGUA
NPRL2-2	CCCAGAAGUCAGCUAUGAA
NPRL2-3	UAGUGGUUGUAGCCUGAAA
NPRL2-4	AUGCCACCCUCUUCCAAUA
NPRL2-5	UGCCACCCUCUUCCAAUAA
NPRL2-6	AUGCGGUGGUUUCCUCCAA
NPRL2-7	UUCCCAGAAGUCAGCUAUG
NPRL2-8	UGGCAAGCACAGCCACAAA
NPRL2-9	CACAGCCACAAAGAUGGUA
NPRL2-10	CUAGUGGUUGUAGCCUGAA
DEPDC5-1	UGCUACCAUGCCCAGCUAA
DEPDC5-2	CCACCAUUGCUAAAGACAA
DEPDC5-3	CUGAAGUGGCAACAUAGAA
DEPDC5-5	AUCACUUGAGCCCAGAAAA
DEPDC5-6	ACCUGGUCUUGGGGGAAAA
DEPDC5-7	CCUGGAUUCUGUGGUUCUA
DEPDC5-8	CAGUAUCUUCACUGGCCUA
DEPDC5-9	AGUAUCUUCACUGGCCUAA
DEPDC5-10	GGUGCCUGUAAUCCCAGAA
TSC2-1	GUUGUGUCACUCCCUUUUA
TSC2-2	CACAACUGUCCAUCAUCUA
TSC2-3	ACCAGUUCCUCCUGACAUA
TSC2-4	AGCCACCGGGUAGAAAGAA
TSC2-5	GGUAUGUGGGCCCUCCUUA
TSC2-6	CACAGGUGAAGGUAGGGUA

TSC2-7	CCCUCUGCAGUCUGGGGUA
TSC2-9	GCUGGUCAGCUGGGCUGUA
TSC2-10	CUGAGUAGAGAGCUCUCUA
GAD1-1	AACCAAUACCAUGGCUCUA
GAD1-2	GGACAAUGCAAGCUCCAAA
GAD1-3	GGCACAUAGUAGGGGAUUA
GAD1-4	GGGCUGUGUUCAAAGUCUA
GAD1-5	CCCAUUUCAUGCCAGUUUA
GAD1-6	CCCAUUUCAUGCCAGUUUA
GAD1-7	CCUCCUUUGCGGUUACAAC
GAD1-8	GAGAAAUUGCCUUGGGCUA
GAD1-9	AGACACAGCUAGGUGGGAA
GAD1-10	GACACAGCUAGGUGGGAAA
GAD1-11	ACCAAUACCAUGGCUCUAA
GAD1-12	UACGUUUCCUGGAGGCUAA
TSC1-1	GGCCACUUUACUUUGUCUA
TSC1-2	GGGUGGUGACACAGAAGAA
TSC1-3	UCCCCAUGAACUAGCAUUA
TSC1-4	CAUACUCAGCCUCGCAAUA
TSC1-5	GGGCACAGAGUGGGAAAUA
TSC1-6	GACCCCAAAGGAAAGAGAA
TSC1-7	ACCCCAAAGGAAAGAGAAA
TSC1-8	GAAGAGUCUGUUCCUGUAA
TSC1-9	CUGUGUUAGAACAGAGGAA
TSC1-10	GGGUCAGAAAUCCAAUGAA

Primers qPCR

Primer name	**Sequence**
CHD2-QPCR_fw	TCTGGGCACCAGAGATTAACG
CHD2-QPCR_rv	TGCCCAGCACAGTCTTATCTT
hGAPDH_fw2	CCATGTTCGTCATGGGTGTG
hGAPDH_rv2	GGTGCTAAGCAGTTGGTGGTG
hCDKN1A_fw	TGCCGAAGTCAGTTCCTTGT
hCDKN1A_rv	GTTCTGACATGGCGCCTCC
hSCN1A3-fw	GCTTTGATACCTTCAGTTGGGC
hSCN1A3-rv	TGTACGTTTTCCCAGCAGCA

qPCR-hAGO2_fw	GATATGCCTTCAAGCCTCCA
qPCR-hAGO2_rv	AACTCTCCTCGGGCACTTCT
qPCR-hDEPDC5_fw	GCTGTGAATGGTTTCCTTGCT
qPCR-hDEPDC5_rv	CTGTCGAATTGAGGCTCGGT
qPCR-hNPRL2_fw	CAGAAGTATCCTGTGCGGGTGA
qPCR-hNPRL2_rv	CATTTTCAAGCCGCTCATCC
qPCR-hNPRL3_fw	CAGCTCATCTGGTGTACTGGG
qPCR-hNPRL3_rv	GGGGCCAGGGGATTCCTAAA
qPCR-hTSC1_fwB	CAGATGAGATCCGCACCCTC
qPCR-hTSC1_rvB	AGCTGCTGCTTTGATCACCT
qPCR-hTSC2_fw	TCCATGACCTGTTGACCACG
qPCR-hTSC2_rv	CGCACATCTCTCCACCAGTT


1. LNP–saRNA approach
2. Cellular uptake
3. Endosomal escape
4. Cytoplasmic release
5. Nuclear entry
6. Chromatin engagement

Chapter 5

Delivery optimization in neurons

Chapter 5

Optimizing saRNA Delivery to Neurons Using Lipid Nanoparticles: Uptake Achieved, Activation Limited

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Abstract

This chapter investigates whether lipid nanoparticles (LNPs) can provide a more efficient and clinically relevant delivery system for *SCN1A*-targeting saRNAs in neurons than conventional lipofection. Because RNA activation requires nuclear entry of AGO2–saRNA complexes, delivery must overcome neuronal uptake, endosomal escape, and nuclear import. We compared Lipofectamine RNAiMAX with four LNP formulations (MC3, SM-102, C12-200, and a cationic "LUNG" formulation) in HeLa cells, AF22 neural stem cells, and differentiated AF22 neurons.

Flow cytometry with fluorescently labelled cargo showed that ionizable LNPs, especially MC3 and SM-102, achieved >90% $Cy5^+$ cells, while LUNG reached approximately 78% in differentiated AF22 neurons, far exceeding RNAiMAX (<20% $Alexa\text{-}488^+$ cells). Using EGFP mRNA as a functional reporter, we further confirmed that SM-102 and LUNG LNPs support endosomal escape and cytosolic delivery, as evidenced by robust EGFP expression in neurons. Thus, for post-mitotic AF22 cells, LNPs clearly outperform lipofection at the levels of uptake and cytoplasmic access.

We then tested whether this superior delivery translated into *SCN1A* promoter activation. In AF22 neural stem (NES) cells, MC3 LNPs successfully delivered *SCN1A*-10, -12, and -14, producing clear and dose-dependent *SCN1A* upregulation, comparable to or exceeding RNAiMAX. However, in differentiated AF22 neurons, none of the formulations yielded robust activation at 50 nM, and only LUNG at 250 nM achieved a modest ~1.5-fold increase. Given that differentiated neurons express baseline *SCN1A*, these findings make inefficient nuclear access a plausible bottleneck for saRNA function in mature neurons, although nuclear localization was not measured directly.

In summary, this chapter shows that optimized ionizable LNPs can efficiently deliver RNA to neuronal cytoplasm and support functional mRNA translation, but saRNA-mediated transcriptional activation in post-mitotic neurons remains limited; inefficient nuclear import of AGO2–saRNA complexes is one plausible explanation. These results position LNPs as a promising backbone for CNS saRNA delivery while highlighting nuclear targeting as a key design priority for future therapeutic development in disorders such as Dravet syndrome.

Introduction

Delivering RNA-based therapeutics to neurons remains a formidable challenge due to both cellular and systemic barriers. Neurons are notoriously difficult to transfect with conventional non-viral methods; lipid-based transfection reagents (*e.g.,* Lipofectamine RNAiMAX) typically achieve low uptake (often <10% of neurons) and can induce significant cytotoxicity [1]. Viral vectors (such as AAV encoding shRNA) can achieve higher transduction but entail high production costs and potential immunogenicity or neurotoxicity [1], [2]. These obstacles have limited the application of RNA interference and related technologies in neuroscience. Thus, there is a pressing need for safe and effective delivery systems that can introduce therapeutic RNA molecules into neuronal cells.

One emerging RNA-based strategy to treat genetic neurological disorders involves upregulating the expression of deficient genes using small activating RNAs (saRNAs). Such an approach could directly address haploinsufficiency-driven conditions such as Dravet syndrome, a severe developmental and epileptic encephalopathy caused by heterozygous loss-of-function variants in *SCN1A*. In Chapter 2, three 21-nt double-stranded saRNAs (saRNA-10, saRNA-12, and saRNA-14) targeting the *SCN1A* promoter were identified and shown to enhance *SCN1A* expression. Because saRNAs are structurally and mechanistically analogous to siRNAs, both require AGO2 and the RNA-induced silencing complex [3] their delivery follows the same multi-step pipeline. In practice, "delivery" encompasses (i) transport of the RNA toward the target neuronal population (*e.g.*, through CSF after intrathecal administration), (ii) cellular uptake via endocytosis, (iii) protection of the RNA against extracellular and intracellular nucleases (degradation being undesired), and (iv) endosomal escape to release the duplex into the cytosol (a highly inefficient but essential step). Consequently, saRNAs face the same delivery barriers as siRNAs and can benefit from siRNA-oriented delivery strategies.

One such strategy involves lipid nanoparticles (LNPs), typically composed of an ionizable lipid, helper lipids (cholesterol and a structural phospholipid), and a PEG-lipid, which together form a protective vesicle for RNA delivery [4]. The first FDA-approved siRNA therapy, patisiran (Onpattro®), demonstrated that an LNP formulation can safely and effectively deliver an RNA drug to patients via intravenous administration [5]. Patisiran's LNP, composed of the ionizable lipid DLin-MC3-DMA (MC3) with helper lipids cholesterol, DSPC, and PEG-lipid, enables hepatocyte delivery of the siRNA targeting transthyretin (TTR), resulting in potent gene silencing in the liver [6]. This success highlighted several key principles of LNP-mediated delivery that are equally applicable to saRNAs: encapsulation of the RNA in a protective lipid vesicle to prevent enzymatic degradation, and the use of pH-sensitive (ionizable) lipids that remain neutral at physiological pH but become protonated in endosomes, triggering endosomal membrane fusion and RNA release into the cytosol [7].

By shielding the RNA's negative charge and size within ~100 nm particles, LNPs facilitate cellular uptake via endocytosis and avoid rapid renal clearance or immune detection [7]. While systemic LNP administration typically results in liver and spleen accumulation due to

apolipoprotein-mediated targeting [5], our therapeutic strategy focuses on direct CNS administration to bypass the blood-brain barrier entirely. Direct intraventricular or intrathecal delivery has shown promise: local injection of siRNA-loaded LNPs in rodents achieved widespread neuronal uptake and gene knockdown in the CNS with minimal toxicity or immune activation [1]. Rungta et al. (2013) showed that intracortical or intracerebroventricular (ICV) injection of siRNA-loaded LNPs in mice achieved potent knockdown of neuronal targets (*e.g. GRIN1*) with excellent uptake and no cytotoxicity or immune activation [1]. More recently, intrathecal delivery to DRG neurons achieved effective *TRPV1* silencing and analgesic effects in rodents [8]. Although an ideal therapy would be delivered systemically with efficient blood–brain barrier passage, such technologies are not yet available. Therefore, in the current landscape, direct CNS administration provides a practical way to bypass the BBB and achieve neuronal exposure. For future clinical translation, direct administration routes such as intrathecal injection or Ommaya reservoir delivery would enable therapeutic access to the CNS while avoiding off-target accumulation in peripheral organs.

In parallel, many *in vitro* studies have confirmed that LNPs can outperform traditional transfection methods in neuronal systems. Rungta *et al.* reported that cultured primary neurons incubated with siRNA-loaded LNPs displayed 100% uptake of the nanoparticles, with punctate LNP signals in virtually all neurons and no detectable cytotoxicity at effective doses. In their hands, LNP delivery achieved robust knockdown of target genes in neurons, whereas lipofection or electroporation typically transfected only a small fraction of neurons (on the order of 1–10%) and often caused cellular toxicity. Thus, LNPs offer a significant performance advantage over Lipofectamine-based reagents like RNAiMAX in delivering small RNAs to neurons *in culture*. Importantly, the superior biocompatibility of LNPs in neurons is attributed to their propensity for endocytic uptake (a pathway neurons readily accommodate) rather than the membrane-disruptive mechanism of cationic lipoplexes, which can trigger cell death. These findings underscore the potential of LNPs to enable saRNA therapeutics for neurological applications, provided that suitable formulations are identified.

The overarching aim of this study was to identify whether LNPs can provide a more effective and clinically translatable delivery strategy for *SCN1A*-targeted saRNAs in neurons, compared with conventional lipofection. To address this, we directly compared Lipofectamine RNAiMAX with a panel of structurally diverse LNP formulations for the delivery of saRNA-10 into differentiated AF22 neurons. Lipofectamine RNAiMAX represents a commonly used benchmark for *in vitro* siRNA/saRNA transfection, whereas LNPs offer a translational route for *in vivo* delivery. We selected a panel of LNP formulations containing different ionizable lipid components to evaluate their efficiency in neuronal RNA delivery: (1) DLin-MC3-DMA (MC3) – the prototypical ionizable lipid from the first approved siRNA LNP, included as a benchmark [5]; (2) SM-102 – an ionizable lipid used in a clinically approved mRNA vaccine (mRNA-1273), representing a newer generation lipid with proven human safety and potency [9];(3) C12-200 – a multi-tail lipidoid that achieved extremely high siRNA potency in preclinical studies (effective gene silencing at doses as low as 0.01–0.03 mg/kg in rodents and primates) [7], included here for its reported efficiency; and (4) “LUNG”

formulation – an LNP incorporating a high fraction of a permanently cationic lipid (analogous to DOTAP-containing LNPs) that was originally designed to target pulmonary tissues [10].

Testing chemically diverse ionizable lipids is important because their structural features influence particle charge state, interactions with endogenous proteins, and endosomal escape capacity, factors that together shape cellular uptake and the subsequent intracellular release required for functional RNA delivery. The rationale for testing this cationic-heavy formulation in neurons is to explore whether increasing surface charge and altering protein adsorption can improve neuronal uptake, given that cationic LNPs bind different endogenous apolipoproteins or ligands than liver-tropic formulations [10]. By comparing lipofectamine against these diverse LNP formulations, we aim to determine which strategy yields the highest saRNA delivery and gene upregulation in AF22 neurons as a step towards the development of an RNA delivery approach suitable for clinical translation.

Methods

Cell Culture

HeLa cells were cultured in Dulbecco's modified Eagle's medium (DMEM) with high glucose supplemented with 10% fetal bovine serum and 1% penicillin/streptomycin. HeLa cells were chosen as an initial model system due to their robust transfection efficiency and well-characterized behavior, allowing for optimization of saRNA delivery conditions.

The induced pluripotent stem cell (iPSC)-derived, long-term self-renewing neuro-epithelial stem cell line (AF22) was obtained from the human iPS core facility at Karolinska Institutet [11]. This cell line was selected as a more physiologically relevant model as it expresses endogenous *SCN1A* and exhibits spontaneous calcium activity upon differentiation, better representing the neural context relevant to our therapeutic targets.

For AF22 NES cells, we used NES medium composed of DMEM-F/12 medium + glutamax (Gibco) supplemented with 1% N2 supplement (Gibco), 0.1% B27 (Gibco), 10 ng/mL Epidermal Growth Factor (EGF) (Thermo Fisher), 10 ng/mL basic Fibroblast Growth Factor (bFGF2) (Thermo Fisher), and 1% Penicillin/Streptomycin (Thermo Fisher). Undifferentiated AF22 cells were cultured on plates coated with 0.1% gelatin solution to maintain their stem cell properties.

To induce neuronal differentiation, AF22 cells were cultured in differentiation medium (Diff) containing DMEM/F12 + glutamax, 1% N2, 1% B27, and 1% Penicillin/Streptomycin. The growth factors were removed to promote differentiation into mature neurons. Additionally, we started differentiated in BrainPhys™ Neuronal Medium (Stemcell technologies) to compare to DMEM/F12 medium for transfection efficacy. Cells were differentiated for four weeks on poly-ornithine and laminin (both 1:500) coated plates to provide appropriate extracellular matrix support for neuronal development. All cells were maintained at 37°C in a humidified 5% CO2 incubator.

Design of saRNAs and EGFP mRNA

saRNA constructs were designed to target and upregulate *SCN1A* gene expression through epigenetic mechanisms. The *SCN1A*-10 saRNA construct was engineered with an Alexa-488 fluorophore coupling to enable direct visualization and quantification of cellular uptake and distribution. This fluorescent labeling allowed us to track saRNA localization without affecting its gene-activating functionality.

Separately, EGFP mRNA constructs were utilized as reporter molecules to assess endosomal escape efficiency of different delivery systems. The expression of functional EGFP protein requires successful endosomal escape and cytoplasmic delivery of the mRNA, making it an ideal readout for comparing the intracellular delivery capabilities of various LNP formulations versus standard transfection reagents.

Standard Transfection Protocol

For baseline comparisons, we employed Lipofectamine RNAiMAX (Thermo Fisher) transfections following our previously established protocol. One day before transfection, cells were seeded in 12-well plates at a density to achieve ~70% confluency (200,000-250,000 cells) on the day of transfection. For HeLa cells in the flow cytometry experiment, medium was changed to Opti-MEM™ (Thermo Fisher) 4 hours prior to transfection to minimize serum interference with lipid complex formation. saRNAs were transfected at a final concentration of 50 nM, and cells were harvested 72 hours post-transfection to allow sufficient time for saRNA amplification and target gene modulation. For GFP mRNA delivery experiments, differentiated AF22 cells were transfected with 0.5 μg mRNA per well using Lipofectamine MessengerMax according to the manufacturer's protocol.

LNP Formulations

To improve delivery efficiency and reduce toxicity, we tested four different lipid nanoparticle (LNP) formulations prepared in the Central Diagnostic Laboratory within the UMC Utrecht: MC3, SM-102, C12-200, and LUNG formulations. These formulations were selected based on their distinct ionizable lipid compositions and its mRNA delivery efficacy was previously tested in different in vitro and in vivo set-ups. MC3 and SM-102 represent clinically validated ionizable lipids, while C12-200 offers high endosomal escape but reported some toxicity in animal studies. The LUNG formulation was specifically optimized for pulmonary delivery applications. LNPs were labeled with a lipid fluorescent dye (DSPE-Cyanine 5) to enable tracking of particle uptake via flow cytometry. Detailed formulation parameters are provided in the appendix.

Lipid nanoparticles were produced using microfluidic mixing with a NanoAssemblr Benchtop system (Precision Nanosystems, Vancouver, Canada). The synthesis involved combining an ethanolic phase containing lipids with an acidic aqueous phase comprising 100

mM sodium acetate (pH 4.0) and mRNA. This formulation took place at a flow rate ratio of 3:1 (aqueous:organic), with a total flow rate of 9.0 mL/min. The lipids, dissolved in 100% ethanol at a concentration of 5-20 mM, consisted of C12-200, Dlin-MC3-DMA, SM-102, DODMA, IM21.7c, DPyPE, cholesterol, DSPC, DSPE-Cy5 and DMG-PEG in different molar percentages. For the experiments with LNPs containing fluorescent lipids the employed molar percentage was 0.2% in detriment of cholesterol. Encapsulation was achieved at an w/w ratio (ionizable lipid/RNA) of 10. Right after production, the LNPs underwent dialysis at 4 °C against an excess of 20mM TRIS 8% sucrose (pH 7.4) using Slide-a-Lyzer™ dialysis cassettes G2 (membrane cutoff 20 kDa) for 16–28 hours. The LNPs were then aliquoted, characterized and stored at -80°C until use.

LNP characterization

Dynamic Light Scattering for size and polydispersity index measurement

Dynamic Light Scattering (DLS) technique using a Zetasizer Nano S instrument (Malvern Panalytical, Malvern, UK) was employed to determine the hydrodynamic diameter of LNPs. A 4 mW HeNe laser emitting at 633 nm was used for the measurements. The LNPs were diluted in Dulbecco's Phosphate-Buffered Saline (DPBS, pH 7.4), and scattering measurements were conducted at an angle of 173° and a temperature of 25 °C for a duration of 10 seconds per measurement, repeated ten times. Each specimen underwent three measurements.

mRNA quantification and encapsulation efficiency assessment

Total RNA concentration was measured using the Quant-It™ Ribogreen RNA Assay kit (Thermo Fisher Scientific) with 2% (v/v) Triton® X-100 in TE buffer. Free RNA concentration was determined separately in TE buffer alone. Lysed LNPs produced fluorescent signal indicating total RNA (μg/μL), while non-lysed samples showed free non-encapsulated RNA (μg/μL). Reference calibration curves were created for both 2% Triton® X-100 and TE buffer to quantify the RNA concentration. The encapsulated mRNA concentration was calculated by subtracting the free RNA concentration from the total, and then used to determine the percentage of encapsulated RNA according to this formula: ((Total RNA – Free RNA) / Total RNA) x 100.

RNA Isolation and Quantitative PCR

RNA isolation and qPCR were performed following protocols established in previous chapters. This consistency ensured comparability across experiments. One-step qPCR was performed using the ViiA 7 Real-Time PCR System (Thermo Scientific) with the following thermal cycling conditions: reverse transcription at 48°C for 15 minutes, initial denaturation at 95°C for 10 minutes, followed by 40 cycles of 95°C for 15 seconds and 60°C for 1 minute.

Each 10 μL reaction contained: 3.75 μL diluted RNA, 5 μL TaqMan RT-PCR mix (Applied Biosystems), 0.5 μL *SCN1A*-FAM assay (Hs00897353_m1, Thermo Scientific), 0.5 μL reference gene *GUSB*-VIC assay (Hs00939627_m1, Thermo Scientific), and 0.25 μL TaqMan Enzyme mix (Applied Biosystems). *GUSB* was selected as the reference gene due to its stable expression across our experimental conditions. The use of multiplex qPCR with different fluorophores allowed simultaneous quantification of target and reference genes, reducing technical variation.

Flow Cytometry

Flow cytometry analysis was performed to quantify transfection efficiency and cellular uptake. In one set of experiments, *SCN1A*-10 saRNA was directly labeled with Alexa-488, enabling detection of transfected cells. Gates for Alexa-488–positive populations were set using untransfected controls to account for background autofluorescence.

In a separate set of experiments, differentiated AF22 cells were transfected with saRNA-loaded LNPs that were labeled with Cy5 on the lipid membrane, enabling flow-based detection of nanoparticle internalization. These experiments therefore reported on particle uptake rather than direct saRNA signal. While the two labeling strategies cannot be directly compared, together they provide complementary information on saRNA delivery and LNP internalization in neurons. For all experiments, uptake was assessed 72 h post-transfection, and prior to analysis cells were washed with PBS to remove residual extracellular material. The culture medium was not replaced after transfection, ensuring continuous exposure under the indicated conditions.

Microscopy

Fluorescence microscopy was performed to evaluate transfection efficiency across different delivery methods. Differentiated AF22 cells were transfected with EGFP mRNA using LUNG LNP, SM-102 LNP, or Lipofectamine RNAiMAX. After 24 hours post-transfection, cells were imaged using the Zeiss Cell Observer microscope with a 10x objective. This timepoint was selected to allow sufficient EGFP expression while minimizing potential cell death from transfection stress. The 10x magnification enabled assessment of transfection efficiency across larger cell populations, providing a representative view of delivery performance. Fluorescence intensity was quantified using Fiji (ImageJ) software, with maximum fluorescence values measured for each condition to assess the peak delivery efficiency achieved by each method.

Statistical Analysis

Statistical analyses were performed using custom Jupyter scripts to ensure reproducibility and transparency. Prior to statistical testing, data normality was assessed using the Shapiro-Wilk test, which informed the selection of appropriate parametric or non-parametric tests. This approach ensured that statistical assumptions were met and conclusions were valid.

For qPCR data analysis, we performed statistics on ΔCt values rather than fold-change values to avoid bias introduced by logarithmic transformation. ΔCt values were calculated by normalizing Ct values of *SCN1A* to the reference gene *GUSB*. ΔΔCt values were then obtained by normalizing against the average ΔCt of non-targeting control samples. Fold changes were calculated using the 2^(-ΔΔCt) method for data presentation only.

Statistical significance was determined using One-Way ANOVA for multi-group comparisons or unpaired t-tests for two-group comparisons, as appropriate for the experimental design. All tests were two-tailed with significance levels indicated as: $p < 0.05$ (*), $p < 0.005$ (**), $p < 0.0005$ (***), and $p < 0.0001$ (****). Results are presented as mean ± standard deviation unless otherwise stated. This statistical approach ensured robust and reliable interpretation of our experimental findings while maintaining consistency with established practices in the field.

Results

Delivery Efficiency of *SCN1A*-targeting saRNA in Multiple Cell Lines

To evaluate the cellular uptake of our *SCN1A*-targeting saRNA (*SCN1A*-10), we performed flow cytometry analysis using Alexa Fluor 488–labeled saRNA delivered via Lipofectamine RNAiMAX across different cell lines and culture conditions (Figure 1a). We tested uptake in both standard DMEM and OptiMEM, the latter lacking antibiotics and serum that could potentially interfere with RNAiMAX-mediated transfection. For differentiated AF22 cells, we additionally employed BrainPhys medium, which more closely reflects the neuronal extracellular environment and is serum-free. Wild-type cells for each condition served as negative controls to establish baseline fluorescence.

We next evaluated four lipid nanoparticle (LNP) formulations as alternative delivery vehicles with greater translational potential (Figure 1b). In these experiments, LNPs were loaded with saRNA and labeled with Cy5 on the lipid component, enabling detection of particle internalization. Uptake was again assessed 72 h post-transfection, with PBS washing prior to measurement.

All LNP formulations achieved high levels of Cy5-positive cells, indicating robust particle internalization. MC3 and SM-102 LNPs reached the highest uptake (90.27% and 91.02%, respectively), while C12-200 and LUNG formulations showed intermediate efficiencies (71.80% and 77.85%). One-way ANOVA analysis revealed no statistically significant differences among formulations ($p = 0.093$).

It is important to note that Cy5 labeling tracks the LNP lipid component, whereas RNAiMAX experiments monitored Alexa-488–labeled saRNA. Because these readouts capture different signals, the percentages cannot be compared quantitatively. Although different fluorophores were used (Alexa-488 on the saRNA for RNAiMAX vs. Cy5 on the lipid component for LNPs), both assays quantify the proportion of cells showing detectable fluorescence. This makes the percentage of positive cells directly comparable as an indication of how many neurons internalized particle-associated material (<20% for RNAiMAX vs. >90% for several LNPs). However, these measurements do not reflect the amount of saRNA delivered per cell or whether the RNA successfully escapes the endosome. Cy5 positivity mainly reports the presence of LNPs, either surface-bound or endosomal, rather than functional cytosolic release of the saRNA.

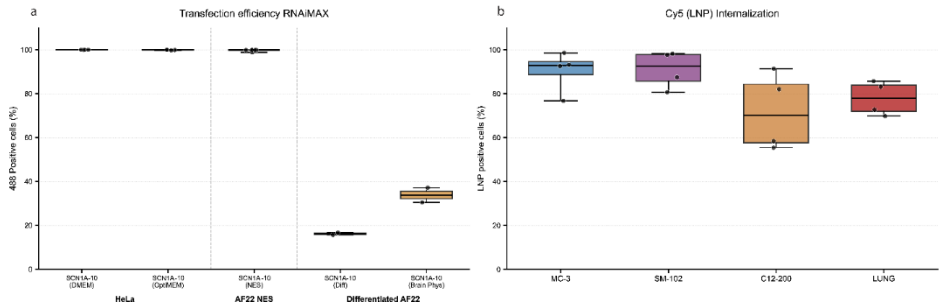

Figure 1. Cellular internalization efficiency of *SCN1A*-targeting saRNA using different delivery methods. *(a) Flow cytometry analysis of transfection efficiency using Lipofectamine RNAiMAX to deliver Alexa Fluor 488-labeled SCN1A-10 saRNA across multiple cell lines (HeLa, AF22 neural epithelial stem (NES), and differentiated AF22) and culture media. (b) Comparative analysis of Cy5-labeled lipid nanoparticle (LNP) formulations for saRNA delivery to differentiated AF22 cells. Box plots show the percentage of Cy5-positive cells following treatment with four distinct LNP formulations: MC3 (90.27%), SM-102 (91.02%), C12-200 (71.80%), and LUNG (77.85%). Box plots display median (center line), interquartile range (box), and minimum/maximum values (whiskers). Individual data points represent biological replicates.*

Functional Validation of LNP-Mediated saRNA Delivery in undifferentiated NES

To confirm that the observed LNP-mediated cellular uptake translated to functional gene modulation, we assessed *SCN1A* expression following delivery of three *SCN1A*-targeting saRNA variants (*SCN1A*-10, -12, and -14) in AF22 NES cells using qPCR (Figure 2a). When delivered at 50 nM using MC3 LNPs, all three saRNA constructs induced *SCN1A* expression above baseline. *SCN1A*-10 demonstrated the most robust response with MC3 LNP delivery, achieving 1.68-fold induction ($p < 0.001$), comparable to its performance with Lipofectamine (1.59-fold, $p < 0.01$). In contrast, *SCN1A*-12 and *SCN1A*-14 showed minimal to no efficacy when delivered via MC3 LNPs (0.99-fold and 1.09-fold, respectively) compared to Lipofectamine delivery, which yielded significant upregulation for both constructs (1.38-fold, $p < 0.05$ and 1.53-fold, $p < 0.01$, respectively).

We evaluated the dose-response relationship of three *SCN1A*-targeting saRNAs (*SCN1A*-10, *SCN1A*-12, and *SCN1A*-14) delivered via MC3 LNPs compared to scrambled saRNA controls in undifferentiated AF22 NES cells (Figure 2b). All three saRNAs demonstrated *SCN1A* upregulation compared to their respective scrambled controls, though with varying efficacy. saRNA-14 showed the strongest activation, achieving 2.28-fold upregulation at 100 nM, followed by 1.67-fold at 50 nM and 1.53-fold at 75 nM. saRNA-10 exhibited its highest activity at 50 nM (1.77-fold) with somewhat lower effects at 75 nM (1.41-fold) and 100 nM (1.44-fold). saRNA-12 showed more modest upregulation, ranging from 1.29-fold to 1.77-fold across the tested doses.

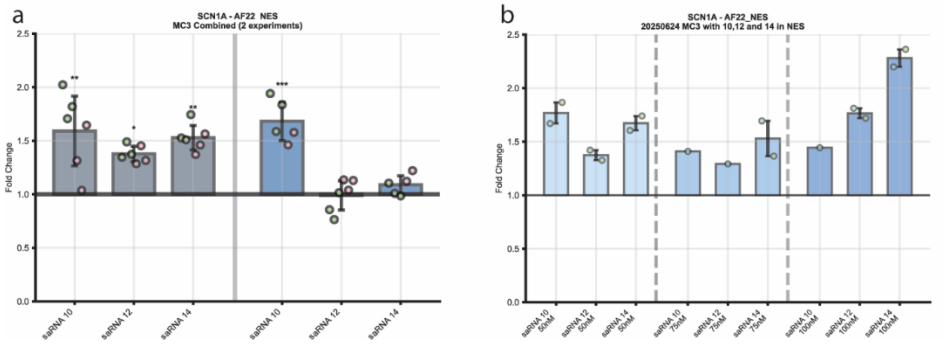

Figure 2. Functional assessment of *SCN1A*-targeting saRNA constructs delivered via MC3 lipid nanoparticles (LNPs) in AF22 neural stem cells. *(a) Expression analysis of SCN1A following delivery of three saRNA variants (SCN1A-10, -12, and -14) at 50 nM using MC3 LNPs. Bar colors distinguish two independent experiments, with individual replicate values shown as circles. Gene expression was quantified by qPCR and expressed as fold change relative to untreated controls. Bars represent mean ± SD. The horizontal line at y = 1.0 indicates baseline expression. Statistical significance: *p < 0.05, **p < 0.01, ***p < 0.001. (b) Dose–response evaluation of SCN1A-10, -12, and -14 delivered via MC3 LNPs. Each construct was tested at 50, 75, and 100 nM, as indicated by the different bar colors. For each concentration, expression values were normalized against a scrambled control at the same concentration. Vertical dashed lines separate the different concentration groups. Bars represent mean ± SD, with individual replicate values shown as circles. The horizontal line at y = 1.0 marks baseline expression relative to untreated controls.*

Verification of Endosomal Escape and RNA Translation

To test whether LNP-delivered RNA cargo could reach the cytoplasm and undergo translation in differentiated AF22 neurons, we performed fluorescence microscopy using EGFP-encoding mRNA as a functional reporter (Figure 3). We specifically chose mRNA rather than saRNA for this validation experiment, as mRNA provides a direct readout of cytoplasmic release and translation without the additional complexity of RNA-induced gene activation. It should be noted, however, that mRNA and saRNA differ substantially in size, structure (single- vs double-stranded), and mechanism of action, which may influence their uptake and intracellular trafficking.

Differentiated AF22 neurons were transfected with 0.5 μg of EGFP mRNA using Lipofectamine RNAiMAX, SM-102 LNPs, or LUNG LNPs. At 24 hours post-transfection, all formulations produced detectable EGFP expression, confirming that these LNPs can mediate endosomal escape and cytoplasmic release in post-mitotic neurons (Fig 3). The maximum fluorescence intensities observed (LUNG: 177 a.u.; RNAiMAX: 165 a.u.; SM-102: 145 a.u.) indicate that individual neurons successfully expressed EGFP following LNP delivery.

These results demonstrate that, although endosomal escape remains a primary limiting factor for LNP-mediated RNA delivery in neurons, the tested formulations are capable of achieving functional escape. Differences in apparent fluorescence between formulations may reflect not only escape efficiency but also kinetic factors: for example, SM-102 has been reported to show slower onset of expression, which may account for its lower values at 24 h despite higher efficiencies at later time points.

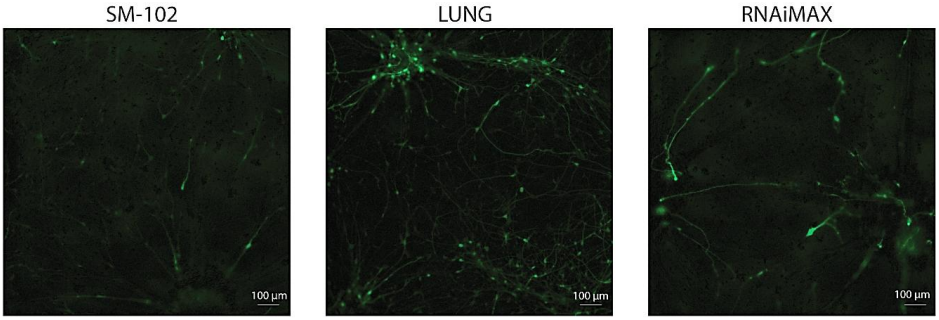

Figure 3: Representative fluorescence microscopy images of differentiated AF22 cells transfected with EGFP mRNA encapsulated in different lipid nanoparticle (LNP) formulations: SM-102, LUNG, and RNAiMAX. *Cells were imaged 24 hours post-transfection using a Zeiss Cell Observer with a 10x objective. EGFP fluorescence (green) indicates successful mRNA delivery and expression. The images demonstrate variability in transfection efficiency and EGFP expression among the tested LNP formulations.*

SCN1A Upregulation in Differentiated AF22 Neurons

We next assessed whether LNP-delivered saRNA could increase *SCN1A* expression in differentiated AF22 neurons, which represents a mixed culture of neuronal cell with predominantly GABAergic neurons that represent the therapeutically relevant cell type for Dravet syndrome (Figure 4). We evaluated five delivery formulations (RNAiMAX, C12-200, LUNG, MC3, and SM-102) delivering *SCN1A*-10 saRNA at 50 nM (Figure 4a) and a Scrambled saRNA as a control. All LNP formulations with *SCN1A*-10 showed very variable efficacy and nonsignificant changes in *SCN1A* expression levels as compared to scrambled.

To test whether higher dosage is needed we performed a second transfection with two different concentrations (50 nM and 250 nM) (Figure 4b). Again, transfection with 50 nM, showed no significant change in *SCN1A* expression, but increasing the concentration to 250 nM of LUNG encapsulate *SCN1A*-10 achieved a 1.46-fold upregulation of *SCN1A*. Further

optimization of formulation and dosing parameters remains necessary for achieving robust and reproducible gene activation in this challenging cell type.

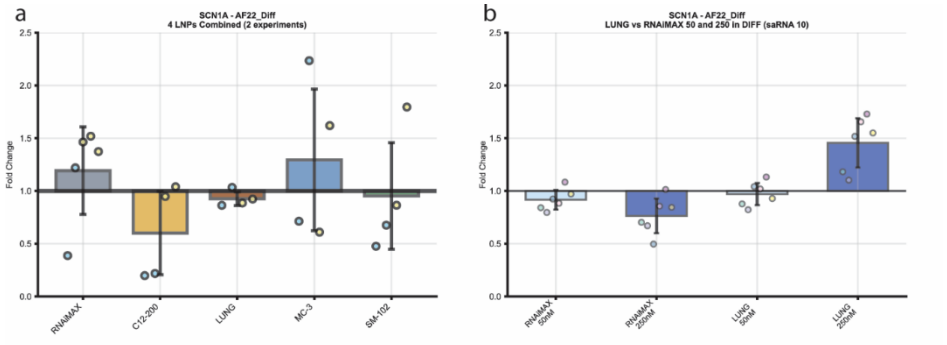

Figure 4. *SCN1A* expression analysis in differentiated AF22 neurons following LNP-mediated saRNA delivery. *(a) Comparison of five delivery formulations (RNAiMAX, C12-200, LUNG, MC3, and SM-102) for SCN1A-10 saRNA delivery at 50 nM. Data are merged from two independent experiments, with different colors indicating each experiment. (b) Dose-response evaluation of LUNG and RNAiMAX formulations at 50 nM and 250 nM concentrations from a single experiment. Data are presented as fold change relative to scrambled control, normalized to housekeeping gene expression. Each data point represents an individual biological replicate. Statistical analysis was performed on ΔCt values. Horizontal lines indicate mean fold change, with error bars representing standard deviation. No statistically significant differences were observed compared to scrambled controls.*

Discussion

The primary aim of this study was to identify lipid nanoparticles (LNPs) capable of efficiently delivering small activating RNAs (saRNAs) directly to post-mitotic neurons. Achieving effective neuronal transfection remains challenging due to limited uptake and intracellular delivery barriers, particularly in differentiated neurons. Our results demonstrate that specific ionizable LNP formulations, particularly MC3, SM-102, and LUNG, substantially enhance neuronal uptake compared with Lipofectamine RNAiMAX: MC3 and SM-102 exceeded 90% Cy5-positive cells, while LUNG reached approximately 78%, versus approximately 16% with Lipofectamine. This improved uptake aligns with previous findings, showing ionizable LNPs outperform cationic lipid-based reagents in neuronal cells [12].

Importantly, successful endosomal escape was confirmed through robust EGFP expression following mRNA delivery via SM-102 and LUNG LNPs. This indicates that our LNP formulations effectively navigate intracellular trafficking, addressing one of the major bottlenecks for nucleic acid therapeutics [13], [14]. The LUNG formulation notably showed superior EGFP expression just 24 hours after transfection, as indicated by higher maximal EGFP intensity compared to the other tested LNPs and Lipofectamine.

Despite excellent uptake in AF22 NES, *SCN1A* activation in differentiated AF22 neurons was modest. At the standard 50 nM dose, we did not observe *SCN1A* induction, whereas at an increased dose of 250 nM with LUNG LNP, *SCN1A* expression was moderately upregulated (~1.5-fold). The dose-dependent response observed is supported by both *in vitro* studies showing that higher intracellular LNP payload enhances efficacy. Han et al. (2025) reported that increasing the dose of mRNA-loaded LNP in neuronal cultures led to proportional increases in reporter gene expression [15]. Similarly, Rungta et al. delivered PTEN-targeting siRNA at 246 nM via LNP to primary rat hippocampal neurons and observed ~80% protein knockdown; importantly, knockdown was still substantial (~60%) at much lower doses (0.7 nM), with greater effects at higher concentrations [1]. These findings collectively suggest that while RNA therapeutics like siRNAs and mRNAs can function at relatively low intracellular concentrations, increasing the dose often enhances their efficacy. Interestingly, MC3 showed a reduced fold change at higher concentration in our study; this may reflect dose-dependent cellular stress, endosomal sequestration, or altered intracellular trafficking at elevated LNP or RNA levels, leading to less efficient nuclear delivery of the saRNA [16]. Given that saRNAs require nuclear entry and act through a distinct mechanism compared to siRNAs or mRNAs, achieving a sufficient intracellular concentration may be even more critical for their functional impact. Future experiments could also explore higher doses with the other LNP formulations (MC3 and SM-102) in differentiated neurons, as only LUNG was tested at the 250 nM concentration in the current study.

The modest activation in neurons contrasts with clear *SCN1A* induction observed in proliferative AF22 neural stem cells (NES), highlighting a crucial difference likely related to nuclear import efficiency of the saRNA–AGO2 complex. Mature neurons have intact nuclear envelopes and potentially lower nuclear transport efficiency compared to dividing cells, restricting nuclear access of the RNA-induced transcriptional activation (RITA) complex [17]. Our data support this hypothesis: despite strong cytoplasmic uptake, limited nuclear localization likely restricted the therapeutic effect. An alternative explanation could involve chromatin accessibility at the *SCN1A* locus. However, baseline qPCR data confirmed *SCN1A* expression in differentiated neurons, indicating active transcription and, thus, accessible chromatin. Although these findings make grossly inaccessible chromatin less likely, they do not exclude locus-specific chromatin constraints. Insufficient nuclear delivery therefore remains a leading hypothesis rather than a demonstrated cause.

Safety and toxicity assessments were beyond the scope of the current study; future experiments will involve formal cytotoxicity and immunogenicity testing, essential for clinical translation. Low immunogenicity has enabled repeated LNP administration in a preclinical mouse study, but repeat-dose safety and immunogenicity require formulation- and route-specific evaluation before clinical translation [18].

This study has several limitations. First, some experiments were only partially explored: for example, higher-dose testing was performed only with the LUNG formulation, and *SCN1A* activation was assessed at a single time point, limiting conclusions about dose–response and kinetics across all LNPs. Second, the number of biological replicates and the observed variability in *SCN1A* fold-changes warrant repetition and expansion to confirm the robustness

of the effects. Third, while we infer endosomal escape and nuclear delivery from EGFP expression and qPCR readouts, we did not directly measure these steps; dedicated assays such as Gal8 recruitment, subcellular fractionation, or live-cell imaging would be needed to quantify endosomal escape and nuclear accumulation of saRNA–AGO2 complexes, which are increasingly recognized as critical bottlenecks for nucleic-acid therapeutics. In future work, systematic dose–response studies for all LNPs, combined with direct measurements of endosomal escape and nuclear localization, as well as testing in disease-relevant neuronal models (*e.g.*, patient-derived Dravet neurons), will be important to validate and extend the translational potential of LNP-mediated saRNA delivery.

In conclusion, our findings demonstrate that optimized ionizable LNP formulations substantially improve neuronal nanoparticle uptake and support cytosolic mRNA delivery compared with traditional transfection reagents. Although nuclear localization remains a limiting factor for robust transcriptional activation, achieving significant *SCN1A* upregulation at higher doses supports further exploration and refinement of LNP-mediated saRNA delivery strategies. Future research should focus on improving nuclear entry efficiency, potentially through modifications in nanoparticle design or saRNA chemistry, to fully realize the therapeutic potential of this approach in neurological disorders such as Dravet syndrome.

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Author contributions

Joep van de Vondervoort designed the study with Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman, conducted and analyzed the neuronal delivery experiments, and drafted the chapter. Martijn Koetsier provided technical support and contributed to experimental execution. Serife Uysal contributed extensively to cell culture and experimental work. Laia Avilés Domínguez contributed to LNP formulation, production and characterization, and to experimental planning. Prof. dr. K.P.J. Braun and Dr. B.P.C. Koeleman supervised the study and contributed to interpretation and revision. All authors reviewed the final chapter.

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Appendix

Detailed information LNPs

Table S1 | Nanoparticle characterization details from all experiments. Size, PdI, and RNA encapsulation efficiency were measured for all the tested nanoparticles. The results are represented by mean ± SD (n = 3 measurements).

Nanoparticles	Size (d. nm)	PdI (0-1)	RNA encapsulation efficiency (%)
Anti-PSMA 1% Cy5-labeled EGFP mRNA	117.3 ± 10.4	0.099 ± 0.016	96.33
Anti-PSMA 0.2% Cy5-labeled EGFP mRNA	118.0 ± 10.7	0.093 ± 0.010	97.57
R2 1% Cy5-labeled EGFP mRNA	112.8 ± 3.8	0.114 ± 0.025	93.24
R2 0.2% Cy5-labeled EGFP mRNA	108.2 ± 3.4	0.116 ± 0.008	93.55
DSPE-PEG 1% Cy5-labeled EGFP mRNA	126.9 ± 5.9	0.273 ± 0.015	94.7
DSPE-PEG 0.2% Cy5-labeled EGFP mRNA	107.8 ± 1.9	0.156 ± 0.003	96.28
Uncoated Cy5-labeled EGFP mRNA	104.7 ± 2.9	0.105 ± 0.014	96.26
Anti-PSMA 0.2% DSPE-Rhodamine BFP mRNA	104.0 ± 1.4	0.125 ± 0.013	97.77
R2 0.2% DSPE-Rhodamine BFP mRNA	116.3 ± 4.7	0.212 ± 0.003	98.15
Anti-PSMA 0.2% DSPE-Cy5 luciferase mRNA	99.9 ± 1.2	0.073 ± 0.016	97.05
R2 0.2% DSPE-Cy5 luciferase mRNA	101.1 ± 1.7	0.081 ± 0.015	96.27

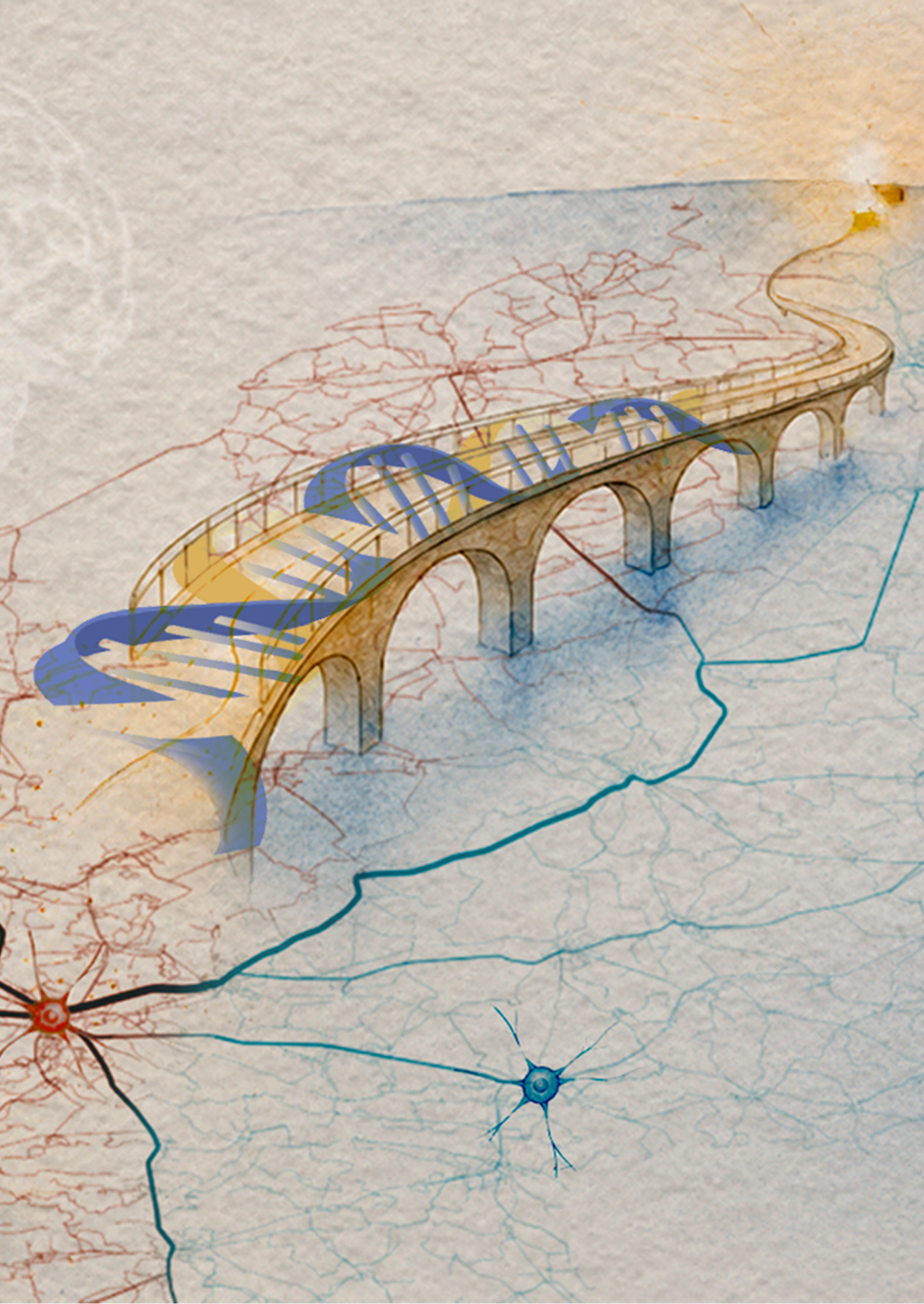
Sequences saRNA and mRNA

	RNA sequence strand 1 (5'-3')	RNA sequence strand 2 (5'-3')
#	*SCN1A* (Located in the promoter region described by Long (2008))	
10	CUUAACUCGCCUUCCAAAA [dT][dT]	UUUUGGAAGGCGAGUUAAG [dT][dT]
#	*SCN1A* (Located in the promoter region by Dong (2014))	
12	AAGGAAGUACAGGGUUCAA [dT][dT]	UUGAACCCUGUACUUCCUU [dT][dT]
14	UGCCUCUCCCAACUUGGAA [dT][dT]	UUCCAAGUUGGGAGAGGCA [dT][dT]
Scrambled control	ACUACUGAGUGACAGUAGA [dT][dT]	UCUACUGUCACUCAGUAGU [dT][dT]

Primers qPCR

Primer name	Sequence
hGAPDH_fw2	CCATGTTCGTCATGGGTGTG
hGAPDH_rv2	GGTGCTAAGCAGTTGGTGGTG

hCDKN1A_fw	TGCCGAAGTCAGTTCCTTGT
hCDKN1A_rv	GTTCTGACATGGCGCCTCC
hSCN1A3-fw	GCTTTGATACCTTCAGTTGGGC
hSCN1A3-rv	TGTACGTTTTCCCAGCAGCA



Chapter 6

General discussion and translational outlook

Chapter 6

General Discussion

In this thesis, we explored small activating RNAs (saRNAs) as a novel therapeutic strategy to upregulate gene expression for monogenic epilepsy. Specifically, we targeted the *SCN1A* gene, which is haploinsufficient in Dravet syndrome, to compensate for the loss-of-function mutations underlying this severe childhood epilepsy syndrome. Our experimental chapters showed that certain saRNA molecules can indeed upregulate *SCN1A* at the mRNA level in cell-based models, as confirmed by RT-qPCR and transcriptomic analyses. These findings provide a proof-of-concept that promoter-targeted saRNAs can (at least partly) reverse haploinsufficiency at the molecular level. In this General Discussion, we critically examine the broader implications of these results and situate them in the current scientific landscape of RNA-mediated gene activation (RNAa). We discuss the need for disease-relevant functional models to determine whether *SCN1A* upregulation translates into phenotypic rescue of neuronal dysfunction. We also address the mechanistic underpinnings of RNAa, evaluating our validation approaches and proposing future strategies to solidify how saRNAs work. Furthermore, we delve into the question of why only some saRNAs are effective, highlighting the importance of sequence, chromatin context, and target site selection – a key challenge in the field. We then consider practical aspects of advancing saRNAs toward therapy, notably how to improve their delivery and efficacy in neuronal systems, exploring advanced delivery vehicles (lipid nanoparticles, viral vectors) and design modifications. We also compare the saRNA approach to other emerging gene upregulation strategies (such as CRISPR-based activators and antisense oligonucleotides), noting the advantages of saRNAs' small size and potential for combination therapies. Through an integrated analysis of our findings and the literature, we aim to chart a path forward for saRNA-based therapeutics for Dravet syndrome and related monogenic epilepsies, emphasizing both the promise and the hurdles that remain.

6.1 Necessity of Disease-Relevant Functional Models

A key question arising from our research is whether increased *SCN1A* expression via saRNA can mitigate the epileptic phenotype characteristic of Dravet syndrome. Dravet syndrome is caused by *SCN1A* loss-of-function mutations in ~80% of patients [1], [2], leading to a deficiency of $Na_v1.1$ sodium channels particularly in GABAergic inhibitory interneurons [3]. This imbalance reduced inhibitory signaling (due to interneuron dysfunction) amidst normal excitatory signaling, results in hyperexcitable neural networks and severe seizures. While our saRNA treatment can successfully elevate *SCN1A* transcript levels in vitro, it remains crucial to demonstrate phenotypic rescue: does restoring *SCN1A* expression also leads to normalized network hyperexcitability and prevents seizure-like activity? Answering this requires a disease-relevant functional model that recapitulates the neuronal network dysfunction of Dravet syndrome.

Patient-derived neuronal cultures offer a promising platform to test whether *SCN1A* upregulation translates to functional improvement. Notably, human induced pluripotent stem cell (hiPSC) models of Dravet syndrome have been developed. For example, hiPSC-derived neuronal networks from Dravet patients (or CRISPR-engineered *SCN1A*± lines) show abnormal electrophysiological activity, such as increased firing rates and network bursting,

especially under stressors like elevated temperature [3]. Multi-electrode array (MEA) recordings have revealed that these patient-specific excitatory networks exhibit mutation-specific hyperexcitability and aggravated responses at febrile temperatures, mirroring a key clinical feature of Dravet syndrome [3]. Such in vitro models recapitulate the foremost clinically relevant features of Dravet providing a valuable assay for candidate therapeutics. We propose leveraging Dravet patient-derived neurons on MEAs (available through ongoing collaborations) as a critical next step. By treating these networks with our *SCN1A*-targeted saRNAs, we can directly test for functional rescue – for instance, normalization of firing rates, reduction in hyper-synchronous bursting, and restoration of network stability.

To illustrate, if saRNA-mediated *SCN1A* upregulation enhances $Na_v1.1$ levels in patient neurons, we would expect an increase in inhibitory interneuron excitability, thereby damping overall network firing. MEA analyses could detect such changes as a decrease in metrics like mean firing rate or burst frequency, moving closer to healthy control levels. Additionally, features like temperature-induced hyperexcitability (a hallmark of Dravet networks [3]) could be mitigated by saRNA treatment, providing evidence of phenotypic correction. In addition to interneurons, *SCN1A* is also expressed, at lower levels, in other cell types such as excitatory neurons and, to a lesser extent, glial cells [4], [5]. Although interneuron dysfunction is widely considered the main driver of the seizure phenotype, the contribution of *SCN1A* haploinsufficiency in these other cell types remains less well understood. Deficits in excitatory neurons may modulate network dynamics and could underlie cognitive and behavioral comorbidities seen in Dravet syndrome, while altered astrocyte or microglial function might further shape network excitability. This represents an unexplored but potentially important disease axis that could broaden the therapeutic window of *SCN1A* upregulation strategies [6], [7].

Importantly, this disease-relevant testing addresses a gap in our current work: while we confirmed molecular efficacy of saRNAs, a functional assay is needed to show that molecular upregulation can ameliorate the disease-related dysfunction. Without a relevant model, we cannot be certain that boosting *SCN1A* mRNA will suffice to restore the delicate balance between excitation and inhibition in a complex neuronal network.

It is worth noting that in genetic epilepsies, even partial restoration of the haploinsufficient gene can have therapeutic impact. In *Scn1a*+/− Dravet mouse models, for example, small increases in $Na_v1.1$ expression (achieved via genetic or pharmacological means) have been shown to improve seizure outcomes and survival [8]. For instance, selectively enhancing the activity of remaining $Na_v1.1$ channels in interneurons was sufficient to rescue mice from seizures and premature death [8]. For our saRNAs, however, mouse models of Dravet cannot be used directly, as promoter sequences are poorly conserved and mouse gene regulation is different from that in humans [9]. Therefore, functional testing of saRNAs beyond cell lines and toward translational relevance requires the use of patient-derived neuron networks (such as Dravet iPSC neurons on MEAs) to link *SCN1A* upregulation with rescue of phenotypes. This will strengthen the case that saRNAs can function as true therapeutic tools, bridging the gap between molecular correction and tangible physiological benefit in monogenic epilepsies.

6.2 Mechanistic Validation: Current Limitations and Future Strategies

At a mechanistic level, we investigated the phenomenon of RNA activation (RNAa) – the process by which saRNAs enhance transcription of target genes. A deeper mechanistic insight is needed to ensure that we are truly modulating *SCN1A* via on-target, Argonaute-mediated chromatin effects (as hypothesized), and not through off-target pathways. It can also reveal why certain saRNAs succeeded or failed, guiding the design of next-generation molecules. In Chapter 3, we performed an AGO2 chromatin immunoprecipitation (ChIP) followed by qPCR at the *SCN1A* promoter, reasoning that if the saRNA guides Argonaute 2 to the target promoter, we should detect increased AGO2 occupancy there. Indeed, RNA activation is known to be AGO2-dependent, as Argonaute proteins are core components of the RNA-induced transcriptional activation complex [10]. These AGO2-ChIP experiments did not provide clear evidence for enrichment of the saRNA binding region at the *SCN1A* promoter. Despite initial indications, both the *SCN1A*- and P21-targeted saRNAs failed to show robust or reproducible enrichment above background levels. The low signal and high background in these assays made it difficult to draw definitive conclusions regarding specific recruitment of AGO2 to the target promoters. Nevertheless, our experiments using AGO2 siRNA mediated knockdown showed that the observed effect of saRNA-10 on *SCN1A* is dependent on AGO2, leaving the question still relevant but in need of additional evidence. To more rigorously validate the mechanism of saRNA action, we propose several complementary strategies:

Fluorescence In Situ Hybridization (RNA-FISH) with Immunostaining

One approach is to directly visualize the saRNA and Argonaute inside the nucleus. By designing a fluorescently labeled saRNA (or an antisense probe against it), we could perform RNA-FISH in cells after transfection. Co-staining for AGO2 would reveal whether the saRNA and AGO2 co-localize in nuclear foci, presumably at or near the *SCN1A* loci. If we observe saRNA–AGO2 complexes accumulating in the nucleus (especially in the vicinity of chromatin), it would strongly support the model that the saRNA guides AGO2 to the genomic target. This method circumvents some of the sensitivity issues of ChIP by relying on microscopic detection of single molecules. Co-localization analysis could be quantified to show that saRNA clusters overlap with AGO2 and with nuclear markers of active transcription (*e.g.*, RNA polymerase II or activating histone marks). Although to our knowledge RNA-FISH has not yet been widely applied to saRNA studies, it is conceptually similar to how one might track nuclear localization of long non-coding RNAs. This experiment would bolster confidence that saRNA engages the RNAa machinery by physically occupying the target site along with AGO2. In Chapter 2, we show that labelled saRNA mainly accumulates perinuclear in the endosome, yet a low percentage may show presence in the nucleus.

Promoter–Reporter Gene Assays

To establish a direct causal link between our saRNA and *SCN1A* promoter activation, a luciferase or GFP reporter assay could be employed. In this approach, the *SCN1A* promoter (including the saRNA target site) would be cloned upstream of a reporter gene. Upon co-transfection of this construct with the candidate saRNA, any sequence-specific activation of the promoter would result in increased reporter expression, which could be quantified by luminescence or fluorescence. While such reporter assays are widely used in promoter biology to test transcriptional regulation [11], [12], they have not yet been systematically applied in the saRNA field. Instead, saRNA studies have primarily validated on the endogenous locus, demonstrating that promoter activation is highly sequence dependent. For example, Meng et al. (2016) showed that mutating the seed region of an saRNA or its corresponding genomic target abolished transcriptional activation [13], confirming that base-pairing to the promoter sequence is essential for RNAa. Likewise, Portnoy et al. (2016) demonstrated that altering the target site in the *CDKN1A* (P21) promoter eliminated saRNA-mediated upregulation [10]. These findings strongly support the rationale for a promoter–reporter assay: by introducing wild-type or mutated *SCN1A* promoter sequences upstream of a reporter, one could directly test whether the saRNA acts through the intended promoter site. A positive readout – increased reporter activity with the wild-type promoter but not with the mutated site – would provide compelling evidence that the saRNA functions via on-target promoter engagement rather than indirect pathways.

Biotinylated saRNA Pull-down (ChIRP) and Complex Analysis

Our AGO2-ChIP experiments may have failed due to weak antibody binding, limiting our ability to detect promoter-associated AGO2. An alternative strategy would be to use a biotin-tagged saRNA duplex for transfection, followed by streptavidin pull-down to capture the saRNA and its interacting partners. With formaldehyde crosslinking prior to lysis, this approach could retrieve both genomic DNA fragments and associated proteins in close proximity to the saRNA. Sequencing of the pulled-down DNA, similar to ChIP-seq, would then reveal whether the *SCN1A* promoter is specifically enriched, providing direct evidence of physical association.

In parallel, mass spectrometry of the co-purified proteins could identify factors involved in saRNA-mediated transcriptional activation. Previous work using this strategy on P21 and E-cadherin saRNAs identified components of the RITA complex, including AGO2, RHA, and CTR9 [10]. Detecting similar factors in our system would strongly support a conserved mechanism of RNA-guided transcriptional activation at the *SCN1A* promoter.

Implementing these experimental strategies will help to overcome current limitations in our mechanistic understanding of saRNA-mediated *SCN1A* activation. By systematically applying visualization techniques, promoter–reporter assays, and pull-down approaches, we aim to clarify the sequence of molecular events, from cellular uptake and AGO2 association, to nuclear localization and recruitment of transcriptional activation complexes at the *SCN1A* promoter, ultimately resulting in enhanced RNA Polymerase II activity. Mechanistic

validation is essential not only for advancing basic scientific knowledge but also for informing the development of saRNA-based therapeutics. Establishing saRNAs act through a specific, on-target mechanism will be critical for demonstrating both specificity and safety. Furthermore, detailed mechanistic insight may reveal opportunities to optimize saRNA design, for example by modifying sequences to improve recruitment of essential co-factors. In summary, rigorous mechanistic studies will provide a robust foundation for the rational development and optimization of saRNA-mediated gene activation as a therapeutic strategy.

6.3 Deciphering Why Only Some saRNAs Are Effective

A striking outcome of our *SCN1A* experiments was the variable efficacy observed among different saRNA candidates. We designed multiple saRNA candidates against the *SCN1A* promoter, yet only a three induced robust gene activation; others had minimal or no effect despite targeting the same gene and *SCN1A-7* resulted in a structural downregulation. This variability raises a critical question: what determines whether a given saRNA will work? Understanding the answer is key to improving saRNA design – currently a somewhat empirical process – and could shed light on fundamental aspects of transcriptional regulation. Based on our findings and existing research, several factors likely contribute to the differential efficacy of saRNAs:

Sequence Features and Seed Region Composition

The nucleotide sequence of the saRNA (particularly in the seed region, the 5′ end of the guide strand that typically mediates target recognition) can dramatically influence activity. Just as with siRNAs, certain sequence compositions may favor Argonaute loading or stable target binding. However, unlike siRNAs (where most sequences targeting an mRNA will silence to some degree), saRNAs show a narrower “hit rate” in practice [14]. For instance, GC content or specific motifs might affect the stability of the RNA–DNA hybrid or the recruitment of particular co-factors. A 2016 study by Meng et al. demonstrated that saRNAs bind to genomic DNA in a seed-dependent manner – mutating a few bases in the target site’s seed region abolished activation, underscoring that sequence-specific base-pairing to DNA is required [13]. It follows that only sequences which form the right RNA:DNA duplex (without inducing secondary structures or off-target binding) will trigger RNAa. Moreover, certain motifs in the saRNA might interact with DNA-binding proteins: for example, a stretch of guanines could potentially form G-quadruplex structures or attract unintended factors, impeding activation. In silico analyses and machine learning on large saRNA datasets may help pinpoint sequence patterns correlating with strong activation versus failure. Until then, empirical testing of multiple candidates remains important.

Epigenomic and Chromatin Context

The state of the target promoter is likely a major determinant of saRNA efficacy. Heterochromatic or closed chromatin characterized by dense nucleosome packing, DNA

methylation, and repressive histone marks can impede access of the AGO2–saRNA complex, whereas euchromatic, permissive, or "poised" promoters may be more susceptible to activation (*e.g.*, nucleosome-depleted regions) [15]. Although direct examples of saRNA altering chromatin accessibility are scarce, one study showed that saRNA binding to its DNA target can affect nucleosome positioning and potentially chromatin remodeling [16].

When saRNAs mis-target or engage in a repressive chromatin context, gene silencing (rather than activation) can ensue, likely due to recruitment of repressive complexes, but this depends heavily on the local chromatin environment. While no specific saRNA study has documented this phenomenon in detail, the principle aligns with general transcriptional regulation paradigms.

saRNA-mediated activation often shows a delayed onset and long-lasting effects (lasting several days), compatible with the notion that the saRNA must overcome chromatin barriers and induce epigenetic changes [10]. Because of this, variability in promoter susceptibility across candidate target sites, driven by differences in nucleosome occupancy, histone modifications, or non-coding RNA interference, may explain why some saRNAs succeed while others fail.

To address these issues, future studies should incorporate assays of chromatin accessibility and state, such as ATAC-seq for open chromatin or ChIP-seq for active (*e.g.*, H3K4me3) or repressive histone marks (*e.g.*, H3K27me3). These data would inform the prediction and selection of optimal saRNA target sites by highlighting accessible regions receptive to regulatory intervention.

Target Site Location and Orientation

The location of an saRNA binding site relative to the transcription start site (TSS) and the strand orientation it targets are both key determinants of RNAa efficacy. Studies noted that only saRNAs targeting promoters within a few hundred base pairs of the TSS activated gene expression effectively, while more distal sites were less potent or inactive [17]. Likewise, targeting too close to the TSS may sterically hinder assembly of the core transcriptional machinery.

In our design, we tiled multiple saRNA candidates across the *SCN1A* promoter, potentially capturing an optimal "window", not too proximal to block initiation, yet close enough to enable recruitment of AGO2 and the RITA complex.

Strand orientation also plays a critical role. While saRNAs are typically designed to target double-stranded genomic DNA, there is emerging evidence that some miRNAs can target promoter-associated noncoding RNAs (ncRNAs) to mediate transcriptional activation. For example, miR-373 can interact with promoter-associated RNA at the E-cadherin (*CDH1*) locus and activate transcription, a phenomenon termed miRNA activation or miRNAa [18]. Likewise, a promoter-proximal antisense transcript at *CDH1* has been shown to guide transcriptional repression via AGO-mediated recruitment of chromatin modifiers in other settings [19]. Thus, the presence and orientation of these transcripts could influence saRNA effectiveness: an saRNA complementary to a promoter-associated lncRNA (antisense strand)

may act via an RNA:RNA hybrid mechanism, whereas one targeting the sense (coding) DNA strand may act through direct RNA-DNA pairing.

Given these complexities, and the still-limited understanding of RNAa mechanisms, empirical screening of multiple saRNAs across promoter regions remains necessary. While the landmark mechanistic study by Portnoy et al. (2016) described AGO2, RHA, CTR9, and the RITA complex as central to RNAa, it did not resolve why some saRNA sequences succeed while others fail [10].

In future work, we plan to systematically tile saRNAs across broader promoter regions and evaluate their activity using reporter assays or high-resolution transcriptomics. This will generate a structured and quantitative dataset of saRNA sequences and their activation outcomes, a prerequisite for building robust predictive models. Using machine learning approaches such as Random Forest or Gradient Boosting, we can analyze complex, nonlinear relationships between saRNA features and functional activity. Relevant features may include primary sequence composition (*e.g.*, k-mer frequencies, seed-region content), thermodynamic properties, predicted RNA–DNA hybridization energy, nucleosome occupancy, local histone marks, and distance to regulatory elements.

Importantly, this framework will not be limited to *SCN1A*. By incorporating data from saRNAs targeting multiple genes, the aim of future studies will be to identify generalizable sequence and chromatin signatures associated with activation, moving toward a more universal design tool. Such predictive models could prioritize high-likelihood candidate saRNAs before experimental testing, thereby reducing empirical screening effort and accelerating therapeutic development. Until sufficient training data is available, however, empirical identification and functional validation of active saRNAs will remain essential for both mechanistic understanding and therapeutic application.

6.4 Enhancing saRNA Delivery and Functional Efficacy in Neuronal Systems

Translating the saRNA approach into a viable therapy for Dravet syndrome (or related monogenic epilepsies) not only requires potent molecules but also a delivery strategy that ensures uptake into the most relevant cells, inhibitory interneurons in the human brain. This challenge was evident in our own work: differentiated AF22 neurons, like most mature post-mitotic neuronal models, proved highly refractory to conventional lipid-based transfection. Early attempts using Lipofectamine RNAiMAX, a cationic lipid widely used for siRNA delivery, yielded only modest uptake and signs of cytotoxicity. This aligns with broader observations: while lipofection is adequate in immortalized dividing cell lines, neurons resist efficient uptake and are unusually sensitive to toxicity [20], [21]. In a direct comparison, Khare

et al. showed that Lipofectamine achieved some siRNA uptake in primary neurons, but at the cost of viability, whereas optimized lipid nanoparticle (LNP) formulations enabled comparable delivery with far lower toxicity. These findings support our interpretation that suboptimal delivery likely contributed to the modest functional outcomes in AF22 neurons, despite the use of an otherwise highly active *SCN1A*-saRNA (as determined in non-neuronal systems).

Lipid nanoparticles (LNPs) therefore represent a more suitable platform for neuronal RNA delivery. They have already proven successful in RNA-based medicines, from siRNA drugs like patisiran to mRNA vaccines. Their advantages are multifaceted: LNPs protect RNA from nuclease degradation, promote endosomal escape, and distribute broadly in tissues when optimized for biodistribution [22], [23].

In our experiments, we observed that one LNP formulation ("LNP-LUNG") at a high dose (250 nM) achieved both uptake and detectable *SCN1A* upregulation, whereas lower doses (50 nM) of other formulations were ineffective. This dose dependency highlights the need for careful optimization of both formulation and concentration. Neurons likely require higher effective intracellular RNA concentrations due to their low endocytotic activity and large cytoplasmic volumes compared to AF22 NES and HeLa. However, dosing must be carefully titrated to avoid lipid or RNA-induced stress. As a next step, we plan to systematically compare LNP formulations across a range of doses (50–250 nM) while monitoring cell health using assays such as LDH release, morphological integrity, and electrophysiological function on MEAs. Additionally, LNPs can be surface-functionalized with targeting ligands, such as peptides, to preferentially direct uptake into neurons. For instance, Han et al. developed an RVG29 peptide–functionalized LNP that, following systemic administration, significantly enhanced neuronal transfection via interaction with neuronal receptors, while minimizing uptake by non-target cells. This strategy substantially improved delivery to brain neurons compared to non-targeted LNPs [24].

Viral vectors, particularly adeno-associated viruses (AAVs), remain attractive for CNS delivery because of their safety profile and capacity for durable expression. In our hands, expressing saRNAs as short hairpins from a U6/Pol III cassette did not yield robust RNAa activity. A more promising alternative is to embed saRNAs in an artificial miRNA scaffold transcribed by Pol II, mimicking endogenous pri-miRNA processing. This approach has strong precedent in the CNS: AAV5-delivered artificial miRNAs targeting HTT (AMT-130) have shown efficient target lowering in preclinical models and are in Phase I/II trials for Huntington's disease, demonstrating durable small-RNA expression in brain tissue [25]. Likewise, AAV9 vectors encoding dual synthetic miRNAs against GRIK2 reduce GluK2 and seizures in preclinical epilepsy models, supporting the general viability of AAV-miRNA designs for neuronal targets [26].

Importantly, AGO2:miRNA complexes can access the nucleus through endogenous pathways, with Importin-8 (IPO8) implicated in their nuclear transport. This suggests that saRNAs delivered via Pol II–driven miRNA scaffolds can, at least in part, engage promoter-proximal RNAa mechanisms without additional engineering. To further bias nuclear activity in a therapeutically realistic manner, rational design of the hairpin structure can be leveraged,

for example, adjusting terminal loop length or introducing subtle internal bulges to fine-tune Exportin-5–mediated export and processing kinetics [27], [28]. In addition, incorporating nuclear-retention motifs such as the MALAT1 triple-helix ENE offers an exploratory route to prolong nuclear dwell time of the pri-transcript [29], [30]. Together, these strategies support prioritizing Pol II miRNA-scaffold AAVs for saRNA expression, while rationally refining hairpin and transcript design to increase nuclear availability and enhance engagement of promoter-proximal RNAa in human neurons.

A particularly promising translational avenue is direct CNS administration using an Ommaya reservoir, a device already in clinical use for intrathecal chemotherapy and antisense oligonucleotide (ASO) delivery (*e.g.*, nusinersen for spinal muscular atrophy) [31]. Preclinical studies in sheep using similar reservoir-based strategies for siRNA delivery have demonstrated broad CNS distribution and excellent tolerability [32]. Combining an Ommaya reservoir with saRNA–LNP formulations could provide a safe and flexible route to bypass the blood–brain barrier (BBB), enabling repeated intraventricular dosing during preclinical optimization.

Another axis of improvement lies in chemical modification of saRNAs. Therapeutic oligonucleotides (e.g., siRNAs, ASOs) routinely use modifications such as 2′-O-methyl, 2′-fluoro ribose, or phosphorothioate linkages to increase nuclease resistance and improve pharmacokinetics. Similar modifications have been shown to be compatible with RNAa: for instance, Place et al. (2008) demonstrated that 2′-O-methyl and 2′-F substitutions preserved gene activation in saRNAs while reducing immune activation [18]. In our own experiments, we also incorporated several of these chemical modifications, including 2′-O-methyl substitutions, into our saRNA designs. However, under our experimental conditions, these modifications did not result in noticeable differences in efficacy compared to unmodified saRNAs.

Another important consideration for boosting specificity and efficacy of saRNAs is strand selection. In RNA interference systems, strand incorporation into Argonaute is strongly influenced by thermodynamic asymmetry of the duplex: the strand with less stable base-pairing at its 5′ end is preferentially loaded as the guide strand, while the opposite strand is degraded [33]. Modifications to reinforce this bias, such as introducing intentional mismatches in the passenger strand seed region, have been shown to reduce off-target silencing and increase the likelihood of correct guide strand incorporation [34]. While strategies like bulky 5′ end blocking of the passenger strand have been proposed conceptually, they are not yet experimentally validated in saRNA systems. Future work will be required to determine whether such approaches can be systematically applied to RNAa.

Beyond stability and strand selection, lipid conjugates represent a promising means to enhance saRNA delivery. Among them, cholesterol-conjugated oligonucleotides are well-studied: attaching cholesterol to the 3′ end of an siRNA or ASO facilitates cellular uptake by exploiting lipoprotein-mediated pathways (*e.g.*, LDLR- and SR-BI–dependent endocytosis) and improves biodistribution, including to the central nervous system [35]. This strategy has already been used in preclinical siRNA therapeutics to enhance uptake into neurons and hepatocytes, highlighting its translational potential. Applying cholesterol or other lipid

conjugates to saRNAs could likewise improve their pharmacokinetic profile, bioavailability, and CNS penetration, especially when combined with direct delivery routes such as intrathecal administration via an Ommaya reservoir.

It is critical to emphasize that our *SCN1A*-directed saRNAs were designed against the human promoter and do not bind the mouse *Scn1a* locus. As a result, rodent efficacy studies cannot test RNAa-mediated upregulation of *Scn1a* with these sequences. That said, delivery technologies, including lipid nanoparticles (LNPs), AAV packaging/administration paradigms, and ventricular access systems (*e.g.*, Ommaya reservoirs), can still be evaluated in vivo for biodistribution, safety, and pharmacokinetics in appropriate animal models (small or large), even when the oligonucleotide is non-functional with respect to the rodent genome. For example, comparative route-of-administration studies in sheep showed that CSF delivery (intracerebroventricular or intrathecal catheter to cisterna magna) enables widespread CNS distribution of chemically engineered siRNAs, and that repeated dosing via an implanted reservoir enhances accumulation in deep structures, establishing principles that generalize to other oligonucleotide modalities (and to reservoir-based dosing) independent of target sequence identity.

In conclusion, solving the delivery challenge is central to advancing saRNA therapy for *SCN1A*-haploinsufficient epilepsies. A multi-pronged path that can provide a realistic roadmap includes: (i) LNP formulation/dose optimization, (ii) neuron-targeted surface functionalization, (iii) AAV vectors for durable intraneuronal expression, (iv) Ommaya-assisted CNS delivery to bypass the blood–brain barrier for repeated dosing, and (v) judicious chemical/architectural modification of saRNAs. While rodent efficacy is precluded by species-specificity of our current saRNAs, human iPSC-derived neurons, MEA networks, and organoids constitute the appropriate efficacy test beds; in vivo animal studies remain valuable for biodistribution/toxicity/persistence of the delivery platforms themselves.

6.5 Alternative and Complementary Therapeutic Approaches

While our focus has been on saRNA-mediated activation, it is important to situate this strategy among other emerging approaches to upregulate gene expression. Two notable alternatives for *SCN1A* augmentation are CRISPR-based gene activation and antisense oligonucleotides (ASOs) designed for *SCN1A*. Each approach has its merits and challenges, and they are not mutually exclusive with the saRNA strategy – in fact, they could be combined for synergistic benefit.

CRISPR/dCas9 Activation (CRISPRa): A powerful method to upregulate genes is to use an inactivated Cas9 (dCas9) fused to transcriptional activator domains (such as VP64, p300 core, or the SunTag system) and guide RNAs targeting the gene's promoter or enhancer. Proof-of-concept of CRISPRa for *Scn1a* has been demonstrated in Dravet syndrome mouse models. Yamagata et al. targeted *Scn1a* in inhibitory interneurons using a dCas9-VP64 system and achieved significant improvements: increased *Scn1a* expression, amelioration of seizures, and even normalization of behavioral phenotypes in Dravet mice [36]. Colasante and colleagues

similarly showed that AAV-delivered dCas9 activators could restore interneuron excitability and attenuate seizures in *Scn1a*+/− mice [37]. These studies confirm that upregulating *SCN1A*/*Scn1a* in the relevant cells is a viable therapeutic strategy. However, a practical challenge for CRISPRa is delivery size: the dCas9 protein (~4.2 kb coding sequence for SpCas9, not including activator domains) plus the requisite guide RNAs and regulatory elements come close to or exceed the cargo limits of AAV vectors. In addition, repeated dosing of CRISPRa-based therapies would likely necessitate immunosuppression, as the bacterial origin of Cas9 renders it immunogenic in humans. This immunogenicity poses a significant barrier to clinical translation, making it unlikely that CRISPRa can be safely and repeatedly administered in patients without unacceptable immune responses [38].

In the aforementioned mouse studies, researchers had to use dual-AAV systems – one AAV for the dCas9-VP64 and another for the guide RNA and a cell-specific regulator – and even then, co-transduction efficiency was limited (~20% of target neurons received both vectors) [37]. They suggested that smaller Cas9 orthologs (*e.g.* SaCas9 or others) could be explored to fit everything in one AAV. This delivery hurdle underscores a key advantage of saRNAs: their small size. An saRNA duplex is only ~21 base pairs (~14 kDa if one considers molecular weight, or ~60 nt in a single-stranded expressed form), which can be delivered as a simple synthetic oligo or easily packaged into a single AAV (for example, as a short hairpin under a U6 promoter, which would be <300 bp). Thus, saRNAs impose minimal cargo demand on delivery vehicles. For instance, an AAV could potentially carry multiple U6-driven saRNAs in one vector, or an saRNA could even be co-delivered with a dCas9 activator if one wanted a combined system. Moreover, as discussed, saRNAs can be chemically synthesized and administered without any vector at all (like traditional ASO/siRNA drugs), which avoids the complexity of gene therapy vectors. In summary, CRISPRa is a potent approach and has provided important validation in vivo, but when considering translation, the simplicity and small size of saRNAs make them attractive, especially for disorders requiring widespread delivery to the brain.

Antisense Oligonucleotide (ASO) – TANGO approach: A different strategy to boost *SCN1A* output is employed by Stoke Therapeutics' ASO called STK-001. This ASO operates via Targeted Augmentation of Nuclear Gene Output (TANGO), which is a mechanism that exploits naturally occurring non-productive mRNA transcripts. In the case of *SCN1A*, it's been found that some transcripts from the healthy allele are prematurely truncated or alternatively spliced in a way that subjects them to nonsense-mediated decay. STK-001 is designed to bind an intronic region of the *SCN1A* pre-mRNA, modifying splicing such that more full-length, productive mRNA is made. In essence, it helps the cell produce more $Na_v1.1$ protein from the existing gene, without altering the gene's DNA. This ASO is currently in clinical trials for Dravet syndrome (the MONARCH and ADMIRAL studies) and represents one of the first gene-upregulation ASO therapies for the brain. Preclinical results have been very promising: in *Scn1a*+/− Dravet mice, STK-001 treatment led to increased *Scn1a* mRNA, increased $Na_v1.1$ protein in the brain, a significant reduction in spontaneous seizures, and improved survival [39]. Notably, recent work also demonstrated that STK-001 rescued the intrinsic excitability of parvalbumin-positive inhibitory interneurons in Dravet mice,

restoring their function to near wild-type levels [39]. This directly validates that augmenting *SCN1A* can reverse the key physiological defect (interneuron dysfunction) in this disease. By early 2025, interim clinical data indicated that STK-001 has been well-tolerated and showed trends of seizure reduction, although full efficacy results are pending.

The TANGO ASO approach and our saRNA approach are conceptually similar in ultimate goal – both seek to elevate the patient's own *SCN1A* expression – but differ in mechanism. The ASO works at the post-transcriptional level (altering RNA processing in the nucleus), whereas the saRNA works at the transcriptional level (enhancing gene promoter activity). Because of these differences, a fascinating possibility is to combine the two approaches for a synergistic effect. In a potential combination therapy, an saRNA could be delivered (by LNP or AAV) to Dravet patient neurons to boost the transcription of *SCN1A*, thereby producing more pre-mRNA, while an ASO like STK-001 could be given to ensure that a higher fraction of those transcripts are spliced/processed into full-length, translation-competent mRNA. In other words, the saRNA increases the "supply" of *SCN1A* transcripts from the gene, and the ASO increases the "yield" of usable transcripts that escape nonsense-mediated decay. Used together, the output of $Na_v1.1$ protein could be greater than with either alone, potentially allowing dose reductions of each for safety. This combined approach could also mitigate variability: if a certain cell type responds better to one mechanism than the other, having both increases the chance that *SCN1A* levels rise sufficiently. Of course, combination therapy would add complexity in clinical development (two molecules instead of one) but given that both saRNAs and ASOs are modular oligonucleotide drugs, co-delivery might be feasible (for example, co-formulating them in the same intrathecal injection). It will be interesting in future preclinical studies to test an saRNA plus a TANGO ASO together in Dravet models to see if additive or synergistic benefits are observed.

In considering all these approaches – saRNA, CRISPRa, TANGO ASO – it is worth highlighting that each has unique advantages. SaRNA therapy leverages an endogenous epigenetic mechanism and is very small in size, making it attractive for delivery and repeated dosing. CRISPRa offers a potentially larger magnitude of activation and very specific DNA targeting, with the trade-off of vector complexity and immunogenicity of the bacterial Cas9 protein. ASO (TANGO) directly fixes an inefficiency in gene expression and has the advantage of already being in human trials, but it relies on the presence of a targetable non-productive transcript. Notably, these approaches are not mutually exclusive and could even target different cell populations: for example, an AAV-CRISPRa might be engineered to target mainly interneurons (with a cell-specific promoter), while an saRNA or ASO delivered broadly could affect both interneurons and excitatory neurons. There may also be scenario-dependent reasons to choose one over the other (*e.g.*, if a patient's mutation is such that *SCN1A* transcripts are very low, a transcriptional boost might be most crucial, whereas if transcripts are made but degraded, an ASO is crucial). Ultimately, the existence of multiple strategies is good news for Dravet syndrome and similar epilepsies, as it increases the likelihood of a successful gene-modulating therapy and allows tailoring to specific needs.

From a development and delivery perspective, saRNAs hold a clear edge in simplicity and size, as mentioned. For instance, where CRISPRa required two AAVs in mice [37], a single

AAV could easily carry an saRNA expression cassette. And where AAV-CRISPRa is essentially a permanent (irreversible) gene therapy once delivered, saRNAs (and ASOs) can be more finely titrated – dosing can be adjusted or stopped if needed, providing a safety valve. On the other hand, CRISPRa might achieve more continuous expression once the vector is delivered, whereas chemically delivered saRNAs or ASOs might require periodic re-dosing (intrathecal injections every few months, for example). As the field moves forward, it will be informative to compare these modalities head-to-head in model systems.

In conclusion, we view saRNA-based activation as part of a broader toolkit of gene upregulation therapies. The success of STK-001 in animal models and the feasibility of CRISPRa in mice provide strong validation that increasing *SCN1A* expression is a disease-modifying strategy for Dravet syndrome [36], [39]. Our work adds to this landscape by providing a new method (RNAa) to achieve that increase, one that could offer practical advantages in delivery and combination potential. The small size of saRNAs and their ability to work synergistically with other approaches (like ASOs) position them as a compelling option in the future gene therapy arsenal. As we advance, lessons from CRISPRa and ASO trials will inform saRNA development (for example, regarding target engagement, dosing, and long-term effects), and conversely, a successful saRNA could complement those therapies or provide an alternative for patients who may not respond to one approach. The ultimate goal is to maximize $Na_v1.1$ restoration in the right cells at the right time and having multiple strategies at hand increases our chances of achieving that safely and effectively.

6.6 Conclusions and Future Perspectives

Through the work presented in this thesis, we have advanced the understanding of small activating RNAs (saRNAs) as a therapeutic modality for monogenic epilepsies. Focusing on *SCN1A* haploinsufficiency in Dravet syndrome, we demonstrated that saRNAs can be engineered to upregulate a disease-critical gene, thereby addressing a fundamental molecular deficit. This constitutes an important proof-of-concept: it shows that epigenetic upregulation via RNA is achievable for a neuronal gene, expanding the therapeutic repertoire for haploinsufficiency disorders beyond the traditional strategies of gene replacement and pharmacological modulation. Our findings contribute to the broader field of RNA therapeutics by highlighting RNA activation (RNAa) in the context of neurological disease, an area where RNAa has so far been studied mainly in cancer and metabolic conditions [10], [40]. Importantly, realizing this potential will require a multi-faceted and integrated approach, addressing the mechanistic, translational, and delivery challenges discussed in this chapter.

Our findings provide a blueprint for next steps. The most urgent need is to test whether molecular *SCN1A* upregulation translates into functional rescue. Human disease-relevant models, such as patient-derived iPSC neurons, brain organoids, and multi-electrode array

(MEA) networks, recapitulate Dravet-relevant hyperexcitability and temperature-sensitive phenotypes [3]. These systems will be indispensable for testing whether saRNA treatment restores inhibitory interneuron excitability and reduces network bursting. Unlike siRNA or ASO-based knockdown, where rodent homologs can be studied directly, our human-specific saRNAs do not bind the murine *Scn1a* locus. Thus, efficacy testing in conventional Dravet mouse models is not possible. Instead, rodent studies can be leveraged to evaluate delivery feasibility, biodistribution, and toxicity of lipid nanoparticles (LNPs), AAV vectors, or Ommaya-assisted CSF administration, while functional efficacy must be demonstrated in human-based neuronal platforms.

Delivery remains the largest barrier to translation. Our experiments confirmed that differentiated neurons (e.g., AF22) are resistant to standard lipofection with cationic lipids, consistent with literature reporting poor uptake and cytotoxicity in post-mitotic neurons [20]. By contrast, LNPs, already validated in siRNA drugs and mRNA vaccines, provide a more promising avenue, offering encapsulation, protection, and improved endosomal escape [41]. Optimizing formulation and dose is crucial. Targeting can be further improved by surface functionalization (peptides/antibodies/sugars) to bias uptake toward neuronal (or interneuron) populations; both primary studies of peptide-functionalized LNPs and recent reviews support that ligand decoration can enhance brain delivery and reduce off-target exposure [42]. Moreover, for clinical translation, bypassing the BBB is often necessary. Direct intrathecal administration, through repeated lumbar punctures or an Ommaya reservoir (ventricular access device) is an established, clinically used route for repeated CNS dosing, and consensus/clinical reviews document its safety and practical considerations, making it a realistic platform for serial administration of saRNA-loaded nanoparticles.

In parallel, AAV vectors represent a complementary strategy for long-term expression. AAV-mediated delivery of small RNAs has already shown efficacy in neurological disease. For saRNAs, an analogous approach would be to embed the guide sequence in short hairpin or microRNA scaffolds driven by U6 or cell-type-specific Pol II promoters. Identifying the correct hairpin structures that promote efficient Argonaute loading and nuclear shuttling will be critical, since the saRNA must localize to the promoter DNA to induce RNAa. Argonaute itself is known to shuttle to the nucleus with bound small RNAs [43], but optimizing nuclear enrichment through scaffold design (*e.g.*, snoRNA frameworks) could further improve efficacy.

Chemical modifications offer an additional lever for safety and performance. Incorporating 2′-O-methyl or 2′-fluoro ribose nucleotides, or phosphorothioate linkages, enhances stability and reduces immunogenicity without abolishing activity [14]. Moreover, strand-selection strategies, such as introducing seed mismatches in the passenger strand, can bias AGO2 loading toward the intended guide strand [10]. Finally, conjugates such as cholesterol moieties at the 3′ end can improve biodistribution and CNS penetration by leveraging lipoprotein uptake pathways [35]. It is worth considering that combination approaches could further enhance therapeutic impact.

Despite these advances, safety concerns must remain central. Overexpression of $Na_v1.1$ above physiological levels could theoretically increase interneuron firing excessively, though in

haploinsufficiency our goal is restoration rather than excess. Off-target transcriptional activation remains a risk; while our RNA-seq analysis did not reveal major non-specific upregulation, extensive validation across cell types will be needed. Immune activation, particularly via TLR7/8, must also be carefully monitored, and chemical modification strategies incorporated to mitigate such responses.

In conclusion, the work described in this thesis has opened a new avenue for treating Dravet syndrome by pioneering the use of small RNAs to activate gene expression. We have demonstrated proof-of-concept at the molecular level and identified both the promise and the obstacles of this approach. Although conventional rodent models cannot be used for efficacy testing because our saRNAs are human-specific, translational evaluation remains feasible through human neuronal cultures, organoids, and advanced delivery platforms.

By integrating mechanistic validation, delivery innovation, and translational modeling, saRNAs could emerge as a novel class of precision therapeutics for haploinsufficiency disorders. Instead of replacing genes or silencing transcripts, saRNAs gently "nudge" the genome to produce more of what is already there, a unique therapeutic logic with wide potential. The next steps, refining delivery, optimizing nuclear-targeting hairpin scaffolds, and demonstrating functional rescue in patient-derived neurons, are challenging but within reach. With sustained interdisciplinary effort across RNA biology, neurogenetics, and therapeutic delivery, saRNAs may one day mature into a powerful treatment option for Dravet syndrome and related epileptic encephalopathies.

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Nederlandse wetenschappelijke samenvatting

Achtergrond en doel

Ontwikkelings- en epileptische encefalopathieën vormen een ernstige groep vroeg beginnende epilepsieën, waarbij zowel de onderliggende aandoening als de epileptische activiteit bijdragen aan ontwikkelingsachterstand, cognitieve problemen en moeilijk behandelbare aanvallen. Een belangrijk voorbeeld is het Dravet-syndroom. Bij de meeste patiënten wordt dit veroorzaakt door een heterozygote loss-of-function-variant in *SCN1A*. Daardoor is slechts één goed functionerende genkopie beschikbaar en ontstaat haplo-insufficiëntie: de hoeveelheid $Na_v1.1$-eiwit is onvoldoende om normale remmende neurotransmissie te ondersteunen. $Na_v1.1$ is vooral belangrijk in inhiberende interneuronen. Verminderde functie van deze cellen verstoort de balans tussen excitatie en inhibitie en draagt bij aan epileptische netwerkactiviteit.

De huidige behandeling bestaat voornamelijk uit anti-aanvalsmedicatie. Deze kan de aanvalsfrequentie verminderen, maar corrigeert het onderliggende tekort aan *SCN1A*-expressie niet en heeft vaak onvoldoende effect op cognitieve en gedragsmatige comorbiditeit. Daarom worden therapieën ontwikkeld die de expressie van het intacte *SCN1A*-allel verhogen. Bekende voorbeelden zijn antisense-oligonucleotiden die RNA-verwerking beïnvloeden en CRISPR-activatie, waarbij een inactief Cas9-eiwit met transcriptie-activerende domeinen naar een promotor wordt geleid. Beide strategieën ondersteunen het principe dat verhoging van *SCN1A* therapeutisch relevant kan zijn, maar kennen ook beperkingen in toepasbaarheid, grootte, immunogeniciteit of aflevering.

Dit proefschrift onderzocht small activating RNAs (saRNA's) als alternatieve strategie. saRNA's zijn korte dubbelstrengs RNA-moleculen die worden ontworpen tegen promoter-geassocieerde sequenties en de transcriptie van een doelgen kunnen verhogen. Verondersteld wordt dat één streng van het saRNA wordt opgenomen door Argonaute 2 (AGO2), waarna een nucleair complex wordt gevormd dat transcriptie-activerende eiwitten naar de doelwitlocatie brengt. Het exacte mechanisme is echter nog onvoldoende opgehelderd, vooral in neuronale cellen. De centrale vragen van dit proefschrift waren daarom: kunnen saRNA's *SCN1A* in menselijke neurale cellen verhogen, welke moleculaire factoren bepalen deze activatie, kan de strategie worden uitgebreid naar andere haplo-insufficiënte epilepsiegenen en welke afleveringsbarrières moeten worden opgelost voor toepassing in neuronen?

Identificatie van *SCN1A*-activerende saRNA's

In hoofdstuk 2 werd een screeningspanel van tien saRNA's gericht tegen promoterregio's van *SCN1A* onderzocht (SCN1A-6 tot en met SCN1A-15). De kandidaten werden getest in HeLa-cellen, waarna geselecteerde kandidaten werden onderzocht in ongedifferentieerde AF22-neuro-epitheliale stamcellen en gedifferentieerde AF22-neuronen. Deze combinatie maakte snelle screening in goed transfecteerbare cellen mogelijk, gevolgd door validatie in een menselijk neuronaal model.

Meerdere saRNA's verhoogden *SCN1A*-expressie. Vooral *SCN1A*-10 liet een reproduceerbare toename zien in HeLa- en AF22 NES-cellen. Ook *SCN1A*-12 en *SCN1A*-14 waren actief. De verhoging werd bevestigd met qPCR en RNA-sequencing. Voor *SCN1A*-10 werd daarnaast een toename van $Na_v1.1$-eiwit gemeten, waarmee werd aangetoond dat de transcriptieverhoging kon worden vertaald naar het eiwitniveau. In gedifferentieerde AF22-neuronen was de verhoging kleiner maar reproduceerbaar. Deze resultaten vormen een eerste moleculair proof-of-concept dat promotergerichte saRNA's een neurologisch relevant haplo-insufficiënt gen kunnen activeren.

Functionele calcium-imaging liet na behandeling met *SCN1A*-10 slechts beperkte veranderingen zien. Het aantal calciumtransiënten per actieve cel nam licht af, terwijl het

percentage actieve cellen niet significant veranderde. *SCN1A*-knockdown had duidelijkere effecten op meerdere calciumparameters, wat liet zien dat de assay gevoelig was voor veranderingen in *SCN1A*. De bescheiden respons op overexpressie is waarschijnlijk mede het gevolg van het gebruikte wildtype, gemengde neurale model. In deze culturen wordt netwerkactiviteit grotendeels door excitatoire neuronen bepaald en ontbreekt een ziekte-gerelateerde uitgangssituatie met *SCN1A*-haplo-insufficiëntie. Daarom kan uit deze experimenten nog niet worden geconcludeerd dat de moleculaire verhoging tot functioneel herstel leidt. Daarvoor zijn patiëntafgeleide neuronen, isogene *SCN1A*-haplo-insufficiënte modellen, organoïden of multi-electrode-arraynetwerken noodzakelijk.

Onderzoek naar het werkingsmechanisme

Hoofdstuk 3 onderzocht of *SCN1A*-activatie aansluit bij het voorgestelde AGO2-afhankelijke RNA-activatiemechanisme. siRNA-gemedieerde verlaging van het AGO2-eiwit verminderde de activatie door *SCN1A*-10. Hetzelfde werd gezien voor een saRNA tegen *CDKN1A*/p21, dat als positieve controle werd gebruikt. Deze resultaten ondersteunen een functionele associatie tussen AGO2-beschikbaarheid en saRNA-activiteit. Een knockdown-experiment bewijst echter niet dat AGO2 rechtstreeks op de *SCN1A*-promotor bindt.

Om directe doelwitbinding te testen werd AGO2-ChIP-qPCR uitgevoerd. Ondanks meerdere experimenten en het gebruik van twee antilichamen werd geen overtuigende AGO2-verrijking gevonden op de *SCN1A*-promotor. Ook bij de p21-controle werd geen duidelijke verrijking gemeten, terwijl de positieve chromatinecontrole wel werkte. De combinatie van een functioneel AGO2-effect en een negatieve ChIP kan verschillende verklaringen hebben. AGO2-binding kan tijdelijk of zwak zijn, via een RNA-intermediair verlopen of niet goed behouden blijven tijdens standaard ChIP-procedures. Het precieze promotorengagement blijft daarom onopgelost. RNA-gerichte chromatinebenaderingen, aangepaste crosslinking, tijdsreeksen en experimenten waarin AGO2 direct op de doelwitlocatie wordt gemeten zijn nodig om dit mechanisme verder te onderbouwen.

Daarnaast werd onderzocht of het antisense-transcript *SCN1A-AS1* een rol speelde. Ribo-Zero RNA-sequencing liet geen detecteerbare expressie van *SCN1A-AS1* zien op de door *SCN1A*-10 getargete locatie in AF22-cellen. Een mechanisme dat in dit model afhankelijk is van blokkade of herkenning van dit transcript is daarom onwaarschijnlijk.

Poly(A)-RNA-sequencing bevestigde verhoging van *SCN1A* door *SCN1A*-10, -12 en -14. *SCN1A*-7 liet juist een tendens tot verlaging zien. De verschillende saRNA's veroorzaakten daarnaast duidelijk verschillende transcriptomische effecten. *SCN1A*-10 combineerde relatief sterke doelwitactivatie met het kleinste aantal differentieel tot expressie komende genen en kwam daardoor naar voren als de meest specifieke kandidaat binnen deze vergelijking. Toch kan RNA-sequencing alleen niet vaststellen welke veranderingen directe off-targeteffecten zijn en welke secundair volgen op verandering van *SCN1A* of neuronale routes. Verdere dosis-, tijds- en sequentiespecifieke controles blijven daarom noodzakelijk.

Uitbreiding naar andere epilepsiegenen

In hoofdstuk 4 werd onderzocht of dezelfde ontwerpprincipes konden worden toegepast op andere genen waarbij haplo-insufficiëntie epilepsie veroorzaakt. Er werden saRNA's ontworpen tegen *DEPDC5*, *NPRL2*, *NPRL3*, *TSC1* en *TSC2*, die remmers zijn van mTORC1-signalering, en tegen het chromatinemodellerende gen *CHD2*. Per gen werden tien kandidaten getest.

Slechts een minderheid van de saRNA's gaf een reproduceerbare, meestal bescheiden activatie in HeLa-cellen. Voor *CHD2*, *DEPDC5*, *NPRL3* en *TSC2* werden enkele potentiële kandidaten gevonden, terwijl *NPRL2* en *TSC1* geen betrouwbare activatoren opleverden. In AF22 NES-cellen

kon geen robuuste activatie worden bevestigd en verschillende saRNA's veroorzaakten juist lichte repressie. Ook de p21-positieve controle was in AF22 variabel. Deze resultaten laten zien dat een sequentie die aan bestaande ontwerpregels voldoet niet vanzelfsprekend functioneel is. De toegankelijkheid van de promotor, chromatinekenmerken, de beschikbaarheid van AGO2 en andere RNAa-cofactoren en de cellulaire toestand lijken mede te bepalen of activatie mogelijk is.

Om sequentiekenmerken van actieve saRNA's te verkennen werd een dataset samengesteld met gevalideerde functionele en niet-functionele sequenties. Globale sequentiesamenstelling maakte geen duidelijke scheiding tussen beide groepen mogelijk. Positiespecifieke analyse identificeerde wel mogelijke voorkeuren op afzonderlijke nucleotidenposities en een lokaal stabiliteitspatroon, met relatief sterke basen in het centrale deel en zwakkere basen richting het 3'-uiteinde. Door de kleine groep positieve sequenties zijn deze bevindingen hypothesevormend en nog niet geschikt als zelfstandig voorspellend algoritme. Een bruikbaar ontwerpmodel zal grotere datasets moeten combineren met informatie over chromatine-toegankelijkheid, histonmodificaties, afstand tot transcriptiestartplaatsen en celtype.

Aflevering in neuronen

Hoofdstuk 5 richtte zich op de grootste translationele barrière: efficiënte aflevering van saRNA's in postmitotische neuronen. Conventionele lipofectie werkte goed in delende cellen, maar gaf lage opname in gedifferentieerde AF22-neuronen. Vier lipid-nanoparticle-formuleringen werden daarom vergeleken met RNAiMAX.

Fluorescente cargo liet zien dat meerdere ioniseerbare LNP's, waaronder MC3, SM-102 en LUNG, in meer dan 90% van de gedifferentieerde AF22-neuronen werden opgenomen. Dit was duidelijk hoger dan bij RNAiMAX. Met *EGFP* mRNA werd bovendien aangetoond dat ten minste enkele formuleringen endosomale ontsnapping en functionele afgifte aan het cytoplasma ondersteunden. In AF22 NES-cellen konden MC3-LNP's *SCN1A*-10, -12 en -14 functioneel afleveren en *SCN1A* dosisafhankelijk verhogen.

In gedifferentieerde neuronen leidde de hoge cellulaire opname echter niet automatisch tot sterke promoteractivatie. Bij 50 nM werd geen robuuste verhoging gemeten; alleen de LUNG-formulering gaf bij 250 nM een bescheiden toename. Omdat mRNA-translatie wel plaatsvond, lijkt opname of algemene endosomale ontsnapping niet de enige beperking. De resultaten wijzen erop dat voldoende beschikbaarheid van het saRNA-AGO2-complex in de celkern een belangrijke resterende barrière kan zijn. Dit moet direct worden getest met subcellulaire lokalisatie, nucleaire RNA-kwantificatie en ontwerpen die nucleaire shuttling bevorderen.

Conclusies en toekomstperspectief

Dit proefschrift laat zien dat promotergerichte saRNA's *SCN1A* op RNA- en eiwitniveau kunnen verhogen en levert daarmee een eerste proof-of-concept voor RNA-activatie als strategie voor haplo-insufficiënte genetische epilepsie. Tegelijkertijd maken de resultaten duidelijk dat moleculaire activatie slechts één onderdeel is van therapeutische ontwikkeling. Het mechanisme van promotorherkenning is nog niet afdoende bewezen, de voorspelbaarheid van saRNA-ontwerp is beperkt, respons verschilt sterk per locus en celtype en efficiënte opname in neuronen leidt nog niet vanzelf tot nucleaire target engagement.

De eerstvolgende noodzakelijke stap is aantonen dat saRNA-behandeling een ziektefenotype kan herstellen. Hiervoor zijn menselijke *SCN1A*-haplo-insufficiënte modellen nodig waarin activiteit van inhiberende interneuronen, netwerkbursts, temperatuursgevoeligheid en andere relevante uitkomstmaten kunnen worden vergeleken met isogene gezonde controles. Daarnaast moeten dosis-respons, duur van het effect, directe en indirecte off-targeteffecten, immuunactivatie en mogelijke effecten van overexpressie systematisch worden onderzocht.

De huidige saRNA's zijn gericht tegen de humane *SCN1A*-promotor en herkennen de muizenpromotor niet. Conventionele Dravet-muismodellen zijn daarom niet geschikt om de werkzaamheid van deze specifieke sequenties te testen. Diermodellen blijven wel relevant voor biodistributie, toxiciteit, farmacokinetiek en vergelijking van toedieningsroutes. Voor functionele werkzaamheid zijn patiëntafgeleide neuronen, isogene iPSC-modellen, organoïden en geavanceerde neurale netwerken geschikter.

saRNA's bieden een compacte en potentieel doseerbare vorm van genactivatie, maar bevinden zich nog in een vroeg ontwikkelingsstadium. De uiteindelijke therapeutische waarde zal afhangen van het vermogen om de juiste hoeveelheid *SCN1A* in de relevante neuronale celtypen te herstellen, zonder aspecifieke transcriptieveranderingen of toxiciteit. De combinatie van verbeterde sequentieselectie, mechanistische validatie, neuron-gerichte LNP's of andere afleveringssystemen en functionele testen in humane ziektemodellen vormt daarom de meest logische route naar verdere ontwikkeling.

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of the biggest challenges of the saRNA-project, getting these molecules efficiently into neurons.

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Joep

Curriculum vitae

Joep van de Vondervoort was born on 5 July 1994 in Wijchen, the Netherlands. He studied Medical Biology at Radboud University in Nijmegen, where he specialised in Medical Epigenomics. During his master's programme, he conducted research at the Department of Mitochondrial Medicine, where he investigated CRISPR/Cas9-based strategies to correct pathogenic mitochondrial variants in patient-derived fibroblasts. He subsequently completed a research internship at Glycostem Therapeutics, focusing on the use of self-delivering antisense oligonucleotides to enhance the anti-tumour activity of natural killer cells. He obtained his master's degree in 2020.

In January 2021, Joep started his PhD research at the Department of Genetics of the University Medical Center Utrecht. His research focused on the development of small activating RNAs as a potential therapeutic strategy for haploinsufficient genetic epilepsies. In particular, he investigated whether promoter-targeted small RNAs could increase the expression of *SCN1A*, the gene most frequently associated with Dravet syndrome. His work combined the design and experimental validation of small activating RNAs with transcriptomic analyses, neuronal cell models, calcium imaging and the development of lipid nanoparticle-based delivery strategies. The results of this research are presented in this thesis.

Since October 2025, Joep has worked as a data scientist and data engineer at D-Data in 's-Hertogenbosch. In this role, he works on data science and data engineering projects in which large and heterogeneous datasets are integrated, analysed and translated into practical applications. His work includes the development of predictive models, the design of reliable data-processing workflows and the translation of complex data-driven questions into scalable analytical solutions.